A Lean Blueprint for the IUT III Corollary 3.12 Corridor

IUT Lean formalization project

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Contents

Chapter 1

Aim and Scope

1.1 Purpose of this blueprint

This blueprint records the first formalization target for the repository: the narrow corridor from IUT III, Theorem 3.11 to Corollary 3.12. It is not a blueprint for the whole proof of the *abc* conjecture, and it does not treat the correctness of IUT as an assumption. Its job is to make the objects, comparison levels, indeterminacies, and real-valued estimates visible enough that Lean 4 can distinguish proved consequences from source obligations.

Definition 1.1 (Stage 1 target)**.** The first target is the implication

$$\text{IUT III, Theorem 3.11} \Longrightarrow \text{IUT III, Corollary 3.12.}$$

Following Mochizuki's 2026 formalization report, we refine this into the route

$$\text{Theorem 3.11} \xrightarrow{\text{APT+IPL}} \text{“Theorem 3.11.5”} \xrightarrow{\text{SHE+IPL}} \text{Corollary 3.12.}$$

The intermediate label "3.11.5" is not a separate theorem in the source papers. It names the reorganized boundary obtained by moving the holomorphic hull plus determinant operation before the final simultaneous comparison of the Θ-pilot and the q-pilot.

Remark 1.2. This blueprint is neutral about the Mochizuki–Scholze–Stix dispute. A completed Stage 1 route would say that the stated source hypotheses imply the Corollary 3.12 inequality. It would not by itself prove that those source hypotheses are constructible from IUT I–III.

1.2 Current repository status

Proposition 1.3 (Three levels of availability)**.** *The current Lean development distinguishes three kinds of availability. First, elementary ordered-real consequences are proved directly. Second, route theorems prove that named source-facing records imply the desired signed comparison. Third, some records are still obligations standing for Hodge-theater, Frobenioid, log-Kummer, hull, determinant,* SHE, IPL, *and* APT *constructions not yet built from the source papers.*

Proof. The code contains no proof placeholders in the current source files, and the project builds through the experiment surface. The remaining gap is not a Lean elaboration gap; it is the mathematical construction of the source records consumed by the route theorems. □

Definition 1.4 (Read order)**.** A reader should use the following order. First read the source-facing statement of the Stage 1 route in Chapter 2. Then read the formal interface in Chapter 3. The actual conditional route through the corollary is in Chapter 4. The unresolved source obligations are collected in Chapter 5. Later consequences are only sketched in Chapter 6.

Definition 1.5 (Two status axes)**.** Each blueprint node should be read with two independent status axes. Its Lean status says whether the displayed declaration checks, proves a theorem, or only records a diagnostic report. Its source status says whether the corresponding mathematics has been constructed from IUT I–III, is a conditional record field, is still a source obligation, or is only a later roadmap node. In this project, a Lean-checked record can still be a source-level obligation.

Chapter 2

The Source Stage 1 Route

2.1 The source objects

Definition 2.1 (Initial theta and theater data). The source construction begins with the initial Θ-data of IUT I: a number field, an elliptic curve, bad places, label data, and the arithmetic hypotheses from which Hodge theaters are built. The mod-l representation, its kernel field, containment of $\mathrm{SL}_2(\mathbb{Z}/l\mathbb{Z})$, and positive bad-place q-parameter orders prime to l are explicit predicates and data rather than free propositions.

Definition 2.2 (Hodge theater histories). A Hodge theater carries an arithmetic holomorphic structure. The Stage 1 source now constructs the domain history of the q-pilot and the codomain history of the Θ-pilot as distinct constructors. Their separation is therefore a theorem of the datatype, not an inequality between caller-supplied strings.

Definition 2.3 (Prime strips and links). The $\Theta^{\times\mu}_{\mathrm{LGP}}$-link between the 0- and 1-columns has a value-group portion and a unit portion. The value-group portion relates q-pilot data at $(0,0)$ to Θ-pilot data at $(1,0)$. At the IUT I boundary, prime strips are indexed by their owner history and project the global orbicurve, selected places, local orbicurves, cusps, and q-orders from the same initial theta datum. The $\Theta^{\pm\ell}$ and Θ^{NF} components are glued by an equality of their typed gluing-role prime strips. The later IPL property remains part of the Theorem 3.11 route.

Definition 2.4 (Topological pseudo-monoids). Following IUT I, Section 0, a topological pseudo-monoid is the embedded image of a topological space in a topological abelian group. Its product is defined exactly for pairs whose ambient product remains in that image; the image need not contain the identity and need not be closed under arbitrary products. The restricted product is continuous, commutative, and associative whenever both composites are defined. Divisibility retains both ambient roots and the power-membership equivalence, while cyclotomicity retains the torsion identification with $\mathbb{Q}/\mathbb{Z}$ and stability under torsion. Continuous morphisms preserve defined products, isomorphisms reflect their domain, and continuous group actions are by these pseudo-monoid automorphisms. For Definition 5.2(v), the invariant carrier is constructed as a sub-pseudo-monoid and is proved to inherit every product defined in the original carrier. A profinite action whose kernel acts trivially descends through a surjective profinite quotient; continuity follows from the compact-to-Hausdorff quotient topology, and pullback of the descended action recovers the original one. These results implement the fixed-point and $\pi_1^{\kappa\text{-sol}}$ factorization mechanisms. They do not construct the ambient overgroup, rational fundamental group, or the coric rational-function carriers supplied by the cited absolute-anabelian reconstruction algorithms.

Definition 2.5 (Source Frobenioid prime strips)**.** The transitive Frobenioids I dependency is retained explicitly: $O^{\triangleright}(-)$ is a contravariant functor on the isotropic linear category, and a characteristic splitting is a subfunctor whose product with the units is objectwise bijective and satisfies the isotropic-hull condition. Thus a split Frobenioid is not identified with a Frobenius-degree trivialization. At nonarchimedean places an $F^{\sim}$-component is an actual split Frobenioid; at archimedean places it also carries the compact/noncompact $TM^{\sim}$ factorization, tied to the selected characteristic splitting. Example 3.2(i) identifies the group represented by D_v with the arithmetic fundamental group of X_v. The source model now uses exactly this group, and the sign-quotient geometry constructs its open embedding into the arithmetic fundamental group of C_v, with injectivity and open image proved. This is the geometric target of Definition 5.2(v), not yet the absolute-anabelian algorithm recovering it from an abstract D_v group. For an actual finite place, the algebraic closure is now the filtered topological-module colimit of its finite Galois local subfields. Restricting this colimit to its nonzero integral elements gives continuous finite-stage inclusions that exhaust the carrier, a multiplicative equivalence with $O^{\triangleright}_{\overline{K_v}}$, and the literal compatible absolute Galois action. Normality of each finite stage now gives a continuous restriction of every Galois automorphism; the colimit universal property produces the corresponding continuous ind-automorphism, and Lean proves that these automorphisms satisfy the identity and composition laws and restrict continuously in both directions to $O^{\triangleright}_{\overline{K_v}}$. In accordance with the meaning of the prefix "ind-" in Absolute Anabelian Topics III, Section 0, Lean also retains the finite Galois integral stages as an actual filtered system of topological monoids. Its transitions are continuous and equivariant; continuity of Krull restriction to the finite Galois group, together with its discrete topology, proves joint continuity of evaluation on every stage. The stage inclusions are injective, exhaust $O^{\triangleright}_{\overline{K_v}}$, and commute with the limit action. The global number field is dense in K_v. Primitive elements, same-degree polynomial approximation, continuity of roots, and Krasner's lemma show that every finite Galois stage is generated by a root of a polynomial over the global field. These polynomial-root witnesses form a countable type and surject onto the stage type. The countable filtered-category theorem therefore constructs a monotone cofinal natural-number subsystem, hence the strict positive-integer presentation used by the paper. Reconstruction from the abstract D_v group remains open. A globally realified strip additionally carries the global Frobenioid, its prime/place bijection, all local characteristic and realified monoids, and commuting currency-exchange isomorphisms ρ_v, compared componentwise with the Example 3.5 model. Holomorphic F-prime-strips form the isomorphism-only category required by Definition 5.2(iii), hence admit genuine capsules, and their associated D-prime-strips are functorial. The F-strip associated to a Definition 3.6 theta-Hodge theater is derived, not supplied independently. Its mono-analytic $F^{\vdash}$-strip is constructed only from an explicit structure-preserving transport of the rational monoids, characteristic splittings, and selected archimedean splitting; equivalences of the bare carrier categories are insufficient. Consequently, selected F-level theta-bridge arrows induce a D-theta bridge, with the Corollary 5.3 uniqueness of selected lifts exposed as a separate theorem. Proposition 2.2's construction of the rational-monoid transport, the Examples 3.2–3.5 model constructions, mono-analyticization, global realification, the remaining reconstruction content of Definition 5.2(v), and clauses (vi)–(viii) remain explicit source obligations.

Definition 2.6 (Capsules and processions)**.** The capsule conventions of IUT I, Section 0 and Definition 4.10 are formalized directly. A capsule-full poly-morphism contains all componentwise isomorphisms over a fixed injection. An n-procession has a j-capsule in position j, and its morphisms use order-preserving injections together with capsule-full poly-morphisms. Capsules and processions carry the identity and composition operations required by the categories used in Proposition 4.11. The positive representatives $1, \ldots, l^*$ are proved equivalent to the nonzero Tate labels modulo sign. Consequently, a Proposition 4.7 evaluation-label bijection derives the nested holomorphic l^*-procession and its exact $l^*!$ indeterminacy. An actual natural mono-analyticization functor maps this to the mono-analytic procession while preserving the cardinality and factorial proofs. Construction of that functor from the source subcategories, derivation of the label bijection from evaluation sections, and bridge functoriality remain source obligations.

Definition 2.7 (Toy degree-preorder realization)**.** The typed source constructs local and global effective-divisor degree preorder categories, the theta-link equivalence, and the $\Theta^{\pm\ell}/\Theta^{\mathrm{NF}}$ gluing equivalences. This degree-level test double does not prove Definition 3.6, Corollary 3.7, Definitions 4.1 and 5.2, Proposition 6.7, Remark 6.12.2, or Definition 6.13 of IUT I. Frobenioid and Kummer structure is not asserted by this model.

Definition 2.8 (Mono-theta environments and inverse-limit group)**.** The source boundary for Etale Theta Definition 2.13 uses genuine topological groups, continuous automorphisms, the proved-normal exact range of inner automorphisms, the full inverse image of a subgroup of the outer automorphism group, a cyclic μ_N-kernel, and the conjugacy class of the image of a continuous injective theta section. The IUT II Definition 2.7 system is an actual functor with a certified categorical inverse limit; its ambient homomorphism has the exterior cyclotome as exact kernel and the tempered subgroup as exact image. The two groups in Definition 2.7(ii) are computed as inverse images, and each of its four subquotients is represented by an actual topological quotient U/N by a normal subgroup. Pullback is the derived quotient $f^{-1}(U)/f^{-1}(N)$: the exterior cyclotome is restricted from the exact kernel in part (i), and the other three objects are pulled back from ambient subquotients. Restricted cyclotomic rigidity is a continuous group equivalence between these derived endpoints. Construction from tempered geometry, the three precise ambient subquotients, the full reduction morphisms, and the link to Corollary 2.8 remain source obligations.

Definition 2.9 (Continuous theta-value germs and Gaussian monoids)**.** Continuous nonabelian H^1 is defined as continuous crossed homomorphisms modulo coboundaries. The displayed direct limit over open subgroups is an explicit quotient of cocycle germs. Pullback along a continuous group homomorphism is constructed on representatives and proved to descend to the quotient. For abelian coefficients, pointwise multiplication descends to commutative group structures on both continuous H^1 and its germ colimit, and restriction is a homomorphism. Given the two paper homomorphisms and equivariance with the $\mu_{2\ell}$- and μ-actions, the first and second restricted theta-orbit families are derived rather than stored as output fields. Identity representatives at the zero label survive both restrictions, and every ambient orbit value is proved to map into the corresponding final orbit. The inertia choice and ambient-subgroup choice carry independent copies of the conjugacy action; a two-variable equivariant evaluation descends to the corresponding product-action orbit quotients. The defining infinite-theta condition is stated exactly: membership means that some positive power agrees with a theta value up to torsion. At the constant boundary, the model arithmetic monoid is literally the nonzero part of the integral closure of the MLF valuation ring in its algebraic closure. Its Krull-topological absolute Galois group and algebraic action are constructed, a model pair has a continuous surjective Galois augmentation, and every general pair is required to be isomorphic to such a model. Following Absolute Anabelian Topics III, Remark 3.1.1, the exact T_{M} certificate uses the equivalent topology-free monoid presentation; an arbitrary topology therefore cannot certify a model. The fraction-field theorem identifies the actual Grothendieck group of this cancellative monoid with $\overline{k}^{\times}$. Algebraic closedness makes this group rootable, and injectivity of divisible abelian groups extends the prescribed map $\mathbb{Z} \to \overline{k}^{\times}$ across $\mathbb{Z} \to \mathbb{Q}$. Thus coherent compatible rational roots are now constructed from the literal model and transported to every model pair; Kummer theories must realize those roots rather than supply arbitrary ones. Two systems agree on integer exponents, so their quotient at q has order dividing the denominator of q. This finite-order element and its inverse are integral, hence construct an actual unit of $O^{\triangle}_{\overline{k}}$; transport through the model isomorphism constructs every root-system comparison rather than assuming it. Applying Galois to the constructed root system and comparing it with the original constructs every integral Galois root ratio from a fixed-point proof. Krull stabilizers of the underlying algebraic integers are open and transport to the general pair. The root ratios are locally constant in discrete units, hence continuous in the pointwise cyclotome topology. The latter is a closed subspace of a product of finite root-of-unity groups, so Lean proves that it is compact, Hausdorff, totally disconnected, and bundles it as a profinite space. The ratios are proved multiplicative, killed on $\mathbb{Z}$, descended through $\mathbb{Q}/\mathbb{Z}$, and assembled into continuous cocycles. Comparing two root systems gives the explicit coboundary between their cocycles. Hence both Kummer maps $M^H \to H^1(H, \mu_{\mathbb{Z}}(M))$ and $M \to \varinjlim_J H^1(J, \mu_{\mathbb{Z}}(M))$ are derived monoid homomorphisms, and the canonical local map is proved compatible with the canonical germ map. The unit Kummer homomorphism is its restriction, never an independently supplied field; injectivity proves that M^{μ}_{TM} is exactly the image of the torsion units. Proposition 3.2(ii) constructs the map but does not assert injectivity. Following Definition 1.5 and Remark 1.5.4(i), Lean now proves that the integral-closure unit group in every finite MLF extension is residually finite: Krull intersection separates a nontrivial unit, while reduction modulo a suitable power of the base maximal ideal gives a finite quotient. For a compact IUT acting group, Lean also proves that the image of each open germ domain under the arithmetic augmentation is open: it has finite index, and compactness makes its image closed. Infinite Galois correspondence then constructs the associated finite fixed extension. The remaining precise source obligation is to extract, from a zero constructed Kummer germ, compatible adjusted roots lying in that extension and thereby derive injectivity. At a finite place, the Definition 4.9 input now ties the selected splitting at an isotropic reference object to an actual MLF-Galois TM-pair, a stable characteristic submonoid, and the genuine Ism Kummer orbit. Lean constructs the bad-place quotient by $\mu_{2\ell}$ and the good-place characteristic factor, forms the literal product with $O^{\times\mu}$, and proves that this is its complete unit group. The reconstruction output must have the original base category and this exact reference rational monoid, so an unrelated Frobenioid cannot inhabit the interface. Derivation of that output from Frobenioids I, Theorem 5.2(ii), remains explicit. At an archimedean place, the exact N-stage replaces the compact factor by S^1/μ_N and retains the original noncompact factor. Lean proves that this noncompact factor has no nontrivial units and constructs the continuous transition which is the torsion-quotient map on the compact factor and the identity on the noncompact factor. A reconstructed stage must be an actual split Frobenioid whose split topological monoid is structure-preservingly isomorphic to this exact product and whose base is equivalent to the original base. Its transition functors preserve the base projection, FSM morphisms, divisors, and Frobenius degrees. Carrier categories, bases, and reference rational monoids then form strict divisibility-indexed diagrams. Every multiplicatively cofinal restriction is proved final, so it preserves and reflects the universal colimit cocone. Construction of these stages and transition functors from the original archimedean TM/Frobenioid data remains explicit. Definition 4.9(vi)–(viii) is now represented by a universe-polymorphic $F^{\vdash\times\mu}$-prime-strip boundary. It fixes the bad/good finite construction at every selected finite place and the complete torsion-quotient system at every archimedean place. The corresponding $F^{\times}$- and $F^{\times\mu}$-components are the actual wide subcategories of divisor-zero isometries. The global record retains the realified Frobenioid, prime/place bijection, and every ρ_v. Its pilot is computed by transporting typed cyclic bad-place generators through these ρ_v's and applying a global reconstruction function, and its arithmetic degree is required to be negative. Supplying this reconstruction from Corollaries 4.6 and 4.10 remains open. At each finite place, the structure-preserving isomorphism layer now transports the topological Galois group, arithmetic action, unit-mod-torsion quotient, every open-subgroup invariant Kummer image, both product generators, the reconstructed split Frobenioid, its base, reference object, and rational monoid. Identity and composition are constructed, and preservation of the genuine Ism-orbit is derived. At an archimedean place, the system isomorphism transports the original split Frobenioid, every torsion quotient, quotient and orientation maps, level transitions, reconstructed stages, bases, divisors, Frobenius degrees, FSM morphisms, split products, and rational transitions. Identity and composition are constructed. The level maps restrict to the isometry coarsifications and form a natural transformation of complete divisibility diagrams with equivalence components. The assembled globally realified isomorphism now combines these place-indexed local equivalences with a structure-preserving equivalence of the global Frobenioids, the prime/place and ρ_v squares, bad-place generator and currency compatibility, reconstruction for every character tuple, and arithmetic-degree compatibility. Identity and composition are constructed. Compatibility of the distinguished pilot character, pilot object, and pilot degree is derived. Following IUT I, Section 0, the place-indexed and globally realified full poly-isomorphisms are literal quotients of all complete representatives. Categorical equivalences are identified by natural isomorphism, while group, monoid, quotient, splitting, prime, and local currency maps remain pointwise data. Identity, composition, unit laws, associativity, representative existence, and exact nonemptiness criteria are proved. Constructing and populating these records from the source evaluation and reconstruction algorithms remains open. A pointed inversion carries an actual decomposition subgroup; restriction along its continuous inclusion derives torsion detection from the precise equality $M^{\times}_{\mathrm{TM}} \cap \mathrm{res}^{-1}(\mathrm{tors}) = M^{\mu}_{\mathrm{TM}}$, and hence derives the splitting. Quotienting an actual constant subgroup by its exact torsion subgroup gives a precise multiplication parametrization; it is proved bijective when every zero-label theta value is constant torsion. Given a zero-to-base action-pair comparison, the zero-to-label comparisons derive a graph proved equal to the diagonal range from Corollary 3.5(i). The zero factor is continuously and multiplicatively equivalent to this graph, equivariant for all transported labeled actions. Multiplication by $(\mathbb{Z}/\ell)^{\times}$ descends to the exact EtaleTheta sign-label quotient, factors through $(\mathbb{Z}/\ell)^{\times}/\{\pm 1\}$, and the resulting action on all nonzero absolute labels is proved transitive. After sign identification, symmetrization retains one topological group/monoid action pair at every nonzero absolute label; its outer transports satisfy identity and multiplication coherence, and action-compatible isomorphisms derive an injective diagonal monoid map. Its actual range is a submonoid continuously and multiplicatively equivalent to the base monoid, and this equivalence intertwines the transported action. Thus the label set is not replaced by a coarse one-point quotient. Its diagonal restriction-isomorphism synchronization is represented first by the exact quotient for the equivalence relation "differ by one global inner action." More precisely, the full compatible diagram $\Pi_X \leftarrow \Pi_v \leftarrow G_v^{\langle|F_l|\rangle}$, together with its equivariant monoid isomorphism, is quotiented by one base-group element that acts simultaneously on the group map and monoid map. This full quotient maps to the monoid-only quotient. Thus the conjugator cannot depend on a label, group element, or monoid element. Construction of the diagram and action from the paper data remains open.

The diagonal unit submonoid and finite Gaussian monoid are constructed inside the heterogeneous direct product of these labeled factors. The base group acts diagonally through the symmetrizing group isomorphisms; constants are equivariant, diagonal units are stable, and profile stability induces an actual action by automorphisms of the Gaussian monoid. For a value profile, the homogeneous finite Gaussian monoid is the submonoid of the direct product generated by diagonal units and natural profile powers; a chosen compatible root system gives the rational-root Gaussian monoid, with root ambiguity stated as diagonal torsion. Root systems differing by this permitted ambiguity are proved to generate the same infinite Gaussian monoid. The specialized profile consists of the actual elements q^{j^2}. The n-torsion of $\mathbb{Q}/\mathbb{Z}$ is proved to have exactly n elements and is explicitly equivalent to the roots-of-unity group of a torsion-cyclotomic monoid. A multiplicative realization of the $\mu_{2\ell}$-action is required to be injective, hence acts freely, and derives its exact torsion-unit range and the theta-value family from the continuous H^1 orbits; the resulting multiplicative orbit set is proved equal to the original action orbit at every label. Any two simultaneous choices from it are proved to differ by independent coordinatewise 2ℓ-torsion. Its representative family constructs a canonical common 2ℓ-multiple Gaussian submonoid. The ambiguity vanishes after taking 2ℓ-th powers, so every selected profile yields exactly this common object. The source Hodge–Arakelov pilot places every q^{j^2} in its actual continuous-H^1 theta orbit and every compatible positive root in the corresponding infinite-theta orbit. It derives both Gaussian monoids, every 2ℓ-element orbit cardinality, the full $(2\ell)^{\ell^*}$ profile count, and zero-label torsion. Multiplicative equivalences preserve the exact constant-torsion subgroup, the two quotient factors, and splitting bijectivity. The finite and infinite Gaussian carriers are proved to be precisely the products $\Psi_{\mathrm{cns}}^{\times} \cdot \xi^{\mathbb{N}}$ and $\Psi_{\mathrm{cns}}^{\times} \cdot \xi^{\mathbb{Q}_{\geq 0}}$. A multiplicative finite-order action proves that a zero-label orbit based at the identity consists entirely of torsion. Thus the actual composite continuous-H^1 restriction constructs both finite and infinite zero-label splittings once torsion detection on the constant subgroup is proved. More generally, a restriction homomorphism that sends theta values to torsion and detects torsion on constants is proved to derive the splitting used in Corollaries 1.12, 2.6, 2.8 and Proposition 3.1. Consequently, once that restriction datum and the Corollary 2.8 Gaussian restriction equivalence are supplied together with the two factor identifications, both Gaussian splittings and their compatibility are derived without accepting a Proposition 3.1 splitting proof as input. Construction of the paper's subgroup/subquotient and decomposition-group homomorphisms, their actions and input orbits from the mono-theta system, the symmetrizing isomorphisms, root choices, independent conjugacy compatibility, and both restriction maps with their torsion and factor-identification proofs remain source obligations.

Definition 2.10 (Toy Hodge–Arakelov degree pilots)**.** This finite degree model computes the positive, l-prime q-order and the square-weight profile expected from q^{j^2}. At each bad place, the calculation uses the theta-root torsor, its base-point coordinate equivalence, typed $\mathbb{Z}/l\mathbb{Z}$ labels, sign classes, balanced-square weights, and coordinate, weighted, and sign-orbit averages. Positivity, normalization, sign invariance, and translation invariance are proved. It does not construct the cohomology classes, restriction algorithm, Gaussian monoids, or Gaussian Frobenioids of IUT II, and therefore discharges no paper-level evaluation claim.

2.2 Theorem 3.11 and Step (x)

Definition 2.11 (Source-shaped Theorem 3.11 interfaces)**.** The source layer now represents a local packet by the genuine dependent tensor product of its $j+1$ place-indexed direct sums. The $I_{\mathbb{Q}}(S_{j+1,j}; v)$ submodule is the span of pure tensors whose distinguished factor lies in the summand indexed by v. Integral regions must be either nonarchimedean lattices or archimedean seminorm unit balls and must prove their rational spans. The $\mathbb{Z}^2$-indexed lattice contains actual F-theta bridges and distinct Hodge theaters; its horizontal theta-links, mono-analytic processions, and vertically coric procession maps are constructed. Ind1 is derived from a functor on the category of processions. Ind2 accepts only the independent actions on individual place summands; the direct-sum action, tensor action, full dependent-product action, and their laws are derived. The complete presentation also carries its bad-place splitting monoids, number-field embeddings, and objects of the four Frobenioid categories. Its exact quotient is the equivalence closure generated by Ind1 and Ind2. A bi-coric bridge member induces a procession isomorphism, and a transport intertwining both actions descends, by a proved relation-induction argument, to an equivalence of these quotients.

The vertical interface uses placewise continuous linear Kummer equivalences, their induced tensor maps, complete localization diagrams of topological group/pseudo-monoid action pairs, number-field ring equivalences, splitting-monoid equivalences, and structure-preserving equivalences of the realified and non-realified split Frobenioids. Ind3 fixes one vertically coric $n, \circ$ container while m varies. At nonarchimedean places it contains both direct Kummer images and forward iterated-log images. The direct finite-place subgroup is no longer supplied as arbitrary packet data: it is generated by pure tensors of the summandwise invariant-unit logarithm images. Each such invariant-unit type is itself the fixed subgroup of an explicit continuous profinite Galois action, rather than an unrelated type carrying an invariant name. The generated packet subgroup lies in the closed tensor lattice and is transported exactly by the vertical Kummer equivalence. A finite log-link step is represented as a partial correspondence between the actual fixed units in every summand. Lean applies the summand logarithms, tensors related tuples, takes the additive closure of the resulting graph, and recursively composes adjacent steps. It proves that an (r)-fold iterate starting at (m-r) ends at (m), and derives containment of the endpoint's Kummer image from its unit-log subgroup membership. Thus neither a direct multi-step map nor a final iterated-containment proof is accepted by the concrete Ind3 constructor. At an archimedean summand the Hermitian seminorm is normalized by π, so its unit ball is exactly Definition 1.1(ii)'s raw complex radius-π disk; there is no second radius-π ball of the normalized metric. Lean uses $\arg(z)i$ to construct a principal logarithm in this ball for every genuine S^1-unit and proves that exponentiating recovers the unit. Choosing one place per tensor factor, the direct-sum and tensor metric equations put the resulting pure unit tensor, and every selected local radius-π tensor, in the packet unit ball. Each unit is also canonically a point of the exact S^1 core of an antitone projective system of open annuli. Lean identifies their intersection with S^1 and proves that they are cofinal among all neighborhoods of S^1, restricts ambient complex differentiability to every level, proves that ambient multiplication and inversion preserve the neighborhood filters, and proves that deleting S^1 leaves the two radial connected components. The component determined by $\mathcal{O}_{\mathbb{C}}^{\triangleright} \setminus S^1$ is selected by a connected-component theorem rather than an arbitrary input. For every connected annulus U_n, Lean now derives the actual group $\mathrm{Aut}_{\mathrm{hol}}(U_n)$ from biholomorphic self-maps and proves identity, composition, inversion, and restriction laws. For a groupification-induced holomorphic automorphism of $\mathbb{C}^\times$, pullback along the exponential covering yields an integer exponent; bijectivity restricts it to 1 or -1, and preservation of the selected inner component rules out inversion. The induced filter germ is therefore proved equal to the identity. An infinite-place log-link step is only a partial correspondence on these semi-germ core-point tuples. Recursive composition derives every positive iterate, its domain and range, the participating endpoint-ball subset, and relational surjectivity onto the source domain. Direct

Kummer containment follows from the full vertical unit-ball image. Thus the concrete Ind3 constructor accepts no arbitrary archimedean subgroup, multi-step domain, preimage set, iterate map, surjectivity proof, or direct containment. The general Riemann-surface and RC-holomorphic local-morphism category of *Absolute Anabelian Topics III*, Definition 2.1 and Corollary 2.3, extension of rigidity from groupification-induced maps to every purely local group-germ automorphism, and co-holomorphic Kummer reconstruction are not yet formalized. Equality to the common measure-derived log-volume implies independence of m. A multiplicative log-link into a rational module provably kills roots of unity. At a bad prime, cross-link the actual unit groups of the splitting monoids map multiplicatively into the log-link domain. Cross-link admissibility is derived from membership in the current and next domains together with an explicit root-of-unity quotient in the next domain; it is not an arbitrary relation field. Admissible inputs have the same additive log image. Horizontal compatibility is no longer an unindexed field of a single column. Its boundary contains two column assemblies, proves that they use the same lattice at indices n and $n+1$, and fixes all four corners of every square. The active clauses (iii)(a)–(b) interface now accepts only the Definition 4.9 $F^{\vdash\times\mu}$-prime-strips. Each triangle corner is computed by one functorial reconstruction algorithm from the mono-analytic transport of its actual site or vertically coric theater. The same algorithm computes every environment strip and comparison from the Proposition 2.1(vi) mono-analytic data and derives the inverse laws. Thus no prebuilt times-μ family enters the active corridor. The ordinary-F square remains only as precursor scaffolding. Following the convention of IUT I, Section 0, a times-μ prime-strip full poly-isomorphism is represented by a componentwise equivalence map modulo natural isomorphism on its categorical components; it is not a strict categorical isomorphism of prime-strip records. Lean proves that identity and composition descend to these classes, together with associativity and cancellation of selected inverse pairs. A prime-strip square quantifies in both directions over all horizontal classes; its chosen upper and lower classes only witness nonemptiness. The Corollary 4.10(iv) is retained as a representative-level rigidity law for every actual adjacent horizontal pair: lifting the underlying mono-analytic map of any output equivalence recovers that output up to structured natural isomorphism. Lean proves from this law that every times-μ horizontal class has a mono-analytic preimage. The corridor accepts neither a clause (a) square, a chosen theta-link member, an arbitrary quotient-level fullness witness, nor chosen Kummer maps. Every mono-analytic strip carries structure-preserving maps to and from the common model with two-sided Section 0 inverse laws. Lean therefore derives the canonical theta-link and both Theorem 1.5 Kummer directions, proves their cancellation, reconstructs the upper and vertical maps, defines the lower map by Kummer conjugation, and proves commutativity and both all-class transport directions. The triangle and environment prime-strip squares range over every m; the environment family carries two-sided natural comparison classes with the triangle strips, and its square is proved class-by-class in both directions by conjugating the triangle square and cancelling those comparisons. These facts validate the exact typed coarsification, reconstruction, fullness boundary, and conjugation mechanism. Constructing the finite/archimedean outputs, proving representative-level reconstruction fullness, and deriving the coherent model comparisons from Corollaries 4.5, 4.6, 4.10 and Theorem 1.5 remain open. The mono-theta square ranges over every m and bad place and uses actual projective systems with levelwise environment realizations. Clause (iii)(d) no longer collapses $\{\pi_1^{\kappa\text{-sol}} \ltimes M_\infty^\kappa\}$ and its localizations to two profinite groups. For every m, absolute label, and selected place, it retains the global action pair, the local M_∞^κ and $M_\infty^{\kappa\times}$ action pairs, the equivariant localization map, and the equivariant topological embedding that realizes the displayed inclusion. These are genuine partial-product pseudo-monoids, not total-monoid surrogates. Horizontal and vertical Kummer isomorphisms preserve both arrows, and site/common coherence is equality of the complete diagram transports. The resulting arrow-category squares transport the full automorphism groups. Constructing these diagrams from Corollary 4.8 and the cyclotomic-rigidity algorithm remains open. Compatible evaluations descend to the equivalence relation generated jointly by Ind1–Ind3.

These declarations remain a partial construction of Theorem 3.11. The tensor packets, vertical Kummer maps, finite integral images, and finite direct unit-log containment are constructed from the typed lattice data. Constructing that lattice data, its summand unit/log inputs, splitting monoids, fields, and Frobenioids from the preceding source theorems remains open, as do the general RC-holomorphic local-morphism classification and rigidity for arbitrary purely local group-germ automorphisms, construction of the adjacent finite and infinite unit correspondences from the Gaussian log-theta lattice, and construction of the indexed horizontal isomorphisms and their equivariance from the actual theta links.

Definition 2.12 (Source full vertical log-links)**.** A selected-place full log-link in the sense of IUT III, Definition 1.1, consists of a continuous rational-linear post-composition between the actual ind-topological logarithmic modules, a compatible ring equivalence of their realized logarithmic fields, and continuous injective realizations of the adjacent target's local units. At a finite place the adjacent relation asserts that the source p-adic logarithm equals a realized target invariant unit. At an infinite place it asserts the same equality for the source principal logarithm and a realized target circle unit. These are partial relations: existence is not assumed, while injectivity proves uniqueness of any target.

A family of these links on all adjacent lattice sites derives the finite packet graph and archimedean semi-germ relation componentwise, and hence constructs the complete $Ind3$ upper-semi input. Constructing the full links from the cited Psi_{cns}, Frobenioid, and poly-isomorphism data remains a source obligation.

Definition 2.13 (Assembled source Theorem 3.11 column and corridor boundaries)**.** At one fixed horizontal column, the source assembly forces the multiradial presentation and the vertical family to use the same common local construction. Its inputs are the absolute-label Gaussian log-theta lattice, mono-analytic log-shell algorithm, finite-stage integral data, full local log-links, splitting and labeled Kummer data, and the multiradial functor. From these Lean derives the adjacent finite and archimedean packet relations, the actual procession action, independent summand action, generated Ind1/Ind2 quotient, vertical packet Kummer maps, Ind3 system, vertical log-volume independence, and roots-only bad-prime logarithm equality. A separate corridor joins two such boundaries on the same lattice at adjacent columns and carries the universally indexed horizontal squares and their coherence with vertical Kummer. Its labeled clause (d) data preserve the full global/local Corollary 4.8 localization diagram and every local inclusion, rather than a collapsed profinite-group arrow.

This assembly is not an existence proof for its upstream inputs. Its role is to replace the unrelated finite adapters by one coherent source boundary and to identify precisely which constructions from IUT I–III remain.

Definition 2.14 (Indeterminacy generator interface)**.** The Stage 1 interface names the three source-facing indeterminacy generators separately: procession automorphism, local tensor symmetry, and upper-semi compatibility. This is a Lean-checked naming interface, not yet a construction of the source Step (x) estimates.

Definition 2.15 (Step (x) averaged corridor record)**.** The averaged corridor record is the conditional Lean shape of Step (x): Ind1 and Ind2 preserve the procession-normalized average, and Ind3 supplies the upper-semi bound. Its Lean theorem proves the resulting averaged inequality from those record fields.

Proposition 2.16 (Step (x) corridor)**.** *The source Step (x) data should produce a corridor*

$$A_0 = A_1 = A_2 \leq U_{\mathrm{Ind3}},$$

where A_0, A_1, A_2 *are the averaged log-volumes before indeterminacy, after* Ind1*, and after* Ind2*, and* U_{Ind3} *is the upper-semi target coming from* Ind3*.*

Proof. In the current code this is an interface-level object. The proof from the source papers must still account for the log-volume invariance under Ind1 and Ind2, the upper-semi orientation of Ind3, and the normalization conventions for averaging over labels. □

Definition 2.17 (Legacy finite M3 procession model)**.** The numerical Gaussian pilots construct a finite coordinate model with two actual $\mathbb{Z}/l\mathbb{Z}$-translation actions. Their generated equality quotient preserves orbit averages and nonempty possible images. Natural-number lifts give the correctly oriented Ind3 upper-semi relation and do not collapse in that quotient. At every model point, the bad-local quotient torsor maps bijectively to the evaluated vertical target; its graph, range, and log-volume image are exact, and its finite discrete image is compact-open. The horizontal object retains the distinct zero/one histories and carries the typed theta link, gluing, Gaussian evaluation, and finite transport laws named IPL/SHE/APT. This definition is a regression model and does not discharge Theorem 3.11.

Theorem 2.18 (Legacy degree-level Theorem 3.11 wrapper)**.** *Initial* Θ*-data constructs a finite degree-level regression wrapper shaped after Theorem 3.11, including model multiradial possibilities and* Ind1, Ind2*, and* Ind3 *behavior. The older qualitative packet is not a field of this source and is available only through an explicitly named legacy adapter. It does not construct the source objects of Definition 2.11 and does not prove Theorem 3.11. Analytic Step (xi) hull and determinant enrichment is outside this wrapper.*

2.3 Step (xi)

Definition 2.19 (Simultaneous holomorphic expressibility)**.** The SHE property says that the relevant output possibilities may be expressed relative to the arithmetic holomorphic structure in the 1-column while keeping the q-pilot log-volume fixed as input data.

Definition 2.20 (Algorithmic parallel transport)**.** The APT property records how the multiradial algorithm transports data through gluings up to indeterminacy. It belongs primarily to the implication from Theorem 3.11 to the reorganized hull-plus-determinant checkpoint.

Definition 2.21 (Hull plus determinant checkpoint)**.** The HDD checkpoint is the composite of descent with holomorphic hull and determinant formation. It is the source-side operation that turns multiradial output regions into objects comparable with the q-pilot at the level of signed real log-volume.

Proposition 2.22 (Step (xi) upper-ray input)**.** *The multiradial construction, followed by holomorphic hull, determinant, and log-volume, should produce a signed upper ray*

$$\mathbb{R}_{\leq -|\log(\Theta)|}$$

of possible output log-volumes, and the q*-pilot reading* $-|\log(q)|$ *should lie in this ray.*

Proof. This is the central source obligation. It is exactly where the comparison must avoid identifying the arithmetic holomorphic histories too early while still obtaining a common log-volume container in which the final inequality is meaningful. □

Proposition 2.23 (Localized Step (xi) C_Θ source)**.** *The current source-derived exact-*$\Xi(P_B)$ *route starts from compact-open log-Kummer map data over the transported theta-region family and constructs the graph, pointwise log-shell, valuation-ball, and Remark 3.9.5 exact-family audit internally. The strongest local-arithmetic valuation-cover boundary then starts from a named localized vector-bundle determinant source: its weighted adjusted local terms give the Step (xi) finite sum for the canonical* C_Θ*-scale, while the local arithmetic upper term is decomposed into the Step (xi) term, a Haar normalization defect, and a main-log term. The localized-determinant constructor removes the separate determinant-source handoff: the constructed Remark 3.9.5 source is projected from the same localized vector-bundle determinant source.*

Proof. Lean first carries the compact-open log-Kummer source through the source-derived exact-$\Xi(P_B)$ global C_Θ audit. At the local-term boundary it identifies the localized determinant source with the formula-matching Step (xi) source by construction, derives the displayed Ob3/Ob4 and valuation-cover handoffs, and invokes the finite-sum closed Corollary 3.12 endpoint. □

Chapter 3

Formal Interface

3.1 Real-line copies and regions

Definition 3.1 (Labeled real-line copies)**.** The formalization treats ordered real-line copies and positive-scale transports explicitly. A comparison cannot silently identify a Θ-pilot real line, a q-pilot real line, and the final signed log-volume target unless a transport has been supplied.

Definition 3.2 (Regions and measures)**.** Region comparisons package a source-to-target transport together with a target region. Region measures are abstract monotone real-valued measures used to model log-volume estimates.

Definition 3.3 (Common target bound)**.** A common target bound records that every choice in a family of transported regions is bounded by the same measured target. This is the formal shape of the post-hull upper-bound interface.

3.2 Algorithmic output and source ledger

Definition 3.4 (Algorithmic output)**.** An algorithmic output packages transported region-family data together with qualitative certificate slots for IPL, SHE, APT, hull plus determinant, and common-container compatibility.

Definition 3.5 (Source obligation ledger)**.** The source obligation ledger records the data required to promote a charted common-container comparison to the signed Corollary 3.12 endpoint. It keeps the chosen output, membership proof, target volume, Θ bound, q-pilot positivity, and normalization proof in one auditable package.

Lemma 3.6 (Ledger three-term comparison)**.** *The ledger proves $q_{\text{signed}} \leq \Theta_{\text{signed}}$ by the explicit chain*

$$q_{\text{signed}} \leq \textit{chosen target volume} \leq \Theta_{\text{signed}}.$$

The second comparison is tied to the charted common bound produced by the supplied certificate; it is not an arbitrary convenient Θ bound.

Proof. The Lean equalities expose the comparison as the ledger's three-term comparison and, more concretely, as membership in the chosen target followed by the common-bound inequality. $\square$

Definition 3.7 (Stage 1 comparison endpoint)**.** A Stage 1 comparison endpoint is the final signed real comparison

$$q_{\mathrm{signed}} \le \Theta_{\mathrm{signed}},$$

tagged by the q-pilot and Θ-pilot sides. The endpoint type is not a source proof; the source route must construct it from the ledger.

Lemma 3.8 (Projection warning)**.** *Once a completed endpoint has been supplied, the formal Corollary 3.12-shaped inequality is a projection from that endpoint.*

Proof. The Lean theorem intentionally does no source mathematics. It prevents us from confusing the endpoint package with a proof that the package is constructible from IUT I–III. □

3.3 Signed real algebra

Definition 3.9 (Corollary 3.12 predicate shape)**.** The bare predicate states the target signed pilot inequality for already-tagged Θ-pilot and q-pilot log-volumes. It is a proposition shape, not by itself a proof that source obligations construct the needed comparison.

Lemma 3.10 (Signed comparison gives the corollary shape)**.** *If $q_{\mathrm{signed}} \le \Theta_{\mathrm{signed}}$, then the corresponding tagged pilot log-volumes satisfy the Corollary 3.12 inequality.*

Proof. This is a direct sign conversion: the stored pilot log-volume value is the positive magnitude, while the source display is written in signed form. □

Theorem 3.11 (Final ordered-real calculation)**.** *Suppose $q_{\mathrm{signed}} \le \Theta_{\mathrm{signed}}$, $0 < -q_{\mathrm{signed}}$, and*

$$\Theta_{\mathrm{signed}} \le C_\Theta \cdot (-q_{\mathrm{signed}}).$$

Then $-1 \le C_\Theta$.

Proof. This is pure ordered-real algebra. The hard source issue is therefore not this calculation, but the construction of the signed comparison consumed by it. □

Theorem 3.12 (Strict and boundary alternatives)**.** *Under the same hypotheses, either $C_\Theta = -1$ or $-1 < C_\Theta$. In the boundary case, the formal corridor also proves equality of the signed q-pilot and Θ-pilot readings and negativity of the Θ-pilot reading.*

Proof. The endpoint is a dichotomy for the final signed real inequality. It should be read only after the source route has supplied the comparison $q_{\mathrm{signed}} \le \Theta_{\mathrm{signed}}$. □

Chapter 4

Route to Corollary 3.12

4.1 Diagnostic finite label and Gaussian layer

Definition 4.1 (Finite label square-weight profile)**.** The current finite model records the j^2-weighted Gaussian degree behavior of theta labels. It separates raw coordinate square weights, sign-quotient full labels, zero and nonzero classes, and several averaged log-volume conventions.

Proposition 4.2 (Diagnostic: no silent scalar absorption)**.** *At the representative level, one label-independent scalar cannot absorb all j^2-weighted theta degrees when the environment degree is nonzero. This is a counter-collapse diagnostic: it shows that the formalization does not silently erase the q^{j^2} issue inside a real-line identification, but it does not discharge the source Step (xi) hull/log-volume comparison.*

Proof. This is represented in the finite-label experiment surface. The exact Lean declarations distinguish representative square weights from descended full-label data and from the final hull/log-volume endpoint. □

4.2 Nonarchimedean upper-semi route

Definition 4.3 (Local Ind3 orientation)**.** The experiment dashboard is a Lean-checked status report. It records that missing real alignment blocks the final route, that nonarchimedean Ind3 entries have the current packet-to-theta orientation, and that archimedean entries have the reverse orientation for this route. It should not be read as a source construction of the Ind3 comparison.

Theorem 4.4 (Nonarchimedean entry reaches the signed comparison)**.** *A nonarchimedean local* Ind3 *entry, together with the required real-line/log-volume alignments and structured* SHE *square-weight transport audit, implies*

$$q_{\text{signed}} \leq \Theta_{\text{signed}}.$$

Proof. The theorem composes the local nonarchimedean alignment with the Hodge-descent packet-to-theta route. It still assumes the relevant source-facing records; it proves that those records have the correct formal shape to feed the endpoint. □

Theorem 4.5 (Strongest current packet-calibrated route)**.** *In the strongest current route, Gaussian identity at the canonical full label* 1*, the finite Kummer-plus-forgetting product-log-volume chain, packet-normalized theta-source equality, packet-normalized* Ind3 *target calibration, and packet-normalized entry-target calibration imply*

$$(q_{\text{signed}} = \Theta_{\text{signed}} \ \wedge\ \Theta_{\text{signed}} < 0) \quad or \quad -1 < C_{\Theta}.$$

Proof. The route first constructs the needed nonarchimedean log-Kummer upper-semi compatibility from the Kummer transfer and mono-analytic forgetting chain. It then derives the target-side equalities through packet-normalized calibration rather than taking the direct theta-to-entry equality as a primitive input. □

Definition 4.6 (Packet-calibrated hypothesis ledger)**.** The strongest current route consumes a large family of source-facing hypotheses: a source package, hull/determinant obligations, a log-volume chart audit, a Hodge-descent transport packet, source and target Gaussian profiles and evaluations, canonical full-label 1 preservation, source and target log-volume equalities, finite Kummer transfer, mono-analytic forgetting, a nonarchimedean entry and membership proof, packet-normalized calibrations, q-pilot positivity, and the displayed C_Θ upper bound. This node is intentionally marked not ready: the theorem checks the conditional route after these inputs are supplied, but the source construction of the inputs remains the mathematical work.

4.3 The Corollary 3.12 node

Definition 4.7 (Θ finite-real endpoint)**.** Corollary 3.12 starts from an extended signed Θ log-volume and must conclude that the endpoint is finite real-valued. The current Lean theorem projects this fact from a statement-endpoint record; the source construction of that record remains open.

Definition 4.8 (q-pilot positivity)**.** The final C_Θ algebra requires $0 < -q_{\text{signed}}$, or equivalently that the signed q-pilot log-volume is negative. The current theorem extracts this from the statement-endpoint record; a source proof must still construct the record from the prime-strip data.

Definition 4.9 (Displayed C_Θ upper bound)**.** The displayed source hypothesis

$$\Theta_{\text{signed}} \le C_\Theta \cdot (-q_{\text{signed}})$$

is a separate input to the ordered-real lower bound for C_Θ. It should remain visible rather than being folded into the signed comparison.

Theorem 4.10 (C_Θ lower bound from signed data)**.** *Once the signed comparison, q-pilot positivity, and displayed C_Θ upper bound have been supplied, the ordered-real conclusion $-1 \le C_\Theta$ is Lean-checked algebra.*

Proof. This is not source mathematics; it is the real-inequality consequence of the signed endpoint data. □

Corollary 4.11 (Conditional Corollary 3.12 endpoint)**.** *If the source constructions of IUT III supply the required multiradial output,* IPL, SHE, APT*, hull-plus-determinant comparison, log-Kummer upper-semi compatibility, and packet-normalized calibrations, then the Corollary 3.12-shaped signed pilot inequality follows. The Lean tag points to the conditional route theorem from the source ledger, not merely to the bare predicate that states the target inequality.*

Proof. The Lean code already proves the endpoint from the named route records. What is not yet proved is that the source papers construct all of those records without collapsing the histories, labels, or comparison levels that the dispute requires us to keep separate. ☐

Chapter 5

Open Source Obligations

5.1 The obligation boundary

Definition 5.1 (Source obligation gap)**.** The source obligation gap is the current formal boundary between proved route algebra and unconstructed source mathematics. It records the missing certification of the algorithmic output, the SHE-certificate alignment, q-pilot positivity, and the normalization condition needed by the ledger.

Proposition 5.2 (Gap promotion is formal)**.** *Once the source obligation gap fields are supplied, they promote to the source-obligation ledger consumed by the current Stage 1 route.*

Proof. This promotion is bookkeeping. It is valuable because it marks exactly which source fields remain mathematical obligations rather than burying them inside the final comparison theorem. □

5.2 Source-native Definition 3.1 foundation

Definition 5.3 (Etale theta quotient input)**.** The construction preceding Etale Theta Definition 2.1 is represented by the actual lower-central and exponent-l theta quotient, its distinct rank-two arithmetic elliptic quotient, linked arithmetic and cuspidal exact sequences, a continuous surjective rank-one profinite Lagrangian quotient on the elliptic quotient that is trivial on the cuspidal decomposition subgroup, its derived open kernel, the direct-product inversion eigenspaces with actual arithmetic-conjugation compatibility, and the theta-root subgroup derived from a Galois splitting. The theta-root realization also records Proposition 2.2(iii)'s normalizing order-two lift, unique admissible coset criterion, induced arithmetic conjugation, and generation statement. Existence of these interfaces from the source geometry and the corresponding finite-etale orbicurve covers remain source obligations. The base stack carries a coherent C_2-action pseudofunctor, invariant maps satisfy the C_2 cocycle, and the quotient has the full bicategorical universal property through factorization and full faithfulness on 2-morphisms. At bad finite places, the local quotient factors through the actual profinite completion $\widehat{\mathbb{Z}}$ and canonical reduction modulo l; the cuspidal involution, invariant splitting, and canonical sign orbit of 1 mod l are explicit.

Definition 5.4 (Source initial theta core)**.** This replacement path binds the elliptic curve and its puncture to a scheme over the field, its origin section, and the complementary open subscheme with its open immersion. The X_F stack is bicategorically equivalent to the representable Yoneda-style pseudofunctor of this punctured scheme, so pullback, identity, and composition coherence are intrinsic; its coherent sign action is identified with an involutive scheme map acting by elliptic negation on rational points. The path also binds these objects to etale stacks, strong-transformation morphisms, certified fundamental groups, exact profinite sequences, compatible open immersions, the distinct arithmetic theta and elliptic quotients over the scalar-extended K-core, cuspidal and Lagrangian quotients, inversion splitting, theta-root subgroup, canonical restriction maps on places with a chosen section, all-K-place core scalar extensions, type-$(1, l$-tors) cover/sign-quotient stacks constrained by the derived open cover group, scalar extensions of those covers at every selected finite and infinite completion, compatible local-to-global fundamental-group and elliptic-quotient embeddings, global and bad-local theta-root/sign-quotient stack interfaces constrained by the derived theta-root groups and open-subgroup chains, and a distinguished complete Q-indexed and sign-orbit-indexed cusp atlas represented by rational cuspidal decomposition exact sequences. The transitive Definition 1.1 construction discards all nonselected inertia, forms $\Delta^{\dagger}$ and its two eigenspaces, derives the doubled cusp 2ϵ, and records the continuous exact quotients $J_X \hookrightarrow J_C$. The associated section is tied to the doubled-cusp decomposition group, its image is proved normal, and the arrowed-cover groups are the preimages of $\mathrm{Im}(\sigma)$ and $\mathrm{Im}(\sigma) \times \mathrm{Gal}(X/C)$. At every finite and infinite place, one sign-quotient scalar-extension package constructs the local X-stack, C-stack, and quotient map definitionally; separate local target pairs and comparison certificates are not supplied. Complete cusp atlases localize at every selected place; at every good finite place the dagger quotient, J-sequences, sections, arrowed stacks, and their fundamental groups commute with scalar extension. The selected local cusp is constructed at the global Q-label inside each localized complete atlas, and its decomposition diagrams and sign label are inherited there; no separate local cusp, atlas equality, or exact synchronization of arbitrary sheet representatives is supplied. The bad-place label is proved to be the sign orbit of $1 \bmod l$. Selected finite and infinite completions are tied to closed decomposition subgroups and compatible embeddings of fundamental exact sequences; each such subgroup is the literal stabilizer of a prolongation of the place to the absolute separable closure. The orbicurve exact sequences use the Krull-topological Galois group of that closure definitionally, rather than accepting an arbitrary profinite group together with an equality certificate. The local embedding is derived from the equivalence with this closed stabilizer, and Lean proves that the Galois maps in both the (X)- and (C)-diagrams have exactly this image; no independently chosen subgroup or subgroup-equality field remains. The finite and infinite place sections assemble into a single section on all places whose inverse is canonical restriction. Lean therefore constructs the source bijection $V_{\mathrm{mod}} \simeq V$, rather than merely storing its two halves, and proves the disjoint decompositions $V = V^{\mathrm{non}} \sqcup V^{\mathrm{arc}}$ and $V^{\mathrm{non}} = V^{\mathrm{bad}} \sqcup V^{\mathrm{good}}$. At bad places, the Tate parameter is an actual contracting local-field element with a geometric Galois-equivariant group-level Tate uniformization, and its positive order is derived from its discrete valuation exponent. This parameter and its prime-to-l order are required at every F-place above the bad moduli locus, not merely at a selected K-lift. The embedded moduli field is identified with the fixed field of the geometric-j stabilizer, its maximal solvable extension is the compositum of finite solvable Galois layers, and embedded K is the fixed field of the mod-l representation kernel. Analytic/topological realization remains open. The cover square carries the full two-pullback universal property. Construction of the scheme realization from a concrete Weierstrass model and finite-etale derivation of the supplied stack realizations, their compactifications, and their algebraic boundary cusps remain open. Thus these records specify the source constructions but do not yet prove their geometric existence. The scalar-extended K-core is constructed from the two extended stacks and the extended quotient morphism; its X-stack, C-stack, and quotient map agree definitionally with those extensions, so no separately assembled target pair or comparison certificates are accepted. The remaining sign action, quotient universal property, and fundamental groups are explicit geometric realization obligations. The global-to-K and selected local quotient maps carry invertible-modification scalar-extension comparisons pinned on components and naturality 2-cells.

5.3 Legacy finite M2–M3 models

Definition 5.5 (IUT I realization and IUT II pilots)**.** This finite boundary no longer accepts arbitrary theater or pilot identifiers, prime-strip identifiers, string histories, pilot values, or Hodge-evaluation conclusions. Initial theta data constructs the typed source, its degree-preorder toy realization, and numerical q and theta pilots. It does not discharge the cited IUT I–II clauses.

Definition 5.6 (Constructed Theorem 3.11 boundary)**.** The finite model no longer accepts a primitive packet, symmetry transport, quotient conclusions, possible-image witnesses, or finite-coordinate exactness. These laws are constructed from the numerical pilots, but they do not instantiate Theorem 3.11. The source manifest classifies the packet, symmetry, and log-Kummer entries as source obligations.

5.4 Obligations to discharge next

Definition 5.7 (Step (x) finite-divisor/log-Kummer package)**.** Step (x) is packaged as one finite-divisor and vertical-log-Kummer source: finite divisor tensor packets, realified Frobenioid Kummer compatibility, mono-analytic forgetting, the vertical IQ target, and source-target log-volume calibration are projected from this data.

Definition 5.8 (Theorem 3.11 IPL/SHE/APT bridge package)**.** The Theorem 3.11 bridge is packaged as one source object containing an input-prime-strip-link construction, a one-column structured SHE transport context, and an APT construction. The selected input link, one-column SHE, and APT transport endpoints are projections of this package. At the concrete packet boundary, this bridge component can now be constructed from the lower constructed qualitative certificate source, and Lean exposes a no-collapse audit recording the forbidden direct SHE domain-to-codomain identification, the permitted non-forbidden APT quotient transport, and the selected IPL source/target match. The constructed-qualitative bridge-alignment source packages that lower certificate with the packet-specific prime-strip and one-column laws; the common-container/SHE alignment is now derived one layer lower from constructed qualitative common-container data, and the canonical-prime-strip bridge source chooses the packet input and selected-output strips directly from the constructed qualitative IPL source. The codomain-canonical bridge source then fixes the packet one-column Hodge theater to be the constructed SHE codomain theater and reads the one-column history from the selected q-choice. The q-pilot-canonical bridge source goes one step lower: it builds the selected concrete q-choice with the constructed SHE q-pilot theater, then derives its codomain placement from the structured SHE theorem $q_{\mathrm{pilot}} \in$ codomain. The q-pilot-class bridge source lowers this further by rebuilding the selected q-choice from a theta-pilot lattice class and representative data for the forgotten finite label, procession state, local tensor state, and upper-semi state. The component-representative bridge source lowers the representative section itself to the log-theta/vertical-log-Kummer/Frobenioid divisor-column component kernel; Lean constructs the section and audits the selected q-choice as the kernel representative of the q-pilot theta class. The source-spine q-pilot component bridge then reads the q-pilot history label, label-forgotten log-theta lattice coordinate, and coric datum from the selected q-choice of the unified Theorem 3.11 source spine; its audit records the projected component bridge and these source-spine equalities. The generated-full-label source-spine bridge lowers the final representative equality one layer further: it matches the source-spine selected q-choice to the concrete projection of the generated full-label representative over the q-pilot theta class, and derives the component-kernel representative equality from the full-label kernel theorem. The generated-q-choice source-spine bridge lowers this again by supplying the generated q-choice itself, proving that its theta class is the q-pilot theta class and that its finite label is the component kernel's representative label; Lean then derives the generated-full-label representative projection. The selected-label source-spine bridge removes this generated q-choice as a public object: Lean constructs it as the kernel representative full-label choice over the source-spine q-pilot theta class, proves that its finite label is the selected source label, and derives the selected q-choice projection from fieldwise source equalities for the q-pilot theater, representative finite label, procession state, local tensor state, and upper-semi state. The component-representative selected-label bridge lowers the fieldwise equalities to the vertical-log-Kummer/Frobenioid component representative source: applying its constructed-representative theorem to the selected q-choice and transporting across the q-pilot theater alignment yields the representative finite label, procession state, local tensor state, and upper-semi state internally. Inside the one-sided multiradial construction namespace, the source lowers this boundary once more: the component representative source is the `volumeSource` projected from the concrete one-sided component package, and the remaining data only synchronize the one-sided selected q-choice with the source spine and the constructed q-pilot theater. The SHE-codomain variant derives the q-pilot theater placement from the source-spine one-column law, the constructed SHE codomain/one-column alignment, and the structured SHE q-pilot/codomain equality. The source-spine-selected variant removes the remaining selected-q synchronization field at this layer by constructing the one-sided component source with `sourceSpine.selectedQChoice`. The source-spine-data variant additionally projects the theta possible-image source and gluing torsor from the unified source spine. The primitive-packet variant lowers this once more: the multiradial record and source spine are projected from the same primitive packet, so their theta possible-image agreement is inherited from the primitive target-region construction rather than supplied as a separate bridge input. The component-kernel variant constructs the former one-sided component `volumeSource` from the Frobenioid divisor-column representative kernel. At this same component-kernel level, constructs the log-theta label-procession upper-semi source from the generated Frobenioid base realified volume. The finite average is no longer a separate field on this slice: Lean uses to prove that the uniform average of the constant $\mathbb{Z}/l\mathbb{Z}$-indexed label family is the base volume, and then transports the result through the constructed representative upper-semi state. One layer lower, vertical log-Kummer/log-link transport now proves constancy for the whole finite log-shell family: identifies the constructed $\mathbb{Z}/l\mathbb{Z}$-labelled log-shell family with the constant family on the base object, and reads the normalized log-volume as the base finite log-volume. The generated full-label kernel layer also carries this finite average internally: builds the constant $\mathbb{Z}/l\mathbb{Z}$-indexed normalized log-volume object, and proves that the generated full-label typed-core log-volume is exactly this finite average. Thus the quotient possible-image audit records the procession-normalized average at the generated source boundary itself, not merely after projection to an ambient representative source. The actual generated F_l-orbit through a selected full-label choice now has its own averaged object as well: indexes the normalized entries by the concrete generated label choices in the orbit, and proves the paper-facing $\mathbb{Z}/l\mathbb{Z}$-average formula. The orbit audit records that this orbit average equals both the selected generated log-volume and the theta-class average. The Ind1/Ind2 part of the same quotient audit now preserves this actual orbit average, not merely the raw scalar log-volume: and derive the averaged equalities from the theta-class invariance of the generated label family. The Ind3 part now acts on the same averaged object: transfers the generated upper-semi inequality to the orbit average itself. Finally, and state the exact no-extra-collapse theorem for the generated equality quotient: two generated choices become equal in the setoid quotient exactly when their theta-pilot classes agree. The underlying quotient/factorization layer is now explicit as well: proves that equality of equality-quotient indices forces equality of the pulled-back possible-image regions, and records the same pullback law at the concrete theta-pilot source boundary. The generated package reindex source pulls the concrete pre-ledger/source package back along the kernel projection from generated full-label choices, provided the selected generated q-choice projects to the pre-ledger chosen output. Thus the remaining mathematical input at this boundary is the selected-q lift and generated record/image construction, not an ambient theorem identifying every concrete Hodge-theater/log-theta choice with a kernel representative. The selected-label package-reindex alignment source now factors this reindex step through the selected-label source-spine bridge: Lean constructs the selected generated full-label q-choice from the source-spine q-pilot class and derives its projection to the pre-ledger chosen output from the single remaining source-spine selected-q/chosen-output alignment. The primitive-packet selected-label reindex alignment source lowers this one more step: the selected-label source is projected from the primitive packet's component-kernel one-sided source, and its source-spine selected q-choice is definitionally the packet selected q-choice. The remaining equality is now packet selected q-choice equals pre-ledger chosen output. The chosen-output primitive packet reindex alignment source removes that equality as an external field by constructing the primitive packet with the pre-ledger chosen output itself. The remaining inputs are the chosen-output one-column/history laws and the bridge/component data indexed by that chosen output. The chosen-output generated full-label quotient corridor source then projects the generated kernel, theta-class image family, gluing torsor, and selected generated q-choice from the same chosen-output packet route. The constructed generated full-label quotient corridor source removes the image-family equality as an external field by rebuilding the generated multiradial record with the full-label image family forced by the chosen-output primitive packet. The remaining generated-corridor boundary is the base generated multiradial record. The structured generated corridor source lowers this further: it constructs that base record from a generated SHE-structured Theorem 3.11 bundle before installing the forced full-label possible images. The remaining generated-corridor boundary is now the generated structured bundle. The reindexed structured corridor source removes that generated bundle as an independent input by pulling back the concrete SHE-structured bundle along the chosen-output full-label projection. The remaining structured boundary is therefore the concrete package's structured Theorem 3.11/SHE bundle. The constructed-qualitative corridor source lowers this boundary again by deriving the concrete structured bundle from a constructed qualitative Theorem 3.11/SHE bundle before reindexing it to the generated full-label index. The direct product-hull/HDD layer now has a matching chosen-output generated source, , which projects the older generated-HDD source from this chosen-output corridor instead of accepting an independently assembled generated quotient corridor at the public route boundary. Its constructed-record strengthening lowers this once more to the constructed generated corridor, so the product-HDD route no longer asks separately for the record/image synchronization. Its structured-bundle strengthening constructs the generated record from the generated structured bundle before lowering to the constructed-record source. Its reindexed-structured strengthening obtains that generated bundle from the concrete structured bundle. Its constructed-qualitative strengthening obtains the concrete structured bundle from constructed qualitative Theorem 3.11/SHE data and records the qualitative and APT transport audits at this boundary.

Definition 5.9 (Hodge/log-theta/log-Kummer source spine)**.** The source spine packages the 0- and 1-column Hodge-theater histories, prime strips, theta-pilot possible images, selected q-choice, and vertical log-Kummer packet alignment. Its Theorem 3.11 bridge and Step (x) payload are the structured packages above, so the preferred concrete route no longer gets the IPL/SHE/APT or finite-divisor/log-Kummer assertions as separate spine fields.

Proposition 5.10 (The spine assembles the source ledger)**.** *Given the source spine, the remaining Step (xi), IUT IV, and additive-Haar layers assemble into the top-level paper-trace obligation record without separately supplying Theorem 3.11 and Step (x) obligation packages.*

Proposition 5.11 (Legacy source-spine route audit)**.** *This legacy paper-trace route consumes one Hodge/log-theta/log-Kummer source spine for the Theorem 3.11 and Step (x) layers, but still takes the old Step (xi) hull/determinant obligation bundle directly. It is retained only as a historical comparison point; the preferred route is the Step-(xi)-constructed route below.*

Definition 5.12 (Step (xi) paper trace)**.** The Step (xi) paper trace names the Corollary 3.12 substeps (xi-a)–(xi-g) and the Remark 3.9.5 Ob1–Ob9 checkpoints. It is the source-facing ledger for passing from Theorem 3.11 possible images to holomorphic hulls, determinants, log-volumes, and the retained prime-strip/log-Kummer comparison.

Definition 5.13 (Paper-derived Step (xi) source)**.** This source object ties the Step (xi) hull/determinant data to the unified Hodge/log-theta/log-Kummer spine. Its selected q-region is required to be the selected Theorem 3.11 possible image, and its Ob7 compatibility is pinned to the same input and output prime strips and one-column log-Kummer coordinate.

Proposition 5.14 (Selected q-region from the quotient family)**.** *The selected q-region used in Step (xi) can be constructed as the possible region attached to the selected Theorem 3.11 equality-quotient choice. Its containment in the possible-image union is therefore the family-union inclusion from the Remark 3.9.5 possible-image source. The paper-derived Step (xi) source has a matching constructor that fixes this choice to the source spine's selected q-choice, rather than taking a preassembled hull record with a separately supplied containment proof. The Theorem 3.11 possible-image source is now witness-bearing: each theta-pilot class image carries a point, Lean pulls these points back to concrete choices, and the equality-quotient source proves nonemptiness of quotient pullbacks and of the selected q-region. Thus Ob5 no longer passes through an unconstrained empty-family predicate at this boundary. The strengthened constructor takes a Remark 3.9.5 minimal-hull system, derives the hull operator from the intersection-of-hulls definition of $\phi(P)$, and proves the Ob1/Ob2 hull absorption facts from that source. The hull source can now also be instantiated from the principal product presentation $\lambda \cdot O$: Lean imports the nonzero-parameter witness, scalar-image description of principal hulls, the family hull's identification with the principal hull at the intersection parameter, and the corresponding principal-hull log-volume equality.*

Definition 5.15 (Symmetry-label principal valuation-ball theta-region source)**.** The symmetry-label packet now carries a source object for the principal valuation-ball theta-region boundary. It combines a source-core Remark 3.9.5 principal valuation-ball cover with the fiber-transport-backed family-union comparison and projects the theta-region-defined principal valuation-ball source and its principal hull. The packet/symmetry-label Step (xi) endpoint now consumes this object directly to project the theta-region principal source and selected q-choice. The possible-image-witness variant constructs the packet-indexed symmetry-label wrapper first, so this principal valuation-ball theorem surface no longer requires a preassembled argument. The corresponding possible-image-witness HDD-source endpoint carries this lowering through the first hull/determinant layer while still auditing the supplied principal-product Ob3/Ob5 measure-Hodge-gap HDD source. Its constructed-HDD companion carries the same lowering through the route that builds this principal HDD source from the theta-region source, selected q-choice, measure calibration, side conditions, and summand-Hodge calibration. Its localized-data companion carries the lowering through the finite-place constructor for the localized Step (xi) C_Θ source, where the determinant source, main-log contribution, Haar-normalization defect, and summation identities are explicit route inputs. Its localized calibration-data companion combines this witness lowering with the theta-region calibration source, so the route also stops exposing the measure/summand calibration pair separately. Its localized calibration-restriction companion carries the same witness lowering through the single local-arithmetic/Ob7 handoff source. Its principal-hull companion also projects the local-arithmetic principal hull from the theta-region source on this witness-derived branch. Its fiber-backed companion also constructs the packet theta-region source from the principal valuation-ball product-hull cover and transported family-union comparison after deriving the symmetry-label wrapper from the possible-image witness. Its theta-region-defined companion projects those two fiber-backed inputs from the single principal theta-region valuation-ball source on the same witness-derived branch. The HDD-source endpoint additionally absorbs the measure and summand-charted Hodge calibrations into a principal-product Ob3/Ob5 measure-Hodge-gap HDD source tied to this theta-region principal hull. The constructed-HDD endpoint builds that HDD source internally from the packet theta-region source, selected q-choice, constructed qualitative bundle, measure calibration, and summand-Hodge data; the localized-data endpoint then constructs the localized C_Θ source from localized determinant, Haar-defect, main-log, and finite-place summation data. The remaining principal-source input is the local-arithmetic restriction boundary.

Proposition 5.16 (Step (xi) determinant comparison from weighted determinants)**.** *The determinant comparison record consumed by the paper-derived Step (xi) source can now be produced from the existing weighted arithmetic-vector-bundle determinant source. The finite adjusted summands, determinant sum, normalization, and positive tensor power are projections from that lower source. The record-canonical Remark 3.9.5 hull/determinant bridge further supplies the cross-comparisons from selected q-region log-volume to normalized determinant and from determinant log-volume to the Step (xi) theta scalar, once the paper-derived selected q-region log-volume is identified with the record bridge q-pilot log-volume. The bridge-backed paper-source constructor derives this log-volume equality from two lower alignments: the paper hull operator is the bridge hull operator, and the bridge q-pilot region is the source-spine selected Theorem 3.11 possible image. The determinant projection can also be constructed from the localized arithmetic-vector-bundle decomposition source: localized hull-region log-volumes are summed, identified with structure-sheaf-adjusted vector-bundle contributions, and projected to the Step (xi) determinant data. The selected q-region-to-normalized determinant inequality is then a theorem of hull containment and the localized family-hull/normalized- determinant equality. The localized projection audit also exposes the adjusted Ob3/Ob5 endpoint, the induced Ob3/Ob5 compatibility source, and the bounded-family hull/determinant log-volume endpoint from the lower source layer. The strict localized route additionally carries an Ob5 quotient/determinant bridge audit: equality-orbit choices give the same quotient possible region and quotient-region log-volume,* *and this same localized family-hull source supplies the Ob3/Ob5 determinant equality and bounded-family tensor endpoint. The record Ob3/Ob5 boundary is now also constructible through the theta-region-defined principal valuation-ball source from typed fiber-transport valuation data: the theta-pilot valuation cover is pulled back to concrete choices, its adjusted possible-region family is identified with the multiradial record possible-image family, and the resulting record Ob3/Ob5 adjusted determinant/log-volume source is threaded into the exact-theta constructed hull/determinant source. The lattice-image-law-backed route retains only the explicit identification of its image law with the concrete fiber-transport law; the fiber-transport backed route derives this comparison internally. The same public Ob3/Ob5 and exact-theta constructed hull/determinant boundary can now be entered from the principal pointwise constructed log-shell metric-zero source and from the compact-open log-Kummer image, correspondence, and map possible-region sources: Lean constructs the local log-Kummer graph or extracts the relation-level image laws, recovers pointwise representatives in the selected compact-open/log-shell valuation ball, proves the converse realization law, and then threads the resulting principal valuation-ball source into the record Ob3/Ob5 and constructed Step (xi) source. At the compact-open realized-exact layer, the literal image formula* $\Theta_v = \mathrm{realization}(U_v)$ *now constructs the log-Kummer map source directly, so this route no longer requires a separate graph source once realized exactness is available. The pointwise local-coordinate source now also projects to compact-open log-shell image exactness: Lean first reconstructs log-shell exactness from local representatives and the converse valuation-ball realization law, then applies compact-open Haar normalization. The same pointwise source now continues through the canonical compact-open graph to construct the possible-region exact, log-Kummer image, correspondence, and map packages consumed by the public Step (xi) route. At the compact-open public boundary, the exact possible-region, log-Kummer map, image, and correspondence endpoints can now consume the Record Ob3/Ob5 valuation-ball source directly; the Ob3/Ob4-to-Step (xi) determinant equality is projected from that source instead of being supplied as an independent endpoint argument. The corresponding choice-indexed arithmetic-formula routes at the exact possible-region, log-Kummer map, image, and correspondence boundaries also construct the raw determinant compatibility from this Record Ob3/Ob5 source; their remaining synchronization datum is the lower equality identifying the adjusted valuation-ball hull operator with the principal product hull. The correspondence route also has an obligations-backed constructor: the package hull/det bridge equality, q-pilot positivity, and normalization are projected from 'IUTStage1SourceHullDetObligations'. Its Hodge-calibrated variant replaces the raw theta/family-hull scalar equality with 'IUTStage1HodgeFamilyHullLogVolumeCalibration', deriving* Θ_{signed} *= family-hull log-volume through the Hodge–Arakelov theta monoid degree. The record-canonical bridge-data synchronization and measure calibration remain explicit lower inputs. At the compact-open map boundary, Lean also provides the side-conditioned constructor and endpoint , where q-pilot positivity and source normalization are projected from 'IUTStage1SourceSideConditions' rather than passed as loose scalar hypotheses. Its stricter source-data-backed form, with endpoint , projects the package hull/determinant bridge equality from 'IUTStage1SourceHullDetData'; only the synchronization of that stored bridge datum with the record-canonical hull/tensor-power construction remains as an explicit lower input. At the choice-indexed compact-open arithmetic-formula and formula-gap boundaries, the map, exact possible-region, log-Kummer image, and log-Kummer correspondence routes also have constructor-sourced bridge variants: Lean projects the package bridge equality from an 'IUTStage1PossibleImageConstructorSourcedHolomorphicHullDeterminantSource', leaving only the identification of the constructor bridge datum with the corresponding Record Ob3/Ob5 principal-product hull/tensor-power construction as a lower synchronization input.*

*The same constructor-sourced bridge form now reaches the choice-indexed arithmetic-degree-controlled, direct p-adic unit-ball Haar, and derived p-adic unit-ball Haar compact-open map, exact possible-region, log-Kummer image, and log-Kummer correspondence endpoints: they still consume the lower arithmetic comparison, local Haar, or derived formula-matching data and the determinant/C_Θ-sum identifications, but derive the hull compatibility and package bridge from the Record Ob3/Ob5 source and constructor bridge. The same calibration now reaches the finite-divisor constructor-backed source: Lean converts the Hodge-calibrated data to the constructor-backed spine, forms the constructor-generated hull/determinant obligations, and builds the possible-image Step (xi) source without reintroducing a raw Θ_{signed}-to-family-hull equality at this boundary. The target-charted exact-theta adjusted-summand source now enters this Hodge-calibrated finite-divisor constructor first and projects the older constructor-backed source only for legacy consumers. On the all-in-one target-charted Hodge/*IPL *possible-image route, the named route bridge audit now retains both the ambient-obligations bridge equality and the record-canonical Theorem 3.11 hull/tensor-power bridge equality, so the public finite-source boundary no longer hides this synchronization behind the obligations bridge projection alone. The same public route audit now also exposes the target-charted reduction to the constructor-built Step (xi) source and the resulting Remark 3.9.5 Ob1–Ob5 bridge audit, so the possible-image hull absorption and determinant compatibility chain is visible at the route boundary rather than only in the lower source namespace. At the constructor-built boundary itself, Lean now has the conversion and endpoint ; the record-canonical bridge equality used by the constructed Remark 3.9.5 source is recovered by , i.e. through the built hull/determinant constructor bridge datum rather than a fresh synchronization argument. For the non-vacuous additive-Haar branch, the residue-module/unit-ball route now has the projected Haar-modulus source and the localized constructor-built C_Θ audit . This audit records the residue-module Step (xi) boundary, the constructed Remark 3.9.5 source projected from the constructor-built source, the localized Step (xi) local-term endpoint, the local-to-global C_Θ audit, and the resulting additive-Haar remaining-payload audit. A parameterized Ob5 quotient audit is now projected through the same route: for any nonempty comparison possible image, Lean records the bounded-family hull inclusions, collapsed upper-semi quotient equality, determinant bound, and local canonical-scale chain from the constructor-built source. The same public route now exposes the Ob6 exact-$\Xi(P_B)$ boundary: it constructs the Remark 3.9.5 possible-image family from the constructor-built hull source, identifies the canonical $\Phi(P_B)$ approximant with the family hull, and derives the selected and comparison log-volume bounds from a source-calibrated exact-$\Xi(P_B)$ log-volume equality rather than from an arbitrary supplied approximant family. On the stricter localized route, the final normalized determinant upper bound is no longer supplied as an independent inequality: it is derived from the identification of the Step (xi) theta scalar with the constructed localized family-hull log-volume. A further constructed-source constructor now projects the bridge and hull operator from the constructed Remark 3.9.5 source itself; on this route the remaining alignment is the identification of the constructed q-pilot region with the source-spine selected possible image. Specializing the constructed source to use the source-spine selected possible image as its q-pilot region makes that alignment definitional; the explicit lower condition at this boundary becomes containment of the selected possible image in the record possible-image union. The Ob7 source link now also has a record-bridge form: the same record-canonical Remark 3.9.5 bridge used for the determinant comparison supplies the q-region and determinant source, while the coric $\Theta^{\times\mu}$ invariant supplies the prime-strip/log-Kummer equality. For same-index constructor-built possible-image data, the equality between record images and source images derives the selected q-region and quotient-family union alignments automatically. The source-spine theta-record wrapper now consumes* *this constructor-built source directly, so the obligations-backed bridge-data synchronization is no longer part of that stricter route boundary. This source-spine wrapper also has a product-formula Ob7 form: finite weighted determinant prime-strip data tied to the same constructor-built determinant source constructs the coric invariant and proves the determinant/global prime-strip equality internally. The Ob7 prime-strip/log-Kummer record and its source-spine label equalities can now also be projected from a single coric* $\Theta_{\mathrm{LGP}}^{\times\mu}$*-style source link aligned with the same quotient hull source. The selected-source constructor has a stricter Ob3/Ob5 determinant form: the record-canonical family-hull compatibility source, and then the adjusted determinant log-volume source, project the old hull/determinant compatibility internally. The same lowering is available at the pointwise source-region route and at the deepest retained* (Ind2) *action-packet route, so those routes no longer need to expose a standalone compatibility record when adjusted Ob3/Ob5 data is present. Their quotient-hull input can now also be lowered from a standalone equality-quotient hull/log-volume compatibility record to the Theorem 3.11 possible-image quotient compatibility for the source images; the Remark 3.9.5 hull-family source is then constructed by applying the selected holomorphic hull operator. The strict source-spine variants go one step lower again: for the unified Theorem 3.11 Hodge-theater/log-theta/log-Kummer source, this quotient compatibility is projected from the theta-pilot possible-image source and the vertical log-Kummer packet-aligned finite-procession source. The same source-spine projection now applies already at the deepest Ob3/Ob4 action-packet determinant route, so its equality-quotient hull/log-volume compatibility is constructed internally before the later adjusted Ob3/Ob5 refinement is used. The concrete one-sided Theorem 3.11 constructor now performs the same lowering from the theta-pilot class possible-image source and the record/source image equality:* (Ind1), (Ind2) *possible-image quotient compatibility is no longer a raw argument at this entry point, while the* (Ind3) *upper-semi comparison remains the typed-core log-volume input. Thus the deepest retained action-packet route no longer exposes a standalone possible-image quotient compatibility argument. The same source-spine route now has an exact-theta determinant form: its Ob4 tensor-power bound is derived from the equality identifying the Step (xi) theta scalar with the record Ob3/Ob5 family-hull log-volume, so the action-packet route need not expose an independent tensor-power estimate. The measured adjusted-source boundary now has a Hodge-calibrated form as well: Lean projects the package measure equality from the measured source and derives the* Θ_{signed}*-to-family-hull equality from 'IUTStage1HodgeFamilyHullLogVolumeCalibration', so neither equality is a standalone downstream constructor argument. On the selected Ob3/Ob4 route, the adjusted Ob3/Ob5 source-region route, and the exact-theta source-spine route, the Ob7 determinant/global prime-strip equality can now be derived from a finite weighted determinant product-formula source: local adjusted summands are calibrated against converted Gaussian local log-volumes, the finite product formula gives the global prime-strip log-volume, and the coric unit character constructs the* $\Theta_{\mathrm{LGP}}^{\times\mu}$ *invariant before entering the existing Ob7 source link. The action-packet theta formula/source-class comparison is also derived from the equality of the record possible-image family with the unified source-spine image family, so the exact action-packet route can consume that family equality instead of a separate class-formula comparison. The newest constructed-record variant installs the source-spine images directly as the record's theta-pilot possible images; for that route the record/source image equality is definitional, and the exposed image-side calibration is that the source-spine images are the package target regions. This image-side calibration has now been lowered from a whole-family equality to a pointwise theta-pilot class-region trace* $\mathcal{R}_{\mathrm{class}}(\mathrm{class}(c)) = \mathcal{R}_{\mathrm{target}}(c)$*; Lean derives the source-spine image/target-region family equality before installing the constructed record. One level lower, Lean can now build this trace from package target-region data: a representative sec-*

tion for theta-pilot classes, theta-pilot-class invariance of target regions, and explicit point witnesses for the representative target regions. The resulting target-region-backed theta-pilot source is installed in the source spine before the exact-theta action-packet Step (xi) route is applied. This target-region class invariance has also been lowered to invariance under the concrete theta-pilot lattice key: Hodge theater, history label, log-theta lattice coordinate, and coric datum. That key-invariance source has now been lowered again to a target-region lattice formula: each package target region is identified with the formula value indexed by the same Hodge/history/lattice/coric data, and Lean derives key invariance, the class trace, the source-image/target-region family equality, and the Step (xi) source-spine route from this formula. The formula is now also constructed from the existing concrete lattice-image law: the record possible-image formula is converted into a package target-region formula by the multiradial record's target-region realization theorem, and the exact-theta Step (xi) source-spine route can consume this lattice-image law directly. The same package target-region formula is now projected further down from typed theta-pilot fiber transport, from the $(\mathrm{Ind1}), (\mathrm{Ind2})$ *equality-quotient pullback, and from the retained post-procession* $(\mathrm{Ind2})$ *action-packet transport source. Thus the target-region formula boundary follows the current deepest local Theorem 3.11 action-packet source rather than stopping at a standalone lattice-image law. The action-packet boundary itself has also been lowered one step: Lean now projects the retained packet/action transport source from place-audited nonarchimedean* Ism *or archimedean order-two action-entry label-transport data, together with explicit identifications of the audited packet tensor states with the post-procession source and target local tensor states. The label-transport source can now be constructed one layer lower from the retained action-entry source, source and target tensor symmetry label equations, and transport of the direct-summand symmetry kind. The same tensor-label data now also projects to the place-audited possible-image/fiber audits for the nonarchimedean and archimedean branches, so the lower action data carries both the packet transport and the equality-quotient/fiber witness information. This boundary has now been lowered again: the tensor-label equalities may be read from a single direct-summand packet* $\mathrm{SymmetryLabelTransportSource}$*, and the corresponding concrete fiber source projects through the tensor-label route internally. Record-level possible-image point witnesses now also factor through the class-indexed Theorem 3.11 possible-image source and the record/source image equality, so the newest symmetry-label fiber source no longer needs independently supplied witness points. The exact-theta selected Step (xi) paper-trace constructor now also accepts this symmetry-label transport source directly and projects the older action-packet transport internally; the source-spine record constructor has the same symmetry-label entry point. The exact-theta source-spine/action-packet route now also has an obligations-backed entry point: Lean projects the package hull/determinant bridge, q-pilot positivity, and normalization from 'IUTStage1SourceHullDetObligations', while retaining only the explicit synchronization between that obligations datum and the record-canonical selected source-spine hull/determinant construction. On the strongest selected route, this source link is itself built from the constructed Remark 3.9.5 bridge and the coric prime-strip invariant, after transporting the determinant compatibility across the record-union/equality-quotient-family alignment. In the same-index source-spine case, this union alignment is now a theorem: equality of the record possible-image family with the Theorem 3.11 source image family identifies the record union with the equality-quotient Remark 3.9.5 family union. The corresponding selected route therefore takes this image-family equality and derives both the selected-q containment in the record union and the quotient-family union alignment internally. One layer lower, the same route can now consume the paper-shaped pointwise trace* $\mathcal{R}_{\mathrm{record}}(c) = \mathcal{R}_{\mathrm{source}}(c)$ *for each concrete Hodge-theater/log-theta choice c; Lean assembles this into the full* `RegionFamily` *equality before applying the quotient-union* *theorem. One layer below that, the pointwise trace is now derived from the existing theta-pilot class-region source, once its class formula is identified with the Hodge-theater source spine's theta-pilot class-image family. The same handoff is now available one level further down: record-level theta-pilot class formulas and Hodge-theater/history/log-theta-lattice formulas project to the class-region source before the Step (xi) route is assembled. Below the lattice formula, the route can consume the same-lattice-key image law; the lattice formula is then built from representative choices before being compared with the source-spine class images. One level further down, the same handoff can consume typed theta-pilot fiber-transport invariance, from which the same-key image law is derived. The current lowest selected route now consumes the* (Ind1), (Ind2) *equality-quotient pullback, or the retained post-procession* (Ind2) *action-packet transport source, and derives the fiber-invariant image law before entering the Step (xi) hull/determinant comparison. The valuation-ball and log-shell lowering now also transports explicit theta-region point witnesses through the local exact-source chain. The class-region, class-formula, log-theta-lattice formula, lattice-image-law, and typed fiber-transport sources themselves now carry these witnesses, so the downstream possible-image and selected-region nonemptiness audits are tied to the same Theorem 3.11 source spine rather than to independent valuation-wrapper fields. The adjusted Ob3/Ob5 determinant-log-volume route now has the same intermediate entry points: class-region, class-formula, lattice-formula, lattice-image-law, and typed fiber-transport sources pass through the adjusted log-volume source's record-canonical determinant compatibility instead of exposing a raw hull-approximant compatibility record. The additive-Haar arithmetic-divisor audit now also consumes such a nonemptiness proof to assemble the witness-bearing top-level Theorem 3.11 possible-image source data. The arithmetic-divisor and record Ob3/Ob5 additive-Haar constructors now derive their all-choice possible-image nonemptiness from the class-indexed Theorem 3.11 image law and record/source image equality, and the same bridge constructs the central Theorem 3.11 possible-image source-data record together with its nonemptiness audit, rather than receiving that nonemptiness as an independent downstream field.*

Proposition 5.17 (Step (xi) source assembles obligations). *The structured Step (xi) source projects to the hull/determinant obligation record consumed by the existing corridor. The old Step (xi) proposition fields are then recovered as theorems from this source data.*

Proposition 5.18 (Preferred Step (xi)-constructed route audit). *The promoted public paper-trace route is now the strict principal-product localized wrapper. It consumes the source spine, a principal product $\lambda \cdot O$ hull source, a localized vector-bundle determinant decomposition, the equality identifying Θ_{signed} with the constructed localized family-hull log-volume, and either the shared $\Theta^{\times\mu}$ log-Kummer Ob7 source or, in the strengthened public wrapper, localized weighted determinant product-formula data that constructs this Ob7 source internally. A stricter public route now constructs the Step (xi)/Remark 3.9.5 paper trace from these same lower sources, so it no longer accepts a standalone paper-trace proposition bundle. A localized-cover public variant fixes the localized determinant source to the same principal-product hull system and quotient possible-image family; Lean projects the localized decomposition and invokes the older route with the hull-operator and possible-region alignments by reflexivity, rather than accepting those alignments as separate public inputs. Its canonical-theta refinement sets the signed theta value to the projected family-hull log-volume, so the theta/family-hull equality is also reflexive. Both localized-cover routes now have additive-payload forms that construct the paper trace, derived Step (xi) source, and remaining non-Step audit from the lower additive-Haar arithmetic-degree payload instead of exposing the full non-Step audit. The underlying preferred route ladder has the same additive-payload lowering, from the raw Step (xi) source through the concrete-localized, theta-equals-family-*

hull, principal-product, cover, canonical-cover, transported scalar-parameter, representable, locally representable, coordinate scalar-image, and real-line coordinate layers. The localized determinant handoff has also been lowered: the Remark 3.9.5 hull/determinant source can be constructed from the localized vector-bundle determinant source, so the local C_Θ constructor no longer takes the determinant-source equality as an independent field in this route. At the deepest finite-extension boundary, the projected localized possible-region equality is now derived from the p-adic source-level possible-region equality by the projection theorem above. The principal-product finite-extension source lowers this comparison again: its p-adic possible-region is definitionally the Theorem 3.11 principal-product possible-image family, and the strongest wrapper invokes the previous route with this equality by reflexivity. The p-adic finite source also has a calibrated hull-cover form: each localized region is tied to a local vector-bundle calibration, and Lean derives the pointwise Ob3 localized-region log-volume identity before constructing the older finite-extension localized decomposition source. Its family-hull-indexed refinement derives the cover identity, pairwise disjointness, and finite-additive volume sum from an index map on the principal-product family hull. A cover-local measure-model refinement lowers this p-adic volume input further: Lean derives the localized cover log-volume sum from the measure model attached to the calibrated p-adic regions, and the public finite-extension wrapper projects this source to the localized hull/vector-bundle decomposition consumed by the older route. A direct-product refinement removes the supplied p-adic family-hull index: Lean constructs the index from the local-factor cell cover of the principal-product family hull, while local-factor separation proves the fiber description and disjointness needed by the measure-backed cover sum. A tensor-measure refinement removes the supplied cover-local measure model: direct-product cell log-volume identities and their finite tensor-cell summation formula construct the measure model consumed by the p-adic direct-product wrapper. A source-core refinement lowers this once more: the tensor-cell log-volumes and the tensor-sum formula are derived from the finite-extension localized hull/vector-bundle decomposition's adjusted determinant log-volume identities. A charted local-ring refinement replaces arbitrary p-adic local-factor regions by per-place vector-bundle chart calibrations and projects their local-ring and direct-summand log-volume equalities. A valuation-ball refinement lowers these chart calibrations further: the p-adic local factors are valuation-ball/additive-Haar calibrations whose regions project to the charted local-ring cover consumed by Step (xi). A principal valuation-ball refinement lowers the p-adic cover geometry again: Lean derives the direct-product cell identity, local-factor separation, and family-hull equality from the SourceCore principal valuation-ball product-hull cover before projecting to the valuation-ball charted route. The p-adic branch now also has principal-hull and hull-formation alignment refinements: the valuation-cover/public-hull equality is derived from the principal valuation-ball hull theorem, and the source-core and valuation-cover possible-region equalities are projected from the concrete `principalProductHullFormationData` *family. Its calibration boundary is now lowered as well: the localized calibration and per-factor valuation calibration are transported from the principal valuation cover across the principal-hull equality, so the public p-adic route no longer accepts these calibration families independently at this level. The compact-open-factor boundary has also been lowered: the principal-product localized-adjusted source now supplies equality with the localized adjusted determinant sum, and Lean derives the old Hodge/family-hull equality from the localized Remark 3.9.5 decomposition theorem. The additive-Haar principal-product branch also has a cover-derived localized hull source: from the cover identity $\phi(\bigcup_i \Theta_i) = \bigcup_\beta H_\beta$ and finite additivity for this cover, Lean derives the old containment and family-hull volume-sum fields consumed by the localized decomposition interface. A calibrated cover source lowers this once more: each H_β is tied to a local vector-bundle calibration, and Lean derives the pointwise localized-region* *determinant identity after aligning that vector bundle with the additive-Haar localization. A family-hull-indexed source lowers the cover boundary again: localized regions are fibers of an index map on the family hull, so Lean derives the cover identity, pairwise disjointness, and finite-additive cover-volume formula. A cover-local measure-model variant lowers the additivity input further: the localized volume sum is derived from the measure model for this finite cover instead of a global finite-additive hull law. A direct-product variant removes the supplied family-hull index map: local factor regions define the cells, local factor separation proves disjointness, and Lean constructs the family-hull index and fiber equality from the cover identity. A tensor-measure direct-product variant also constructs the cover-local measure model from direct-product cell log-volumes and the tensor-cell summation identity. The strict positive determinant contribution has been split off into a non-unit-ball additive-Haar branch: its principal-product source fixes the possible-region family definitionally, constructs the paper trace internally, and derives the positive summand from* $1 < \mu(U)$. *This branch now has a concrete finite-extension realization: the selected factor is the inverse base-prime dilation* $p^{-1}O_K$, *and Lean proves* $\mu(p^{-1}O_K) = p^{[K:\mathbb{Q}_p]} > 1$. *Lean also audits the old unit-ball positive boundary as contradictory, since the unit-ball projection forces* $\mu(O_v) = 1$ *and normalized Haar log-volume* 0. *The public audit has now been lowered once more to raw adjusted-localization calibration data. A stricter public route now computes each raw ratio as the raw adjusted log-volume divided by the nonzero Gaussian global log-volume; the Ob3 weighted determinant sum and the determinant/global prime-strip equality prove the finite ratio normalization. The newest wrapper lowers the Ob7 prime-strip input itself: common environment and Gaussian prime-place coordinates, together with one valuation-indexed coric character, construct the prime evaluation,* $\Theta^{\times\mu}$ *lift, and coric invariant internally. A further anchored wrapper constructs the local-global collection, its environment anchor, and the local environment/Gaussian evaluation from global-to-local restrictions whose local realified objects are now forced by restricted global prime log-volumes. In the diagonal finite valuation-index route, the environment primes, Gaussian primes, and places are all indexed by the same type, so the prime-place equivalences are identity maps constructed internally. The preferred non-vacuous public diagonal wrapper now consumes the additive-Haar compact-open factor source over the principal product hull system, an audited Hodge–Arakelov theta evaluation, global-to-local restriction data calibrated at the Hodge canonical full-label Gaussian degree, and the scalar equality between the Hodge theta-monoid degree and the localized adjusted determinant sum. The environment/Gaussian realified Frobenioid evaluation is constructed internally from zero-divisor ordinary/theta/log-shell copies whose unit log-volume is the Hodge canonical degree; normalization proves that the Gaussian theta divisor log-volume is this Hodge canonical degree. The Hodge evaluation identifies the canonical degree with the theta-monoid degree. The finite Hodge summands are no longer a public family: they are fixed internally to be the localized weighted adjusted determinant contributions, and the scalar equality proves that their finite sum is the theta-monoid degree. The structure-sheaf log-volume is identified with a distinguished direct summand of each localized arithmetic vector bundle. Each direct-summand log-volume is projected from the normalized log-volume of a local compact-open tensor factor constructed from additive Haar data. The additive-Haar route now has a strict compact-open factor constructor: the lower input* $1 < \mu(U)_{\mathbb{R}}$ *gives strict positivity of the normalized Haar log-volume, and the norm-square projection gives the nonzero selected direct-summand norm. The concrete inverse-dilation variant supplies this strict input from the local-field source* $p^{-1}O_K$, *whose compact-open property and Haar mass* $p^{[K:\mathbb{Q}_p]} > 1$ *are checked in Lean. The same inverse-dilation public route now has a family-hull-indexed calibrated localized hull-cover wrapper, so the principal-product family-hull containment, log-volume sum, and pointwise localized-region determinant identities can* *be derived from the index-fiber presentation, finite additivity, and local vector-bundle calibrations rather than supplied as localized source fields. Its stricter measure-model wrapper replaces the global finite-additivity input by the log-volume measure model attached to this cover. The direct-product wrapper goes one step lower by deriving the family-hull index map from local factor regions and their separation. The tensor-measure direct-product wrapper replaces that supplied cover-local measure model by direct-product cell log-volume identities and their tensor-cell finite-sum formula. The source-core tensor-cover wrapper lowers this boundary again: its tensor cells are the localized weighted adjusted determinant summands already present in the additive-Haar source-core decomposition, and the tensor-cell finite-sum formula is derived from the source-core localized log-volume theorem plus the direct-product cover identity. The charted local-ring wrapper lowers the local-factor interface: the factors are now regions of per-place local-ring/vector-bundle calibration objects, and Lean projects the chart log-volume/direct-summand equalities before invoking the source-core tensor-cover route. The valuation-ball charted wrapper lowers this once more: the local-ring calibrations are constructed from valuation-ball/additive-Haar factor calibrations, and the charted direct-product cover is projected from the valuation-ball regions, separation, and family-hull identity. The principal valuation-ball wrapper lowers this cover boundary again: the direct-product cell identity, local-factor separation, and family-hull equality are now projected from the SourceCore principal valuation-ball product-hull cover. What remains at this boundary is the alignment of that cover with the public Theorem 3.11 possible-region family and the additive-Haar source-core localized determinant decomposition. The principal-hull-aligned wrapper also derives the valuation-cover/public-hull-system equality from the lower cover's own principal-hull theorem plus the equality between its principal hull system and the public Theorem 3.11 principal-product hull system. The hull-formation possible-region wrapper lowers the possible-region equality by aligning the source-core and valuation-cover families with the concrete family inside ‘principalProductHullFormationData‘; the public ‘principalProductPossibleRegion‘ equality is then a projection. The valuation-cover transported-calibration wrapper also removes independent localized-calibration and valuation-factor families: Lean transports them from the principal valuation cover across the derived hull-system equality. The p-adic constructed source-core wrapper lowers this once more: it builds the finite-extension source core from transported valuation-cover fibers and finite-extension unit-ball Haar factors, then proves the projected vector-bundle and localized-region equalities consumed by the older transported-calibration route. The remaining lower matching datum is the factor-level equality between each projected finite-extension valuation ball and the transported valuation-cover factor. A finite-extension-backed valuation-cover wrapper now lowers the source of the valuation-cover factors themselves: the lower principal valuation cover is projected from finite-extension p-adic Haar data before entering the constructed source-core route. A hull-formation-selected intrinsic cover constructor now fixes the lower p-adic cover's hull system to the public principal-product hull system and its possible-region family to `principalProductHullFormationData`; the intrinsic possible-region comparison is therefore constructed rather than supplied, while the remaining principal-hull comparison is the lower equality between the p-adic principal hull source and the public hull system. A public-principal specialization now constructs this p-adic principal source as the `ULift` copy of the public principal-product source and proves the product-hull intersection law by transport through the public intersection theorem. Lean now also proves the `ULift`-invariance of the induced holomorphic hull system by reducing record equality to the `isHull` predicate and `logVolume` field. Source-core now lowers the public principal-product hull source itself to a scalar-parameter direct-product hull source. Product-parameter representability is now itself lowered to coordinate scalar-image data: once local coordinate points can be extended* *to product points, Lean proves that each coordinate product region is the image of the local integer factor under the supplied coordinate scalar map, uses the coordinate region itself as the local scalar parameter, assembles these choices into the global product scalar parameter, and constructs the intersection scalar selector from the represented direct-product intersection parameter. These data construct the full principal product-hull system and its minimal intersection law. Lean also transports this constructed product-carrier hull source across a carrier equivalence to the public Step (xi) carrier and exposes a localized theta-equals-family-hull route that consumes this coordinate scalar-image transported source instead of an independently supplied public-carrier principal source, local representability source, or global product representability field. In the one-coordinate real-line case, the carrier bridge is now canonical: a point of a labeled real-line copy is equivalent to its real coordinate and to the product* $\mathrm{Unit} \to \mathbb{R}$*. The remaining principal-hull boundary is to derive the coordinate scalar-image landing/preimage laws from Hodge-theater/log-theta lattice inputs, and to construct the localized determinant/Ob7 data from the same source. For the selected valuation-ball family-hull route, Lean now avoids the stronger all-coordinate-subsets interface: a nonzero local scalar multiplication source* $\lambda_v \cdot O_v$ *constructs the selected scalar-parameter valuation source, and the unit-ball valuation wrapper inherits the same selected-scalar source. The endpoint records equality of the selected scalar image with the valuation-ball direct-product cell union and the adjusted Ob3/Ob5 determinant handoff. The selected-scalar exact route now constructs the possible-image exact-*$\Xi(P_B)$ *source from increasingly local cover data. Lean first proves* $P_B = \bigcup_\beta \prod_v B_{\beta,v}$ *from indexed inclusions* $\prod_v B_{\beta,v} \subseteq P_i$*, then lowers those inclusions to a single local factor and finally to the compact-open valuation ball* $B_{\beta,v}(r_{\beta,v})$ *using the additive-Haar normalization audit. The nonzero-scalar and valuation-unit-ball wrappers inherit the same radius-to-exact-*Ξ *route. The exact-*Ξ *calibration is therefore derived rather than stored independently at this boundary. Lean also proves that holomorphic hull operators, possible-image families, and exact* $\Xi(P_B)$ *witnesses transport across carrier equivalences; the selected radius-cover exact-*Ξ *source can therefore be moved from the direct product carrier to any equivalent public carrier without adding a new compatibility assumption. A public adapter now specializes this transported exact-*Ξ *source to the Theorem 3.11 quotient family: from an equivalence between quotient indices and selected valuation-ball indices, pointwise equality of possible regions, and equality of hull operators, Lean derives the public exact-*Ξ *log-volume calibration. A choice-level refinement lowers the quotient-index equivalence itself: concrete Hodge-theater/log-theta choices map to selected valuation-ball indices, the map is invariant under the Theorem 3.11 equality quotient, and representatives over selected indices construct the required quotient equivalence by* Quot*-lifting. The local selected-q Ob5–Ob6 handoff now consumes this exact-*Ξ *calibration directly: Lean constructs the canonical* $\Phi(P_B)$ *family and the exact* $\Xi(P_B)$ *family before the global* C_Θ *comparison, with closed-family and product-hull provenance wrappers. The remaining paper task is to keep lowering the localized determinant and Ob7 inputs beneath the strongest preferred route. The cross-hull matching between these transported valuation-cover fibers and the public-hull finite-extension factors is now derived at the intrinsic wrapper: Lean transports the lower cover's finite-extension factors across the same hull-system equality used for the valuation-cover calibration fiber, then proves the old factor-matching field by a generic transport lemma. The coordinate scalar-image and real-line public routes now also construct their Step (xi) paper trace internally from the same constructed principal hull, localized determinant, theta-family-hull, and Ob7 data. Their strengthened additive-payload variants also reconstruct the derived Step (xi) source and the remaining non-Step audit from the additive-Haar arithmetic-degree payload, so the coordinate public boundary matches the principal-product boundary on this point. The remaining principal-hull work below this wrapper is the derivation of the* *realized theta-region local image laws from the lower analytic Frobenioid/log-Kummer data. Lean now derives the projected theta-region scalar-image equality from those realized local image laws: the local realization gives the whole theta region, while the coordinate projection of that realization is the local scalar multiple. The exact-image direct-product formula can also be specialized to each projected theta region, constructing the package-level coordinate scalar-image source without a separate projected equality. The coordinate valuation-ball exact-*Ξ *audit now converts this coordinate source to the selected scalar-parameter valuation cover and transports the selected radius-cover exactness to the public carrier. A stricter exact-image variant constructs the coordinate landing/preimage laws internally from this local image formula before entering the same public exact-*Ξ *route. Below it, a Hodge-theater/log-theta lattice-key constructor uses the package target region formula, a public carrier equivalence, and coordinate projections of the theta region to build the scalar-parameter direct-product hull source. The inverse-dilation wrapper lowers the analytic input further: transported valuation-ball factors are identified with constructed inverse base-prime unit balls, and Lean derives* $1 \leq \mu(U)_{\mathbb{R}}$ *from* $1 < \mu(p^{-1}O_K)_{\mathbb{R}}$*. The selected strict determinant summand is now projected from this same valuation-cover inverse-dilation family, at the chosen positive index and summand, so the public boundary no longer accepts a separate selected-factor inverse-dilation source and equality. A theorem-produced boundary audit records both the all-factor and selected-summand inverse-base-prime unit-ball equalities from that same family. The Hodge/family-hull calibrated wrapper lowers the theta scalar input further: its source supplies* $\deg_\Theta = \mathrm{vol}_{\log}(\mathrm{Hull}_{\mathrm{fam}})$*, and Lean derives the older localized adjusted determinant equality by composing with the localized hull decomposition theorem* $\mathrm{vol}_{\log}(\mathrm{Hull}_{\mathrm{fam}}) = \sum_{\mathrm{loc}} \mathrm{vol}_{\log}$*. The summand-calibrated wrapper lowers this once more by taking* $\deg_\Theta = \sum_\beta \mathrm{vol}^{\mathrm{adj}}_{\log,\beta}$ *as source data and deriving the Hodge/family-hull equality. Its public route is . Conversely, the localized-adjusted inverse-dilation source now projects to this finite summand source by unfolding* localizedAdjustedSum *as the finite sum of weighted adjusted log-volumes: and . The constructed Gaussian/Hodge inverse-dilation wrapper now also projects directly to the finite Hodge-summand source by exposing the constructed Hodge theta-summand family and its pointwise equality with localized weighted adjusted log-volumes: . It still has the localized-adjusted projection ; however, its public finite-summand route now enters the summand-calibrated wrapper directly, not through this localized-adjusted projection: . The Haar-modulus-backed wrapper now exposes the same expansion through , whose audit is . This audit now records the direct constructed Hodge-summand boundary, rather than only the localized-adjusted expansion audit. The additive-Haar constructed Gaussian/Hodge layer now projects into this same finite-summand route via , with audit . The direct-summand calibrated wrapper lowers the structure-sheaf input as well: it supplies the localized equality with the chosen direct summand, and Lean derives the additive-Haar normalized compact-open equality by the projection theorem for the constructed additive-Haar localized source. The non-vacuous Haar-modulus/additive-Haar wrapper now also implements the source-level direction needed by the compact-open construction: it supplies the equality with the normalized additive-Haar compact-open factor and derives the structure-sheaf/direct-summand equality by . The corresponding public wrapper is , with audit . Its finite-summand endpoint is ; the residual-payload route now invokes this endpoint, so the IUT-IV arithmetic-source wrapper inherits the finite Hodge-summand expansion. The normalized CTheta-derived route is now also exposed through the unit-ball Haar-character wrapper with full audit . The residue-module normalized route now factors through this unit-ball normalized route, and its full audit records the projected unit-ball normalized full audit before the intrinsic Haar-modulus normalized boundary. The same finite-summand route is now exposed through the intrinsic finite-extension Haar-modulus wrapper , the unit-ball Haar-character* *wrapper , and the residue-module quotient wrapper . The unit-ball route has the CTheta-derived finite-summand endpoint and full audit . The residue-module/unit-ball route has the corresponding endpoint and full audit ; both project the non-normalized additive-Haar arithmetic residual from the IUT IV ledger instead of taking an independent obligations package. The corresponding additive-Haar arithmetic-degree payload has also been split at this boundary: kept the lower Theorem 1.10/IUT IV and IPL/SHE/APT assumptions, while reconstructs the Step (xi) determinant/calibration slots from the normalized finite-summand source audit and proves the old remaining-payload audit for the constructed obligations. The non-normalized additive-Haar/Gaussian-Hodge companion now uses the parallel summand audit and exposes the finite-summand IUT-IV arithmetic-source route . This non-normalized branch now has constructor-coupled, finite-place, and localized Step (xi) C_Θ entry points as well; the strongest residue-module/unit-ball wrapper is . The residue inverse-base-prime version of this non-normalized branch now has the corresponding finite-summand endpoint . Its audit records both the residue radius-p inverse-cover projection and the inherited Haar-modulus/additive-Haar finite-summand expansion boundary. It now also exposes constructor-built, coupled, finite-place local-global, and localized Step (xi) C_Θ handoffs; the deepest localized non-normalized wrapper is . The direct projected Gaussian/Hodge residue source, and the finite-Hodge-summand refinement beneath it, have been lowered to the same finite-place and localized constructor-built C_Θ surface through ; the summand-level endpoint is . Hence this branch no longer stops at an independently supplied IUT-IV arithmetic ledger after the finite Hodge-summand expansion has been constructed. The normalized residue inverse-base-prime branch reaches the same finite-summand endpoint through ; its audit records that the direct-summand handoff is reconstructed from the normalized compact-open Haar structure-sheaf equality before projecting to the finite Hodge-summand expansion. Its IUT IV, constructor-built, and coupled arithmetic wrappers now construct the normalized additive-Haar arithmetic obligations from this finite-summand audit directly, rather than delegating through the older direct-summand public wrapper. Constructor-built, coupled, finite-place, and localized Step (xi) C_Θ entry points now thread this normalized residue endpoint as well; the deepest localized wrapper is . On the non-normalized finite-summand side, the intrinsic finite-extension Haar-modulus, intrinsic unit-ball Haar-character, and residue-module/unit-ball wrappers now construct their additive-Haar p-adic arithmetic-degree obligations at their own projected source boundary, rather than obtaining that construction by delegation to a lower generic wrapper. Their finite-place local-global and localized Step (xi) C_Θ entry points now use the corresponding finite-place/localized arithmetic residual payload constructors directly, and the same direct residual threading is now used by the normalized structure-sheaf/direct-summand entry points, rather than first collapsing to the coupled IUT-IV arithmetic-source projection. The same no-external-scale constructor entry points are now exposed one layer higher in the intrinsic and unit-ball Haar-character normalized wrappers, with localized source data entering through . The newer boundary removes the Step (x) and* IPL/SHE/APT *fields from the residual as well: contains only seven arithmetic/formula fields, and reconstructs the older residual payload from the vertical log-Kummer package and Theorem 3.11 bridge package. The IUT IV source boundary now removes even this standalone seven-field residual when a object is available: projects the seven Theorem 1.10/finite-place/local-to-global arithmetic fields from the C_Θ ledger, while and replace the final local-to-global endpoint by the constructor-built Step (xi)/IUT IV audit once the constructor source's canonical scale is identified with the IUT IV finite-place scale. This source-specific residual has a stricter coupled-source variant as well: constructs the IUT IV finite-place ledger with the constructor-built canonical scale as its finite-place scale and requires the lower arithmetic gap estimate* $C_{\Theta,\mathrm{can}} + 1 \leq \mathrm{upper} - \mathrm{main}$. *The finite-place* *source lowers this once more: records local canonical scales, arithmetic/log contributions, and Haar normalization defects, proves by summing the local Step (xi) bounds, and constructs the coupled IUT IV arithmetic source via . Its endpoint is now recorded by the localized C_Θ handoff, so the public evidence surface names the finite-place summation/gap layer before the signed comparison is projected. The localized local-term source lowers the local inequality itself: takes the localized Ob3/Ob4 vector-bundle determinant source as the local canonical scale, decomposes each arithmetic upper contribution as Step (xi) log-volume plus Haar defect plus main-log term, proves by ordered-ring arithmetic, and constructs the finite-place source via . The arithmetic-divisor-backed additive-Haar source now reaches this same boundary directly: and project the local analytic and Theorem 1.10 C_Θ data from the matched local-degree, formula-gap, and local-global audits, while and now factor those projections through the typed additive-Haar formula-gap arithmetic-divisor source. Thus the Proposition 1.4/1.5 endpoints, distinguished formula-to-gap comparisons, archimedean formula-to-gap comparisons, and divisor nonnegativity bounds are read from that lower source rather than reassembled at the finite-divisor boundary. The top-level paper-trace audit now uses these same projections for its iutIV_cTheta source data, eliminating the previous duplicate inline local-analytic/Theorem 1.10 assembly. The source-level preservation theorem now states directly that the Proposition 1.4/1.5-backed formula source satisfies the older local-case endpoint after the distinguished and archimedean formula-to-gap comparisons are supplied. Meanwhile constructs the localized constructor source after synchronizing the constructor determinant with the localized Step (xi) determinant. Its source-level handoff feeds the residual-payload audit and derives the constructor-built arithmetic-gap and C_Θ comparisons. Consequently and and and and and no longer expose a standalone finite-place scale equality at the public boundary. The direct non-normalized additive-Haar route now has the same finite-place and localized constructor-built C_Θ entry points, via and . The older source-specific residual also reaches the base public normalized and additive-Haar route boundaries via , , and , and their intrinsic finite-extension lifts and , and through the corresponding unit-ball Haar-character normalized and finite-summand wrappers and , and now through the residue-module/unit-ball normalized and finite-summand wrappers and , with the normalized residue-module route also exposing the coupled no-standalone-scale-equality endpoint , together with the direct finite-place local-to-global endpoint , and the localized local-term endpoint , while exposes the corresponding public route. The same CTheta-derived arithmetic boundary is now lifted through the intrinsic finite-extension and residue-module/unit-ball normalized public routes. This audit now also records the paper trace constructed from the same additive-Haar compact-open factor source and exposes its Ob7–Ob9 log-Kummer, vertical-shift, and realified-log-volume components. On the determinant side, the record Ob3/Ob5 compatibility can now be reconstructed from the paper-facing family-hull equality* $\mu_{\log}(\varphi(P_B)) = \widehat{\deg}_{\mathrm{norm}}$ *by and its concrete quotient-source lift . The exact-theta variants and additionally derive the Ob4 tensor-power bound from* $\theta_{\mathrm{signed}} = \mu_{\log}(\varphi(P_B))$*, using . The same exact-theta lowering is now available at the constructor-built possible-image Step (xi) boundary via . Its minimal-hull-system and product-hull variants derive the compatibility package and tensor-power bound before constructing the record-canonical hull/tensor bridge; the bridge equality, measure synchronization, q-positivity, and normalization are still explicit lower inputs. The endpoint records the resulting product-hull provenance, selected-hull containment, reconstructed compatibility, derived tensor-power bound, and signed comparison. The obligations-backed variant projects bridge equality, q-positivity, and normalization from ; the remaining synchronization identifies its bridge datum with the record-canonical product-hull/tensor-power datum. The current hardened non-vacuous public wrapper lowers this boundary again: the restriction local-global object, ordinary-log-volume/Hodge-degree* *calibration, direct summand data, and Hodge/family-hull scalar equality are projected from the constructed Gaussian/Hodge localized-adjusted source. The current public wrapper lowers this boundary further: the Gaussian/Hodge localized-adjusted Ob7 source is constructed from additive-Haar compact-open factors, the selected strict compact-open mass is derived from the inverse base-prime factor, the Haar-modulus valuation-cover side is projected from the intrinsic finite-extension valuation-cover source, and the inverse-dilation Haar-modulus family is derived from residue-module quotient data:* O_v/p_vO_v*, the residue-field cardinality* p_v*, and the finrank comparison* $\dim_{\kappa_v}(O_v/p_vO_v) = [K_v : \mathbb{Q}_p]$ *construct the coset decomposition,* $\mu(p_vO_v) = p_v^{-[K_v:\mathbb{Q}_p]}$*, and the unit-ball Haar-character source. The normalized-Haar structure-sheaf/direct-summand handoff has also been pushed through this intrinsic finite-extension boundary. The generic handoff is projected by ; the strict residue-module wrapper is , with audit . The restriction-calibrated source lowers the Hodge-calibration input at this boundary: the local-prime comparison is derived by from the environment-anchored local-global restriction package, and the localized constructor-built* C_Θ *public wrapper is . The short public surface exposing this restriction-calibrated boundary is , with audit . Its hardened CTheta-derived arithmetic variant is ; its remaining-payload audit is . The combined full-boundary audit bundles the residue-module/unit-ball Step (xi) source audit, the projected normalized additive-Haar audit, the transported-factor handoff into the localized determinant layer, the IUT IV arithmetic-residual audit, and the definitional equality with the projected normalized public route. The local factor handoff is now audited at the point where the determinant layer reads it: records both the transported valuation-ball factor equality and the induced equality of the localized additive-Haar vector-bundle factor with the inverse base-prime unit-ball source constructed from the residue-module quotient data. It also projects these equalities to the determinant-scale invariants consumed downstream: the compact-open subset is* $p_v^{-1}O_v$*, its Haar mass is* $p_v^{[K_v:\mathbb{Q}_p]}$*, and its normalized log-volume is strictly positive, for both the transported factor and the localized additive-Haar factor. The equality boundary has been split in two lower stages. First, the component-matched source ; it supplies equality of the defining additive-Haar compact-open data fields, and the constructor recovers the full source equality by additive-Haar extensionality. Second, the scale-component source matches the transported factor directly against the finite-extension inverse-dilation mass source: the local ring, residue prime, finite degree, realization, Haar measure, compact open* $p_v^{-1}O_v$*, and base-prime scale map. Its constructor derives the generic component-matched source, and its audit recovers the same determinant-factor handoff. Third, the valuation-ball scale source matches the transported local factor before additive-Haar projection, using . The source-core projection turns valuation norm, selected valuation ball* $p_v^{-1}O_v$*, Haar measure, realization, and pointwise base-prime dilation data into the additive-Haar scale-component match. Its Step (xi) constructor and audit recover the same determinant-factor handoff from this lower valuation-ball source. The valuation-ball source has now been lowered one stage further: takes the constructed finite-extension inverse-dilation mass source together with the positive-radius equality* $p_v^{-1}O_v = \{x \mid v(x) \leq r\}$*, constructs the valuation-ball additive-Haar source, and proves . The residue wrapper therefore derives from the constructed inverse-base-prime valuation-ball source rather than an independently supplied component match. The remaining lower input is now only the transport equality to the local factor when the finite extension carries the normed-algebra compatibility. The radius identity is proved by : using* $\|\mathrm{algebraMap}(p)\| = \|p\|_p = p^{-1}$*, the inverse base-prime unit ball is the radius-*p_v *valuation ball. The constructor supplies the inverse base-prime valuation-ball source without a separate radius field, and the residue wrapper exposes the resulting handoff audit. Lean now also records the obstruction that this wrapper cannot be populated against the current* *intrinsic valuation-unit-ball cover at any actual local factor: derives contradiction from the radius comparison $1 = p_v$. The replacement factor cover is now implemented in : its local factors are constructed from inverse-dilation mass sources, SourceCore proves their radius is p_v, and the cover projects to the generic valuation-ball tensor packet. Step (xi) also exposes the hull-formation-selected public principal-product version , which keeps the inverse-dilated branch separate from the legacy radius-one unit-ball branch. The residue-module handoff is now also attached to this inverse cover by : it identifies the cover's inverse-dilation mass source with the residue-module constructed mass source and derives the additive-Haar inverse-base-prime unit-ball equality at each radius-p_v local factor. The preferred-public normalized additive-Haar route now has the inverse-cover projection as well, via and the route . Its audit records that the route's Haar-modulus valuation-cover argument is definitionally the radius-p_v residue-module inverse-cover projection, not the legacy unit-ball intrinsic cover. The same inverse-cover source now also has a non-normalized additive-Haar constructed Gaussian/Hodge route, . This route consumes the localized structure-sheaf/direct-summand handoff and projects the Haar-modulus valuation cover from the radius-p_v residue-module inverse cover before invoking the generic additive-Haar compact-open Gaussian/Hodge constructor. It now has the same localized constructor-built C_Θ entry point, , so its additive-Haar arithmetic-degree obligations are reconstructed from the localized constructor data rather than an independently exposed IUT IV ledger. A stronger residue inverse-cover wrapper, , now takes the constructed Gaussian/Hodge localized-adjusted source over the same residue-derived localized hull vector bundle and projects from it the Hodge local-global comparison, direct-summand data, and localized-adjusted theta equality needed by the older additive-Haar route. Its public route is . The summand-form residue wrapper, , lowers this boundary one step further: it derives the localized-adjusted theta equality by summing finite Hodge summands and identifying each summand with the corresponding localized weighted-adjusted determinant term. Its public route is . A companion restriction-local-global wrapper, , projects the older public source's global object, localization, local object identifiers, and Hodge/local-prime equality from . Its public route is now lowered to the localized constructor-built C_Θ surface: . The constructed Gaussian/Hodge residue source now also factors through this wrapper via and the localized public route . The summand-form residue source now factors through the same restriction layer as well; its audit records both the finite-summand theta expansion and the projected local-global restriction boundary, and its public route is . The inverse-dilation principal-product branch now lowers the finite Hodge-summand formula again through : the total localized-adjusted theta equality is derived by summing normalized Hodge weights whose pointwise contributions equal the localized weighted adjusted determinant terms. The stricter normalized-summand wrapper no longer accepts these weights independently: it defines each weight as the constructed Hodge summand divided by the nonzero theta-monoid degree, proves that the weights sum to* 1*, and projects to the square-weight route. The raw-positive variant derives this nonzero theta input from local determinant positivity: nonnegative adjusted raw local log-volumes and one strictly positive adjusted raw local log-volume imply a positive localized adjusted sum, hence a nonzero theta-monoid degree. The direct-summand raw-positive refinement now derives those raw positivity facts from the Ob3 structure-sheaf summand comparison: the structure sheaf is one nonnegative direct summand, and a separate non-structure summand is strictly positive. The square-weight branch now projects through this direct-summand raw-positive route as well, so its determinant presentation and its positivity normalization are checked by one boundary audit. The Haar-modulus inverse-dilation and additive-Haar public branches now use the same strict projection: their compact-open valuation-cover constructors feed the direct-summand raw-positive normalization branch instead of stopping at localized adjustment. The intrinsic p-adic finite-extension, unit-ball*

Haar-character, and residue-module/unit-ball wrappers now expose the same strict projection, so these lower arithmetic branches no longer land only at the older localized-adjusted finite-summand endpoint. The remaining migration below this branch is therefore to construct the underlying local square/direct-summand determinant factors and the pointwise local determinant identities from the lower Hodge-theater and log-theta-lattice data. The same scale-component match is now exposed below the residue wrapper as well: the intrinsic Haar-modulus source reconstructs the Haar-modulus valuation-cover equality from inverse-dilation mass components, the unit-ball source projects to it, and the residue scale source projects to the unit-ball scale source via . This handoff is now a field of the additive-Haar residue audit, the normalized residue audit, and the combined CTheta full-boundary audit, so the strongest public audit exposes the determinant-factor match directly rather than leaving it as a side declaration. The older public wrapper is , with audit . The finite-extension valuation-unit-ball route is only a normalized legacy boundary: radius 1 *and* $\mu(O_v) = 1$ *construct weak nonnegativity, and Lean proves that the selected unit-ball normalized log-volume is* 0*. The strict-positive unit-ball route is therefore quarantined as vacuous, including at the exact intrinsic endpoint . The p-adic finite norm-square/direct-summand localized-adjusted source , uses the structure-sheaf direct-summand identity, a nonzero norm-square summand, and the localized adjusted determinant sum to derive the Hodge/family-hull comparison before entering the restriction/Hodge calibrated Step (xi) route. The same normalization audit now proves that the selected unit-ball norm-square summand is forced to be zero: . Thus this p-adic unit-ball branch is also quarantined rather than accepted as a non-vacuous determinant route. Its family-hull indexed measure-model, direct-product measure-model, tensor-measure direct-product, source-core tensor, charted local-ring, valuation-ball, principal-hull aligned, transported valuation-cover, constructed source-core, finite-extension-backed, and intrinsic finite-extension cover wrappers remain compatibility infrastructure. The explicit norm-square compatibility route consumes the concrete finite-extension valuation-cover source, the localized-adjusted norm-square/direct-summand source, and the explicitly lower-level additive-Haar remaining-payload audit; Lean constructs the assembled non-Step-(xi) payload internally, rather than accepting a preassembled Step (xi) audit. It expands to the hardened endpoint , and the norm-square public-surface audit carries the compatibility boundary audit. The goal-scope evidence audit packages the surface audit, closed-input audit, norm-square determinant boundary, lower additive-Haar payload, compact-open legacy quarantine, and norm-square unit-ball quarantine without declaring the global goal complete. Its packet-facing refinement projects the Theorem 3.11 source data from the concrete packet and the Remark 3.9.5 principal hull from the packet/symmetry-label theta-region valuation-ball source, so those two objects are no longer independent public inputs at the final norm-square evidence boundary. The restriction-calibrated packet refinement lowers this boundary further: it projects the intrinsic finite-extension valuation cover and localized norm-square/direct-summand calibration from the pointwise restriction-calibrated local arithmetic source before entering the packet-facing final evidence audit, while preserving the unit-ball norm-square contradiction as diagnostic evidence. The localized-*C_Θ *packet refinement removes the remaining standalone IUT IV* C_Θ *ledger, additive-Haar arithmetic-degree obligations, and additive-Haar remaining-payload witness from that packet-facing boundary: all three are projected from the constructor-built localized Step (xi) local-term source and then threaded through the restriction-calibrated packet evidence. The pointwise-localized packet refinement lowers this once more: the constructor-built localized* C_Θ *source is no longer a public input, but is constructed from the pointwise localized Step (xi) determinant source, local main-log terms, Haar normalization defects, and finite-place summation identities before the packet evidence derives the signed comparison. Its IUT-IV arithmetic-divisor packet refinement* *removes that pointwise localized source from the packet boundary: Lean projects it from before deriving the same localized handoff, signed bound, dichotomy, and packet evidence. The Theorem 1.10 valuation-ball packet refinement lowers the IUT-IV input one layer further by projecting the arithmetic-divisor localized source from before entering the arithmetic-divisor packet refinement. The local-analytic valuation-ball packet refinement lowers the formula-gap matched source itself: the packet route now projects it from before applying the Theorem 1.10 valuation-ball packet evidence. The synchronized local-analytic packet refinement constructs that local-analytic source from the constructor-built localized Step (xi) determinant/scale synchronization source, the local-analytic arithmetic-divisor ledger, and the finite Haar/main-log comparison data. The arithmetic-realized synchronized variant lowers this comparison datum: the endpoint now accepts the explicit Theorem 1.10 identity* $\mathrm{StepXI}_v + \mathrm{Haar}_v = a_v(D_v + C_v) + E_v - M_v$, *and Lean unfolds the local arithmetic-upper definition to recover the older Step (xi)/Haar/main-log equality internally. The formula-gap synchronized packet refinement projects this local-analytic ledger from the valuation-ball formula-gap source, retaining its Proposition 1.4 log-shell and Proposition 1.5 metric comparison records before constructing the synchronized packet evidence. Its arithmetic-realized companion performs the same recovery of the Step (xi)/Haar/main-log equality from* $a_v(D_v + C_v) + E_v - M_v$ *after first projecting the local-analytic ledger from the formula-gap source. These pointwise localized, IUT-IV arithmetic-divisor, Theorem 1.10 valuation-ball, local-analytic, synchronized, and formula-gap packet audits now also expose the constructed finite-place local-global* C_Θ *endpoint, so the finite-place summation/gap layer remains visible through the strongest packet-facing route rather than only at the lower localized handoff. The pointwise Theorem 1.10 valuation-ball packet route consumes the corresponding pointwise localized source instead, projecting the determinant/scale synchronization, formula-gap source, Haar defect, Step (xi)/Haar/main-log identity, and finite Haar lower bound internally. This route is now exposed as its own audit record: the record carries the pointwise Theorem 1.10 endpoint, the internally constructed determinant/scale synchronization endpoint, the projected formula-gap packet evidence, and the resulting signed* C_Θ *comparison. The localized local-term packet route constructs that pointwise source from the localized Step (xi) local-term* C_Θ *source and a valuation-ball formula-gap source, leaving only the local arithmetic/main-log matching at this boundary. The compatible refinement packages this matching as : the Theorem 1.10 valuation-ball local main-log function is identified with the localized Step (xi) main-log summand, and the local arithmetic upper contribution is identified with the Step (xi) adjusted determinant log-volume plus Haar defect plus main-log summand. Thus this compatibility is now an audited source layer rather than two anonymous route arguments. The arithmetic-realized refinement derives that compatibility from : the Step (xi) adjusted determinant plus Haar defect is matched with the expanded local Theorem 1.10 arithmetic-degree expression* $a_v(D_v + C_v) + E_v - M_v$, *and Lean then recovers the local arithmetic-upper equality by unfolding the local arithmetic contribution and adding back the main-log summand before passing into the formula-gap synchronized packet audit. Lean now records this as a named arithmetic-realized packet audit, projecting the compatible packet evidence together with the localized* C_Θ *source, the local-term/IUT IV handoff audit, the signed* C_Θ *bound, and the Corollary 3.12 dichotomy directly; the route therefore no longer has to pass through a separate compatibility-only citation before reaching the signed endpoint. The formula-realized refinement consumes the pointwise Hodge measure/summand formula source, constructs the theta-region and pointwise calibration records internally, and then enters the same arithmetic-realized packet audit. The local-arithmetic-degree refinement removes the packaged arithmetic-realization record at this boundary: it constructs that source from the main-log matching and the pointwise identity* $\mathrm{StepXI}_v + \mathrm{Haar}_v = a_v(D_v + C_v) + E_v - M_v$ *before recov-*

ering the localized C_Θ handoff and signed endpoint. The synchronized local-arithmetic-degree refinement goes one level lower: from determinant/scale synchronization, the valuation-ball formula-gap ledger, the Haar defect, and the same local degree identity, Lean constructs the IUT IV arithmetic-divisor localized source and the localized local-term C_Θ source internally. Its constructed-synchronization companion builds that synchronization record from the localized Step (xi) determinant source and the constructor-built determinant/canonical-scale comparison equalities before entering this route. The norm-square localized determinant refinement projects this localized source from direct-summand norm-square vector-bundle data before using the same constructed synchronization route. The p-adic finite-extension refinement projects that norm-square determinant from unit-ball valuation-Haar compact-open local data before entering the same endpoint. A pointwise Theorem 1.10 refinement also projects the determinant/scale synchronization and the arithmetic-degree Step (xi)+Haar identity from a single localized valuation-ball source. The short preferred endpoint is now the constructor-built localized C_Θ handoff projects the arithmetic residual and remaining additive-Haar payload from the constructor-built possible-image hull/determinant source and localized Step (xi) finite-place local-term source; the longer name is retained as a definitional alias. Older public-looking `preferredPublic...` *declarations remain implementation or compatibility layers below this surface audit. The constructor-built localized C_Θ goal-evidence audit records, at the short preferred surface, the constructor-built Remark 3.9.5 Step (xi) boundary, localized finite-place C_Θ endpoint, local-term-to-IUT-IV C_Θ handoff audit, local-to-global C_Θ audit, and the additive-Haar remaining payload derived from that localized arithmetic source. The localized handoff audit identifies the constructor canonical scale with the finite-place scale, proves the summed Step (xi) Haar-defect arithmetic gap, and derives the global C_Θ bound used by the ordered-real dichotomy. The restriction-calibrated public route audit now also carries the inverse-dilation strict-selected audit from the same residue source, so the strict selected Haar mass, positive normalized compact-open log-volume, and selected nonzero direct-summand norm are visible at that public boundary. The stronger goal-evidence package records this route audit together with the projected constructor-built localized C_Θ goal evidence. The product-hull localized C_Θ route lowers this once more by constructing the Step (xi) source from the Remark 3.9.5 product-hull system, Ob3/Ob4 adjusted determinant source, and exact theta/family-hull log-volume equalities before entering the short preferred route; its audit records the product-hull provenance and short-route localized C_Θ evidence together, and now also records the constructor-sourced Step (xi) endpoint where the obligations object is definitionally produced by the legacy finite-route hull/determinant constructor. Its obligations-backed variant projects the bridge equality, q-pilot positivity, and normalization from one object; the remaining explicit synchronization is the equality with the record-canonical product-hull/tensor-power datum. The stricter canonical-HDD wrapper now moves this synchronization below the preferred-route surface into the source object . Its endpoint records, at the common-container/HDD layer, that the package HDD hull/determinant bridge first equals the constructor-generated obligations bridge and then the record-canonical product-hull/tensor-power bridge; the route itself consumes this lower source object and the localized C_Θ source only. One layer below this, Lean now has the common-container update constructor and the product-hull audit . This constructs a package whose HDD hull/determinant bridge is definitionally the record-canonical product-hull/tensor-power bridge while preserving the chart and signed q/Θ endpoints. Lean now also transports the full Theorem 3.11 source record over this package via ; the audit records that the transported record has the constructed coric HDD bridge, preserves the possible-image family, retains the target-union equality, and keeps the separated Hodge-theater histories. The canonical-HDD preferred route now enters the standard short route* *through , i.e. through the constructor-built Step (xi) source over the transported record. The direct canonical-HDD source removes the preassembled input at this boundary: it carries the product-hull, adjusted determinant, exact theta/family-hull calibration, q-positivity, and normalization data, while Lean installs the record-canonical HDD bridge definitionally, transports the Theorem 3.11 record, and generates the hull/determinant obligations from the resulting constructor. The generated-full-label specialization now pins that direct-HDD source to the generated Theorem 3.11 quotient corridor: its audit records the generated corridor audit, the direct-HDD audit, equality of the Step (xi) q-choice with the generated selected q-choice, and equality of the record possible-image family with the generated theta-class pullback. The corresponding public route is definitionally the direct canonical-HDD route applied to this generated source. The chosen-output generated full-label source lowers this boundary by rebuilding that generated-HDD source from the chosen-output generated quotient corridor; its public route is definitionally the generated full-label route applied to this chosen-output source. The constructed-record variant first rebuilds the generated record's possible-image family from the chosen-output packet theorem and then lowers to the chosen-output route. The structured-bundle variant constructs the generated multiradial record from a generated* SHE-*structured bundle before entering the constructed-record route. The reindexed-structured variant derives the generated bundle from the concrete structured bundle before entering the structured-bundle route. The constructed-qualitative variant derives that concrete structured bundle from constructed qualitative Theorem 3.11/SHE data before entering the reindexed-structured route. The combined restriction-calibrated generated-full-label route now requires both hardened inputs at once: the generated full-label/direct-HDD Step (xi) constructor, and the restriction-calibrated local-global Frobenioid source deriving the Hodge local-prime comparison. Its audit records the generated full-label audit together with the direct-HDD restriction-calibrated route audit. The constructed-qualitative principal-product source lowers this boundary by requiring the principal-product Ob3/Ob5 measure/Hodge-gap HDD source to be indexed by the constructed-qualitative generated record and by synchronizing its q-choice with the selected generated q-choice. The corresponding restriction-calibrated public route enters the principal-product route through this coupled source. The direct-HDD audit now also exposes , so the inverse-dilation strict-selected branch remains visible after passing to the product-hull/HDD constructor layer. The current product-hull goal-evidence package records this non-vacuous principal-product route separately from the older quarantined norm-square compatibility audit. Its constructed-qualitative strengthening promotes the same evidence over the coupled constructed-qualitative principal-product source, so this evidence surface no longer ranges over a generic constructor record. The companion audit couples that current route to the restriction-calibrated norm-square/direct-summand projection that feeds the Ob7 construction layer: it constructs the projected p-adic finite localized hull source, records the structure-sheaf/direct-summand synchronization and selected nonzero norm-square input, and exposes the restriction/Hodge-calibrated Ob7 projection. The same audit now also exposes the constructor-built Step (xi) boundary, the localized finite-place* C_Θ *endpoint, and the local-to-global* C_Θ *handoff produced by the localized Step (xi) source. This remains diagnostic because the unit-ball norm-square branch is still quarantined from the inverse-dilation strict-selected branch. The structured-SHE current evidence audit first constructs the principal-product HDD source via , so the source-obligation gap is produced from the structured Theorem 3.11/SHE bundle and side conditions before the current Ob7-coupled evidence is applied; it now projects the same localized* C_Θ *endpoint and handoff fields through the structured-SHE boundary. The theta-region current variant lowers the same source further to the principal valuation-ball constructor: the principal hull, Ob3/Ob4 data, adjusted-summand identity, and pre-ledger measure synchronization are read from the theta-region* *source before the current product-hull/Ob7 audit boundary is constructed, and the localized C_Θ endpoint and handoff are exposed at this theta-region boundary as well. The constructed theta-region public route enters the restriction-calibrated public surface only after Lean constructs this principal-product HDD source from the theta-region data. The goal-facing primitive completion audit now types the localized Step (xi) source through , whose audit records that the source gap and transport guard are induced by the primitive Theorem 3.11 constructor. It then packages the route audit with the constructed theta-region norm-square/Ob7 evidence from the same inputs. The localized calibration-data route lowers the remaining public measure/summand calibration pair to : the source records the pointwise principal-hull log-volume formula and the finite Hodge adjusted-summand identity, then projects the older calibration records before constructing the localized C_Θ handoff. The witness-derived version performs the same calibration projection after constructing the symmetry-label wrapper from possible-image witness data. The companion local-arithmetic route consumes instead of separate restriction-calibrated, structure-sheaf/direct-summand, and nonzero norm-square Ob7 inputs. Its endpoint now also exposes the derived signed C_Θ inequality and Corollary 3.12 dichotomy obtained from the localized Step (xi) local-to-global handoff, so these facts are no longer merely implicit in the handoff audit. The witness-derived companion performs this local-arithmetic projection after constructing the symmetry-label wrapper from possible-image witness data. The stronger removes the separate principal-product hull argument on this path: the principal hull consumed by local arithmetic is projected from the packet/symmetry-label theta-region source. The witness-derived companion performs the same principal-hull projection after constructing the symmetry-label wrapper from possible-image witness data. The fiber-backed variant constructs that theta-region source internally from a principal valuation-ball product-hull cover and the packet-backed fiber-transport family-union source. Its witness-derived companion first constructs the symmetry-label wrapper from possible-image witness data and then applies the same fiber-backed theta-region construction. The current strongest theta-boundary route lowers this again by taking a single theta-region-defined principal valuation-ball source and projecting both of those objects internally. Its witness-derived companion performs the same single-source projection after constructing the symmetry-label wrapper from possible-image witness data. The pointwise metric variant lowers the same route to the principal constructed-log-shell/metric-zero valuation-ball source. Its witness-derived companion constructs the symmetry-label wrapper from possible-image witness data before consuming that pointwise metric source. The calibration-refined pointwise route projects the theta-region measure/summand calibration from that pointwise valuation-ball boundary instead of taking it independently. Its witness-derived companion performs the same calibration projection after constructing the symmetry-label wrapper from possible-image witness data. The Hodge-formula pointwise route goes below the packaged pointwise calibration source. It consumes , and the projection constructs the old calibration record from the hull-volume formula, the Hodge-Arakelov theta-degree charting formula, and the adjusted-summand formula. The local-arithmetic-refined pointwise route constructs the restriction-calibrated norm-square Ob7 handoff from the pointwise residue-module/unit-ball/additive-Haar source. Its witness-derived companion performs that pointwise Ob7 projection after constructing the symmetry-label wrapper from possible-image witness data. The localized Step (xi) witness-derived route continues this lowering through the proof-carrying localized determinant, finite-place summation, and Haar-defect source. The witness-derived IUT IV arithmetic-divisor route projects that localized source from the IUT IV arithmetic-divisor local evaluation ledger over the same pointwise surface. The witness-derived Theorem 1.10 valuation-ball route projects that arithmetic-divisor ledger from the valuation-ball additive-Haar formula-gap source on the same witness-derived surface. The witness-derived*

Hodge-formula route constructs the pointwise measure/summand calibration from explicit formulas before entering that Theorem 1.10 valuation-ball route. The witness-derived local-analytic Hodge-formula route checks the local-analytic valuation-ball source on the same witness-derived surface before projecting to the Theorem 1.10 valuation-ball route. The local-analytic Hodge-formula route keeps this proof-carrying pointwise Ob7 source as the public local-arithmetic boundary while projecting the formula-gap matched Theorem 1.10/IUT IV source from the local analytic valuation-ball source. The product-handoff refinement constructs that packet-facing pointwise source from the lower principal-product restriction-calibrated Ob7 handoff. This Ob7 handoff is a diagnostic, not the non-vacuous route: the checked contradictions and project the existing unit-ball norm-square zero computation against the supplied nonzero selected summand. The compact-open companion constructs the principal pointwise valuation-ball source from compact-open log-Kummer data and still exposes only the single pointwise local-arithmetic source, not its two internal p-adic projection proofs. Its product-handoff variant keeps the compact-open log-Kummer boundary while lowering the local arithmetic input to the same principal-product handoff. The Hodge-formula plus restriction-calibrated local-arithmetic route remains as the constructor route from the lower restriction-calibrated residue-module/unit-ball/additive-Haar source; at that lower surface the structure-sheaf/direct-summand synchronization and nonzero norm-square projection facts are still explicit local arithmetic bridge fields. Its compact-open formula-gap companion reconstructs the local-analytic localized Step (xi) wrapper from the pointwise formula-gap Theorem 1.10/IUT IV source before invoking the same restriction-calibrated compact-open handoff. In parallel, the inverse-base-prime pointwise route now attaches the same concrete pointwise theta-region principal hull to the residue-module inverse-base-prime/additive-Haar branch and obtains the local C_Θ signed comparison without the quarantined unit-ball nonzero-norm input. The strengthened inverse-base-prime pointwise endpoint now has the same lowered Hodge-formula and Theorem 1.10 valuation-ball/IUT IV interface as the old norm-square branch: the pointwise calibration and localized Step (xi) C_Θ source are projected internally before entering the inverse-base-prime signed route. The next endpoint lowers the local arithmetic source itself to the finite Hodge-summand/Gaussian-Hodge projection layer, constructing the restriction/additive-Haar wrapper internally from the residue-module summand-projected inverse-base-prime source. Its local-to-global side has also been lowered to the valuation-ball local analytic arithmetic-divisor layer: the formula-gap matched Theorem 1.10 valuation-ball source is now constructed internally from the Proposition 1.4 valuation-ball additive-Haar local analytic source and its local formula-gap bounds. The strongest valuation-ball local-analytic source now also has constructor-projected C_Θ paper-trace data: , , and . Thus the local-analytic and Theorem 1.10 C_Θ source-data records are no longer independent inputs at this wrapper; they are read from the valuation-ball additive-Haar arithmetic-divisor source itself. The route threads this constructor into the public pointwise surface, so the assembled local-analytic Step (xi) C_Θ wrapper is no longer a public input at that boundary. The non-vacuous inverse-base-prime branch now has the same route-level lowering via ; it constructs the local-analytic wrapper before applying the summand-projected additive-Haar comparison. The compact-open log-Kummer route propagates the same constructor through the possible-region source, so the assembled wrapper is not public there either. The strongest pointwise branch now also accepts the compact-open log-Kummer map possible-region source directly and constructs the principal pointwise log-shell/valuation-ball theta-region source internally. The current compact-open endpoint lowers the inverse-base-prime arithmetic input once more: instead of taking the assembled summand-projected inverse-base-prime source, it receives the residue-module valuation cover and finite Hodge-summand Gaussian/Hodge construction separately and builds the summand-

projected source internally. A companion route also constructs the local-analytic Step (xi) C_Θ wrapper at this valuation-cover/Hodge-summand boundary from the localized determinant source, Haar defect, and valuation-ball additive-Haar local analytic arithmetic-divisor data. The newest compact-open variant lowers the Hodge side below that finite-summand source: it receives the localized-adjusted Gaussian/Hodge source and reconstructs the finite Hodge-summand source by fixing the summands to the localized weighted-adjusted determinant contributions. Its companion route performs the same local-analytic Step (xi) C_Θ construction at this lower localized-adjusted boundary. The square-direct-summand refinement lowers this Hodge input once more by replacing direct nonnegativity and strict positivity with square-root presentations of local direct-summand log-volumes and a nonzero selected root. Its companion moves the local-analytic Step (xi) C_Θ construction below these square-root data. The norm-square refinement then projects that square-root presentation from the norm-square localized hull decomposition. Its companion moves the same local-analytic Step (xi) C_Θ construction below that norm-square projection. The compact-open norm-square refinement moves the public Hodge datum down to local compact-open tensor factors whose projected log-volumes generate the norm-square decomposition. Its companion moves the local-analytic Step (xi) C_Θ construction below those compact-open tensor factors. The additive-Haar refinement lowers this branch to Haar-normalized compact-open local factors and projects the compact-open norm-square source from their normalized log-volume identities. Its companion performs the local-analytic Step (xi) C_Θ construction at this lowest compact-open Hodge boundary. A synchronized companion replaces the two separate constructor/localized determinant and canonical-scale equalities by the bundled . Its pointwise-source strengthening consumes this synchronization, the formula-gap Theorem 1.10 valuation-ball source, the Haar defect, the local Step (xi)/Haar/main-log identity, and the finite Haar lower bound through a single pointwise localized Step (xi) Theorem 1.10/IUT IV source. The additive-Haar source now constructs this synchronization source from the arithmetic-degree calibration, and its Record-Ob3/Ob5 wrapper derives the constructor-built/localized determinant equality through the record Ob3/Ob4 determinant handoff. The same wrapper now carries an endpoint/residual handoff audit for the resulting localized Step (xi) C_Θ source; its compact-open synchronized route audit also exposes the derived signed C_Θ inequality and Corollary 3.12 dichotomy before the definitional comparison with the established synchronized concrete endpoint. A normalized additive-Haar public route now consumes this Record-Ob3/Ob5 source directly and reconstructs the localized constructor-built C_Θ route internally. The compact-open additive-Haar concrete route has the same Record-Ob3/Ob5 synchronized companion: after the packet constructs its canonical HDD record, Lean indexes the Record-Ob3/Ob5 source over that canonical record and projects the localized determinant-scale synchronization before invoking the synchronized concrete endpoint. The compact-open packet-to-HDD construction is now also named as a separate Lean boundary: from the compact-open log-Kummer map and pointwise calibration source it constructs the principal HDD source and its canonical constructor-built source, and the synchronized concrete route now consumes this named boundary internally instead of rebuilding it inline. At the lower valuation-ball boundary, packages the Record-Ob3/Ob5 valuation-ball source over the named HDD constructor-built source. Its projection audit supplies the component Record-Ob3/Ob5 source, the constructor-built/localized determinant-scale synchronization endpoint, the Theorem 1.10 valuation-ball local-analytic arithmetic-divisor source, the finite Haar defect function, the total Haar-defect lower bound, and both the synchronized-source and component-source forms of the pointwise Step (xi)/Haar/main equality. The latter is obtained by transporting the local weighted adjusted log-volume across the component/valuation-ball formula-source equality. These local inputs are also packaged by , with an audit showing *that the route-input fields are the projections from the named-HDD boundary. The same route-input layer exposes the source-data determinant equality and, once the source data is identified with the constructor-built HDD source, the component synchronization endpoint. The route inputs are therefore derived from the valuation-ball source and the single constructor-built/record Ob3/Ob4 determinant equality rather than supplied separately at this boundary. The compact-open log-Kummer theta-region source also derives the record possible-image nonemptiness used by this projection: the principal transported theta-region cover supplies theta-class points, and the record/fiber-theta-region synchronization rewrites them to the record* Θ*-possible-image family. The component Record-Ob3/Ob5 source now has a direct compact-open constructor, , whose audit records this nonemptiness field as the compact-open log-Kummer-map theorem rather than as an independent component hypothesis. In the constructor-built source-data case, the named-HDD boundary is also constructed without a separate constructor-built/record Ob3/Ob4 determinant equality: that equality is projected from the Record-Ob3/Ob5 valuation-ball source whose source data is the constructor-built HDD source. The synchronized route-input bundle is now constructed directly from the compact-open log-Kummer source and this valuation-ball source, so the named-HDD boundary is not exposed as a separate input at that layer. The same route-input layer now has a stricter p-adic-defect/main constructor: Lean reconstructs the Record-Ob3/Ob5 valuation-ball source from the p-adic-defect/main controlled component source, then builds the synchronized route-input bundle and audits its component source, determinant/scale synchronization endpoint, formula-gap endpoint, and finite Haar-residual lower bound. Thus this compact-open handoff no longer has to expose the stronger valuation-ball source when the lower p-adic-defect/main source is available. A further source-facing variant starts from the local-degree formula Record-Ob3/Ob5 valuation-ball source: it reconstructs the p-adic-defect/main controlled source using the valuation-ball projection-preservation theorem, then enters the same compact-open route-input constructor and audits the synchronized route together with the local Step (v), Step (vi), and Step (vii) formula package. The canonical-HDD compact-open endpoint now performs the same lowering one level earlier: the direct canonical-HDD audit transfers theta-possible-image nonemptiness from the primitive Theorem 3.11 record to the canonical HDD record, Lean builds the canonical-HDD named boundary from the compact-open log-Kummer source and Record-Ob3/Ob5 valuation-ball source, and the compact-open inverse-base-prime public route consumes this constructed boundary immediately. Thus the canonical-HDD possible-image witness and named boundary are no longer public inputs at the compact-open valuation-ball handoff. Lean can also construct this valuation-ball Record-Ob3/Ob5 source from the lower Record-Ob3/Ob5 arithmetic-divisor component source, a Theorem 1.10 valuation-ball formula-matching source, and the equality identifying its additive projection with the component formula source. The same canonical-HDD valuation-ball source now also projects the synchronized route-input bundle directly: the component Record-Ob3/Ob5 source, localized determinant/scale synchronization, Theorem 1.10 valuation-ball local-analytic source, Haar defect, and pointwise Step (xi)/Haar/main equality are read from the constructed canonical boundary before the inverse-base-prime comparison is invoked. The p-adic-defect/main compact-open variant pushes this same handoff one source layer lower: Lean projects the arithmetic-degree controlled local source and synchronized controlled component directly from the p-adic-defect/main Record-Ob3/Ob5 source before invoking the controlled-component canonical-HDD route. Its product-handoff companion lowers the p-adic/Haar arithmetic side at the same time: the public route now takes the product-level Ob7 handoff and the p-adic-defect/main Record-Ob3/Ob5 source, then reconstructs the controlled component and arithmetic-degree local source inside the p-adic-defect/main canonical-HDD product-handoff route; the compact-open public route now factors through* *that stricter handoff directly, instead of first re-entering the older controlled-component product route. The canonical-HDD route audit now also reconstructs the inverse-base-prime summand source and the pointwise local-analytic IUT IV Step (xi) C_Θ source from this route-input bundle, recording their endpoints together with the localized C_Θ handoff, signed comparison, and dichotomy at the compact-open handoff itself. The reduced route now lowers the p-adic/additive-Haar input side to the restriction-calibrated residue-module/unit-ball source. The handoff route lowers this one step further to the product-level restriction-calibrated Ob7 source, with the structure-sheaf/direct-summand synchronization and nonzero norm-square projection facts carried as fields of that local-arithmetic source. It still constructs the same local-analytic C_Θ handoff from the Record-Ob3/Ob5 valuation-ball source before invoking the restriction-calibrated Step (xi) endpoint. The strongest combined wrappers and now expose both lowerings at once: they take the component Record-Ob3/Ob5 source, the valuation-ball formula-matching refinement, their projection equality, and either the inverse-base-prime valuation-cover/additive-Haar inputs or the product-level Ob7 handoff, then construct the Record-Ob3/Ob5 valuation-ball source internally before invoking the same signed Step (xi) endpoint. The product-handoff variant now lowers this one step further: it takes the arithmetic-degree-controlled valuation-ball local arithmetic source, constructs the valuation-ball formula-matching record internally, and then applies the same component-to-valuation-ball bridge. This route-input layer now also constructs the pointwise local-analytic Theorem 1.10/IUT IV localized Step (xi) source consumed by the lower C_Θ route, using the bundled localized synchronization, valuation-ball local-analytic source, Haar defect, synchronized pointwise equality, and total-defect bound directly. The corresponding route-input audit records the constructed source endpoint, localized C_Θ handoff audit, signed C_Θ comparison, and Corollary 3.12 dichotomy at this layer. The compact-open route-input audit also records the internally constructed inverse-base-prime summand source, the same local-analytic C_Θ source and signed endpoint, and the definitional equality between the compact-open route and the corresponding pointwise route. This source now also projects the localized C_Θ constructor source and its signed endpoint: , , , and . The current product-hull goal-evidence surface now has the corresponding lowered audit and constructor theorem ; its localized C_Θ handoff, signed bound, and dichotomy fields are projections from the Theorem 1.10 valuation-ball local-analytic source rather than independent public inputs. At the synchronized Record-Ob3/Ob5 route-input boundary, the formula-realized product-handoff route takes the product-level Ob7 handoff plus the bundled valuation-ball formula-gap data, constructs the pointwise Ob7 source and pointwise Theorem 1.10 localized Step (xi) source internally, and then invokes the formula-realized pointwise evidence audit. Its compact-open and named-HDD lifts , , and move the same construction through the compact-open log-Kummer source and through the component/formula bridge, so the valuation-ball source is rebuilt before the product-handoff formula-realized audit is invoked. The arithmetic-degree-controlled variant lowers this final input at the same surface by deriving the valuation-ball formula-matching source from the controlled local-arithmetic source before invoking the formula-realized product handoff. The comparison-source refinement lowers this boundary again by constructing the controlled local-arithmetic source itself from the valuation-ball formula-gap p-adic/arithmetic-degree package and the synchronized Proposition 1.4/1.5 comparison source. The local-degree formula refinement lowers the comparison source itself to , i.e. to the Step (v) tensor formula, Step (vii) archimedean coarse formula, and arithmetic-degree coefficient lower bound. The component-source route threads the same controlled/product handoff through the Record-Ob3/Ob5 arithmetic-divisor component surface, removing the older inverse-base-prime/local-analytic package from that public route. Its formula-gap comparison refinement no longer takes the arithmetic-degree-controlled* *local-arithmetic source as a public input. It constructs that source from the valuation-ball formula-gap p-adic/arithmetic-degree package and the synchronized Proposition 1.4/1.5 comparison source, then uses the resulting constructed formula source for the Record-Ob3/Ob5 matching equality. Its local-degree formula companion exposes the same Step (v)/(vii) formula source at the concrete Record-Ob3/Ob5 component boundary. The combined concrete route keeps the controlled local-arithmetic source as the valuation-ball input and projects the comparison source from local-degree formulas, so the assembled valuation-ball formula source and comparison package are both constructed below the public component handoff. The p-adic Haar-index route lowers that controlled source as well: it constructs it from the valuation-ball local analytic arithmetic-divisor evaluation, the unit-ball Haar normalization defect, the Step (xi) arithmetic-degree calibration, and the distinguished/archimedean formula bounds. The direct controlled-product version stops before the local-degree comparison reconstruction, so it needs only the Record-Ob3/Ob5 equality with the p-adic Haar constructed formula source and not the reconstructed formula-matching equality. The diagnostic records why this remaining direct synchronization is not cosmetic. Its projected-local-degree companion removes the separate local-degree formula source and local-degree matching proof from this p-adic Haar surface; the route projects that formula source from Record-Ob3/Ob5 data and derives the matching via . The projection now recovers the Step (v)/(vii) local-degree formula source directly from the Record-Ob3/Ob5 arithmetic-divisor component chain, with recording its formula-core synchronization, and recording the same synchronization after projection to the arithmetic-degree comparison package. The same projection now feeds the p-adic defect/main source directly via , with recording that the local-degree source and comparison matching are projected from the Record-Ob3/Ob5 arithmetic-divisor component data. The stronger valuation-ball Record-Ob3/Ob5 source now packages the component/formula-source equality itself and projects the same p-adic defect/main source via ; its audit records that no additional component-to-valuation-ball equality is supplied at this p-adic handoff. This handoff now factors through the explicit arithmetic-degree-controlled projection , so the p-adic source consumes a named constructed local-arithmetic object rather than an anonymous reconstruction inside the final projection. The local-degree formula valuation-ball Record-Ob3/Ob5 comparison source packages the formula-matched valuation-ball source, the Step (v)/(vii) local-degree formula source, and the controlled component indexed by the constructed local arithmetic source; the bridge projection sends this source directly to the canonical residual package. The canonical residual frontier is nevertheless lower: it takes the matched local-degree object valuation-ball source as the derived boundary, projects the local-degree formula valuation-ball source from its typed Step (v)/(vii) object data, and then reconstructs the p-adic-defect/main source before entering the canonical residual package. The synchronized controlled component source constructs that matched-object source by . The companion and constructor lower the measure/summand calibration at this same surface: the pointwise Hodge formula source is public, while the calibration source is projected internally before entering the localized C_Θ handoff. The theorem lowers the restriction-calibrated local-arithmetic source at the same current product-hull/Hodge-formula boundary. It constructs that record from the residue-module/unit-ball Haar-character valuation-cover source, the local-global realified Frobenioid restriction package, the structure-sheaf normalized Haar identities, the distinguished non-structure summand, and the theta-degree/localized-adjusted sum comparison before invoking the localized C_Θ handoff. The theorem lowers the assembled Theorem 1.10/IUT IV local-analytic source at this same surface. It constructs that source from the localized Step (xi) determinant source, determinant and canonical C_Θ-scale synchronizations, the valuation-ball additive-Haar local analytic arithmetic-divisor source, local Haar defects, the local Step (xi)/Haar/main-log identity, and the finite defect lower bound. The* *theorem lowers this determinant side one step further: the public input is the constructor-built localized Step (xi) determinant/scale synchronization source, while the localized determinant source and the determinant and canonical C_Θ-scale equalities are projected internally. The theorem keeps this synchronized determinant surface and lowers the IUT IV input to the valuation-ball additive-Haar formula-gap matched source; the local analytic arithmetic-divisor source is projected from the named Proposition 1.4 log-shell and Proposition 1.5 metric comparison data. The companion consumes the synchronized determinant/scale data, formula-gap source, Haar-defect function, local Step (xi)/Haar/main-log identity, and finite Haar lower bound through one pointwise Theorem 1.10 localized Step (xi) source. The synchronized route-input bundle now also preserves the valuation-ball formula-gap Theorem 1.10 source together with its local-analytic projection equality. At the source-audit level, projects a Record-Ob3/Ob5 valuation-ball arithmetic-divisor source to the closed valuation-ball formula-gap source, and records the inherited finite-place Haar summation, arithmetic gap, signed C_Θ bound, and dichotomy without adding a raw numerical comparison premise. The named-HDD route boundary exposes the same closed audit via , so this C_Θ handoff is visible at the compact-open/canonical-HDD public boundary. The canonical-HDD compact-open route audit records the same DOD evidence together with the reconstructed synchronized route-input endpoint, the internally constructed local-analytic Step (xi) C_Θ source and handoff audit, the derived canonical-scale bound, signed C_Θ inequality, source-bridge inequality, Corollary 3.12 dichotomy, and definitional equality to the compact-open handoff. The pointwise inverse-base-prime public route now consumes this route-input bundle instead of taking the local-analytic localized Step (xi) source as an independent input or endpoint: it constructs the formula-gap IUT IV localized source directly from the preserved formula-gap field used by the signed C_Θ comparison. The compact-open log-Kummer/valuation-cover route reaches the same handoff after deriving the principal pointwise valuation-ball and inverse-base-prime summand sources internally. Its named-HDD boundary variant constructs the synchronized route-input bundle from the valuation-ball boundary before invoking this compact-open handoff. The restriction-calibrated local arithmetic branch now has the same local-analytic Theorem 1.10 valuation-ball/IUT IV lowering as the inverse-base-prime branch, and a compact-open wrapper derives its pointwise valuation-ball source from the log-Kummer map source internally. The evidence-level compact-open formula-realized companion now constructs both the pointwise valuation-ball source and the restriction-calibrated Ob7 handoff before entering the pointwise Theorem 1.10/IUT IV formula-gap packet evidence. Its product-handoff variant takes the typed Ob7 handoff itself and no longer exposes the restriction-calibrated source, structure-sheaf/direct-summand equality, and selected norm nonzero field separately. Its local-arithmetic-degree strengthening also constructs the pointwise Theorem 1.10/IUT IV localized source from the localized local-term Step (xi) source, formula-gap valuation-ball source, main-log matching, and local arithmetic-degree identity. Its synchronized local-arithmetic-degree variant removes that localized local-term source from the compact-open public boundary by constructing it from determinant synchronization, the Theorem 1.10 formula-gap valuation-ball source, Haar defects, and the finite-place arithmetic-degree identity. The constructed-synchronization compact-open variant lowers one further boundary by constructing determinant/scale synchronization from the localized Step (xi) determinant source and the two constructor-built comparison identities. The explicit-formula compact-open variant removes the packaged pointwise measure/summand formula record by constructing it from the volume/hull-log-volume, charted-theta/Hodge-degree, and finite adjusted-summand identities. The determinant-calibrated refinement now lowers the summand identity itself: the public input calibrates the Hodge theta-monoid degree to the normalized determinant, and Lean derives the determinant log-volume and* *adjusted-summand equalities from the record-native determinant source. The realized-exact compact-open variant constructs the compact-open log-Kummer map source from realized compact-open theta-region image exactness. Its restriction-calibrated variant also constructs the product Ob7 handoff from the normalized local-arithmetic source, structure-sheaf/direct-summand synchronization, and nonzero norm-square summand, while deriving the Theorem 1.10 formula-gap valuation-ball source from the local-analytic arithmetic-divisor source. The p-adic finite-extension variant additionally projects the ordinary localized Step (xi) determinant source from the unit-ball/valuation-Haar compact-open norm-square localized determinant layer before entering this same route. A synchronized p-adic finite variant now bundles this projected finite-extension determinant together with the constructor-built determinant and canonical C_Θ-scale sum identifications into one source object before invoking the local-upper endpoint; these identifications remain lower source data, but no longer occur as two unrelated public route arguments. Its formula-gap refinement consumes the Theorem 1.10 valuation-ball additive-Haar formula-gap source and projects the local-analytic arithmetic-divisor ledger internally for the residual calculation. The residual calculation is now also packaged as : this source carries the synchronized p-adic finite determinant, the formula-gap ledger, and the finite residual Haar lower bound together, then projects to the localized Step (xi) C_Θ source and the signed dichotomy. The concrete public wrapper consumes this bundled source directly. The lower wrapper also reads the restriction-calibrated source, structure-sheaf comparison, and positive summand norm from the product-level Ob7 handoff source. The formula-source refinement projects the measure identity, Hodge theta charting identity, and finite adjusted-summand formula from . The local-chart refinement replaces the public realized compact-open image source by compact-open theta-region coordinates, their realization law, and the converse realization containment, from which Lean reconstructs the log-shell image and realized-exact sources. The graph refinement derives that local-chart source from the local log-Kummer graph/correspondence, its left-totality, realization compatibility, and compact-open soundness laws. The log-shell image refinement reconstructs the graph and local-chart sources from the canonical compact-open image containments between theta-region cells and realized local shells. The possible-region exact refinement moves this boundary to the Remark 3.9.5 valuation-cover possible-region equality and reconstructs the downstream compact-open source stack from it. The log-Kummer image refinement derives possible-region exactness from local compact-open lifts and the converse realization soundness law. The correspondence refinement extracts those image laws from the local Kummer correspondence relation, left-totality, realization compatibility, and compact-open soundness. The map refinement constructs the correspondence as the graph of the named local Kummer realization map and its possible-region section. The current strongest wrapper combines this map boundary with the determinant-calibrated pointwise formula source. Its local-arithmetic-source refinement constructs the Ob7 handoff from the pointwise restriction-calibrated residue-module/unit-ball/additive-Haar source. Its local-upper variant defines each local Haar defect as the residual between the IUT IV local arithmetic-upper contribution, the localized Step (xi) determinant term, and the main-log summand; the expanded finite-place arithmetic-degree equality is then derived from the definition of the IUT IV local arithmetic-upper contribution. The later same-index theorem uses that expanded local identity together with the total Haar lower bound to recover the formula-gap residual inequality. The local-analytic residual refinement lowers this boundary once more by deriving the formula-gap residual source from the Theorem 1.10 valuation-ball local-analytic arithmetic-divisor ledger. The localized residual refinement no longer takes the p-adic determinant/scale synchronization record; it constructs it from the finite-extension localized Step (xi) determinant source and the two constructor-built determinant/canonical-scale comparison facts. The construc-*

tor pushes the determinant source boundary below this again: the p-adic localized Ob3/Ob4 determinant source is projected from the principal-product Theorem 3.11 localized hull-vector-bundle decomposition before entering the local-analytic residual route. Its source-hypothesis variant removes the direct side-condition package from the public endpoint: the endpoint consumes the source-labeled q-pilot positivity and normalization hypotheses and constructs the record at the primitive current-product-hull handoff. The public principal-product residual route threads that constructor through the endpoint: the residual input is tied to the packet's source-data-with-target-regions object and to the principal hull projected from the internally constructed theta-region valuation-ball source before being converted to the older localized residual source. Its realized-exact variant lowers the public boundary from a supplied compact-open log-Kummer map source to compact-open realized exactness; Lean projects the map source before entering the principal-product residual handoff. The underlying principal-product localized source now keeps the finite-place index and the principal-hull parameter in separate universes, matching their mathematical roles. Its local-arithmetic-handoff refinement constructs the pointwise restriction-calibrated Ob7 source from the product-level local arithmetic handoff before entering the same principal-product residual endpoint. The formula-gap residual refinement lowers the IUT IV input again: it takes the valuation-ball additive-Haar formula-gap residual source and projects the local-analytic residual ledger internally. The projected-principal-product refinement now constructs the p-adic principal-product finite formula-gap residual package from the product-level local-arithmetic handoff, the determinant/scale equalities, and the formula-gap residual data before entering that endpoint. Its local-analytic companion promotes the Theorem 1.10 valuation-ball local-analytic arithmetic-divisor ledger to the formula-gap matched IUT IV source internally before the same principal-product projection. The formula-gap principal-product source now also exposes . This first projects the formula-gap Theorem 1.10 data to the local-analytic ledger and then applies the principal-product canonical-scale synchronization theorem; the corresponding audited endpoint and dichotomy are and . At the underlying principal-product localized local-analytic source, Lean now defines the finite-place Haar-normalization defect as

$$U_v - \mathrm{StepXI}_v - M_v$$

from the IUT IV local arithmetic-upper contribution U_v, the projected localized determinant term, and the main-log contribution. The audited lemmas and prove the local Step (xi) arithmetic-gap equality and recover the total Haar lower bound from this source-side defect. The same projected-principal local-analytic source, and its formula-gap refinement, now expose the constructor-built localized C_Θ source directly via and . Thus the projected-principal layer itself supplies the localized C_Θ endpoint, its IUT IV handoff audit, the signed bound, and the dichotomy. A separate q-normalized p-adic localized source, , lowers the generic scale boundary: it assumes the paper-facing identity $\theta_{\mathrm{signed}} = (\sum_v \mathrm{adeg}_v^{\mathrm{adj}})(-q_{\mathrm{signed}})$ and derives the canonical finite-sum scale equality by dividing by the positive q-pilot log. Conversely, derives this normalized theta identity from the determinant/scale synchronization source, and constructs the q-normalized source from the canonical-scale synchronized local-analytic source. The projected-principal specialization applies the same derivation to the principal-product p-adic localized determinant projection, and constructs that q-normalized projected-principal source from the canonical-scale synchronized principal-product source. The reduced endpoint now consumes this q-normalized projected-principal source instead of a direct canonical-scale equality. The determinant side is now lowered by the adjusted-determinant synchronized q-normalized source : it identifies the constructed principal HDD Ob3/Ob4 adjusted determinant source with the adjusted determinant source projected from the p-adic finite localized Step (xi) object. The p-adic finite principal-product *hull-decomposition source now also computes the family-hull log-volume as the finite sum of projected weighted adjusted local Step (xi) log-volumes, so the q-normalized theta identity can be derived from the family-hull comparison rather than supplied directly as an adjusted-sum formula. The constructor now performs this lowering for the q-normalized principal-product local-analytic source itself. The audited theorem derives the weighted determinant equality used by the localized C_Θ handoff. Its local-arithmetic-degree constructor keeps that full adjusted-determinant synchronization while deriving the residual Haar lower bound from the packaged local arithmetic-degree residual source. The source constructor now derives the full adjusted-determinant equality itself from equality of the localization family, anchor, and positive tensor power. The localization family synchronization has also been lowered to the localized vector-bundle source layer: identifies adjusted localization entries from the underlying bundle, structure-sheaf term, raw adjusted log-volume, and weight, while promotes pointwise localized vector-bundle equality, anchor equality, and tensor-power equality to equality of adjusted-determinant sources. The lower component-field criterion removes even the opaque localized-entry equality: equality of the underlying localized bundles, structure-sheaf log-volumes, raw adjusted log-volumes, determinant weights, anchor, and positive tensor power is enough to identify the adjusted-determinant sources. This has now been pushed below bundle equality: the local-ring/direct-summand criterion derives the same adjusted-determinant synchronization from local-ring labels, direct-summand log-volume families, structure-sheaf log-volumes, raw adjusted log-volumes, weights, anchor, and positive tensor power. The norm-square refinement derives the projected adjusted-determinant synchronization from the corresponding direct-summand norm families. This criterion is now pushed to the Haar-normalized local-field boundary: the additive-Haar theorem reconstructs the raw adjusted log-volume from the normalized Haar log-volume sum and the structure-sheaf term, and the unit-ball and p-adic finite-extension projection theorems and carry the same synchronization through valuation-unit-ball and finite-extension-over-$\mathbb{Q}_p$ data. The same full adjusted-determinant source now projects directly to the projected-weighted local-analytic source via ; in the same-index adjusted-determinant case, the summand, anchor, and positive tensor-power fields are obtained by projecting the full Ob3/Ob4 equality through rather than by supplying the three weighted shadow equalities separately. The same-index source now has the stronger family-hull constructor : it keeps the full adjusted-determinant synchronization and local arithmetic-degree residual source, but derives the q-normalized adjusted-sum identity from the principal-product family-hull comparison and the p-adic finite localized hull decomposition. The refined constructor pushes this one step lower: the unscaled θ_{signed}-equals-record-family-hull equality is derived from the pointwise determinant/Hodge calibration. The compatibility between that record family-hull log-volume and the principal-product family-hull expression is now itself derived from the adjusted-sum q-normalization, determinant/Hodge calibration, and p-adic finite hull decomposition by . Thus the remaining scalar obligation at this boundary is the paper-level construction of the adjusted-sum q-normalized theta/log-volume comparison, not a separate record-family q-scale law. The family-hull side-condition endpoint uses the family-hull constructor before proving the signed Corollary 3.12 dichotomy and the companion $C_\Theta \geq -1$ bound. The refined side-condition endpoint now composes the record-family-hull q-scale constructor into the same signed and $C_\Theta \geq -1$ outputs, so this checked same-index surface no longer exposes either the direct q-normalized adjusted-sum theta identity or the raw q-normalized theta/family-hull equality. The heterogeneous projected-weighted source covers the endpoint case with distinct determinant and local γ indices by comparing the projected weighted determinant summands, anchor, and positive tensor power over the common β-index before deriving the weighted determinant equality. Its constructor lowers the residual Haar input* *at this heterogeneous boundary to the packaged local arithmetic-degree residual source. The adjusted-determinant source also exposes the companion $C_\Theta \geq -1$ endpoint by projecting through the same q-normalized local-analytic handoff. The public route now consumes this heterogeneous source and projects the q-normalized residual handoff internally, so the determinant equality is no longer a public input on that route. Its side-condition variant takes* IUTStage1SourceSideConditions *directly and reconstructs the named side-hypothesis record internally before entering the same heterogeneous route. Its lower-bound companion proves $C_\Theta \geq -1$ from that same public local-analytic projected-weighted source, while gives the direct side-condition-source version. Internally this lower bound is exposed by , so the signed and lower-bound endpoints now live at the same non-same-index determinant boundary. The heterogeneous formula-gap refinement lowers the IUT IV input on this same boundary: it consumes the Theorem 1.10 formula-gap arithmetic-divisor source, projects the local-analytic ledger internally, and exposes both and . Its constructor now derives the formula-gap residual Haar inequality from the local arithmetic-degree residual payload after projecting the Theorem 1.10 source to the local-analytic ledger. The endpoint theorem then reads off both the signed dichotomy and the $C_\Theta \geq -1$ companion from that constructed source. The public realized-exact local-arithmetic route has the corresponding lowered wrapper , which constructs the local-analytic projected source internally before entering the heterogeneous endpoint. Its lower-bound companion exposes $C_\Theta \geq -1$ at the same formula-gap public boundary. The side-condition versions and reconstruct the side-hypothesis record internally, so neither the signed endpoint nor the lower-bound companion has a separate side-hypothesis public surface at this boundary. The source-gap companions and lower the same projected formula-gap public source through* IUTStage1SourceObligationGap*. The constructor-level route lowers that public boundary again: the projected formula-gap source is constructed internally from the principal-product local Step (xi) decomposition, the projected weighted determinant summand/anchor/tensor-power identifications, the q-normalized theta identity, the Theorem 1.10 formula-gap valuation-ball source, and the residual Haar lower bound, and the theorem returns the signed endpoint together with $C_\Theta \geq -1$. Its local-degree refinement removes the raw residual Haar lower bound from this 'SideHypotheses' constituent surface: Lean derives that finite-sum inequality from the packaged local arithmetic-degree residual payload before applying the existing projected formula-gap route. The canonical manifest is one layer lower than the formula-gap wording in the preceding constructors: the preferred same-index q-normalized route consumes a local-analytic valuation-ball Theorem 1.10 source. When a formula-gap package is required, Lean constructs it by ; thus the formula-gap package is no longer the canonical IUT IV public assumption, while the local-analytic Theorem 1.10 ledger remains a source obligation. The side-condition variant lowers this public surface one step further by taking* IUTStage1SourceSideConditions *directly and reconstructing the named side-hypothesis record internally before entering the constituent formula-gap route. The local-degree side-condition form combines both lowerings: it takes direct side conditions and derives the residual Haar lower bound from the local arithmetic-degree payload. The source-gap variants and lower these two non-case-bounded constituent surfaces through* IUTStage1SourceObligationGap*: the direct form keeps the residual Haar lower bound, while the local-degree form keeps the packaged arithmetic-degree residual payload, and both project q-pilot positivity and normalization from the source gap. The case-bounded side-condition form lowers this residual boundary once more: the valuation-ball distal/non-distal/archimedean case source is projected internally to the local arithmetic-degree residual payload before the projected formula-gap endpoint is invoked. Its source-gap form keeps this strongest case-bounded residual surface while projecting the q-pilot positivity and normalization hypotheses from* IUTStage1SourceObligationGap*, so the case-split projected formula-gap* *endpoint no longer exposes the raw side-condition record once the source gap is available. Its structured-SHE successor constructs that source gap from the concrete packet's primitive constructed Theorem 3.11/SHE bundle and the Stage 1 side conditions before entering the same projected formula-gap case-bounded endpoint. Thus the assembled source-obligation gap is not a public input at this boundary; the remaining lower sign/normalization input is the side-condition package itself. The hull/determinant-obligations variant projects those side conditions from* IUTStage1SourceHullDetObligations*, so the projected case-bounded route can share the same source object that carries the Remark 3.9.5 hull/determinant data and the q-positivity/normalization witnesses. The same-index concrete source now records the stricter case in which the local-arithmetic handoff, the p-adic localized Step (xi) determinant source, and the IUT IV local-analytic ledger share the endpoint β-index. Lean now projects the q-normalized principal-product residual source from this concrete source and derives the signed dichotomy through , without passing through the older public wrapper. The shared-β public endpoint now exposes exactly this route at the public theorem boundary: it consumes the same-index constructed source directly and no longer asks separately for the localized C_Θ handoff, determinant equality, or raw signed comparison. Its side-condition variant rebuilds the named side-hypothesis record from* IUTStage1SourceSideConditions*, so the shared-β route no longer exposes the side-hypothesis bundle at its signed endpoint. The accompanying route audit records the q-normalized endpoint, localized C_Θ handoff audit, derived IUT IV signed upper bound, and resulting dichotomy from the same concrete source. The principal-pointwise packet boundary also has a p-adic unit-ball arithmetic-divisor source, . It reads the Haar defect from the finite-place unit-ball index source and projects both the older localized arithmetic-divisor C_Θ source and the generic Step (xi) synchronization source, so the packet-facing handoff no longer needs an independent defect function at this layer. Its endpoint is pinned by . The preferred public packet route now consumes this source directly through . The processional p-adic refinement constructs this p-adic handoff from the Theorem 1.10 valuation-ball local-analytic source and the local Step (xi)-to-arithmetic synchronization; its projection and endpoint are checked by and . The corresponding public packet route is . Its synchronized constructor obtains the localized determinant source and the two constructor-built determinant/scale identifications from one Step (xi) synchronization source, and the synchronized public route exposes that lower boundary directly. The processional synchronization itself is also lowered by : it combines the local p-adic unit-ball Step (xi) arithmetic identity with the Theorem 1.10 processional-estimate source, using only the equality that the p-adic normalized place is the distinguished nonarchimedean place. The packet-facing constructor and public route therefore no longer expose the bundled processional p-adic synchronization record as an independent public input. The normalized-place refinement removes the remaining equality between a separately chosen distinguished place and the p-adic normalized place: its projection chooses the p-adic normalized place definitionally. The constructor and public route therefore expose no separate normalized-place comparison at this boundary. The localized-local-term constructor lowers this once more: it reads the localized Step (xi) determinant object and determinant/scale identifications from the localized local-term C_Θ source, and the corresponding public route exposes the local-term boundary directly. The equality-free normalized version of this localized handoff is now checked by and its constructor-audit export : the local-term source supplies the localized determinant and scale data, while the p-adic Step (xi) identity and p-adic-normalized processional estimate reconstruct the bundled processional p-adic source without a separate normalized-place equality. The pointwise Theorem 1.10 constructor lowers this boundary again by projecting the localized Step (xi) determinant data and local-analytic Theorem 1.10 ledger from the pointwise valuation-ball IUT IV localized source. Its public route therefore exposes the* *processional p-adic synchronization over that pointwise source directly. The intermediate Haar-residual constructor now lowers this pointwise boundary before normalization: the p-adic Step (xi) synchronization is reconstructed from the expanded p-adic unit-ball Haar-index arithmetic-degree residual source, and the public wrapper threads that reconstruction to the signed endpoint. The normalized constructor now removes that bundled synchronization at the constructor layer: it combines the pointwise valuation-ball localized source, the local p-adic Step (xi) synchronization identity, and p-adic-normalized processional-estimate data. The public endpoint wrapper now threads this normalized constructor directly to the signed Corollary 3.12 route. The local-analytic pointwise refinement removes the formula-gap valuation-ball packet from this p-adic public boundary: the localized determinant data and Theorem 1.10 ledger are read from the pointwise local-analytic localized source. The matching public route therefore exposes only that local-analytic source plus the processional p-adic synchronization at this layer. The Haar-residual refinement removes that synchronization from the pointwise local-analytic public surface: Lean derives it from after proves the local synchronization equation from the expanded residual identity. The public wrapper exposes the local-analytic pointwise source plus the expanded p-adic Haar-residual source, not the bundled p-adic synchronization record. This bundled Haar-residual source is now also reconstructed by from the local p-adic residual identity and the processional-estimate ledger, with one equality aligning the estimate's distinguished place to the p-adic normalized place. The packet constructor and public wrapper expose those separated ingredients directly at the signed endpoint. Its normalized constructor removes the bundled synchronization at the strongest pointwise constructor surface: the local-analytic pointwise source supplies the Theorem 1.10 ledger and localized Step (xi) object, while the p-adic-normalized processional estimate fixes the distinguished place definitionally. Its public wrapper therefore derives the endpoint from the pointwise local-analytic source, the p-adic Step (xi) identity, and normalized processional estimates without a bundled processional p-adic source at the public boundary. The hybrid constructor and packet route combine these two lowerings: the p-adic Step (xi) synchronization is constructed internally from the expanded Haar-residual identity, while the distinguished-place alignment is definitional in the normalized processional estimate. The public endpoint exposes only the pointwise local-analytic source, the expanded p-adic Haar-residual source, and the p-adic-normalized processional-estimate data at this p-adic public boundary. The case-source variant lowers this p-adic boundary further: it keeps the case-by-case procession-normalized identifications and the p-adic normalized-place kind, but drops the stronger distinguished $+1$ processional upper-bound field. In the corresponding endpoint , the extra unit contribution is supplied by the p-adic Haar-residual lower bound instead of by a separate processional-estimate hypothesis. The p-adic unit-ball/local-analytic endpoint lowers the same boundary once more. Its input is the p-adic unit-ball localized Step (xi) source, a valuation-ball local-analytic Theorem 1.10 ledger, their local arithmetic-evaluation equality, and the normalized processional case data. Lean projects the pointwise local-analytic source by and derives the expanded p-adic Haar-residual identity by . The normalized case record now also projects directly to , so the distinguished p-adic finite-place weight and the derived global Haar lower bound are visible at this boundary rather than only through a later residual wrapper. The pointwise-residual refinement replaces the normalized processional case input by the lower processional pointwise-residual source plus the p-adic normalized-place alignment, while still deriving the local-analytic and Haar-residual projections from the p-adic unit-ball source. The Haar-defect refinement lowers this once more: Lean projects the pointwise residual theorem from the processional Haar-defect residual source, so the one-unit contribution is read from the Haar-defect residual lower bound before the same p-adic unit-ball/local-analytic handoff is consumed. The p-adic unit-ball*

Haar-index refinement specializes that source to finite-place p-adic unit-ball Haar-defect data and projects the generic Haar-defect source internally. The public endpoint now threads this lower p-adic Haar-index input through the signed Corollary 3.12 route. The endpoint contracts for both projected sources are now pinned by and , so the old local-analytic and expanded residual packages are not merely definable from the p-adic unit-ball source; their checked endpoint payloads are reconstructed there. The localized source itself now also exposes the same arithmetic calculation: identifies the local defined residual with the p-adic unit-ball Haar index, and sums these identities to recover the residual lower bound at this public p-adic/local-analytic boundary. The same reconstructed local-analytic handoff now also gives the lower-bound companion , so this p-adic unit-ball boundary proves $(-1 : \mathbb{R}) \le C_\Theta$ *without taking the older local-analytic source as a separate input. This lower-bound projection is now lifted through the stronger processional p-adic source by , whose proof uses the embedded valuation-ball local-analytic ledger and the p-adic unit-ball projection. The goal-evidence audit adds the relative nonvacuity statement: once this same-index source is inhabited, the route-audit proposition is inhabited and the signed endpoint follows. The remaining existence problem is therefore the foundational construction of the underlying Hodge-theater/Frobenioid source data, not an additional supplied Step (xi) comparison at this public boundary. The scale-synchronized companion replaces the q-normalized theta identity at this same-index local-analytic boundary by the canonical principal-product localized residual source. Its map applies the principal-product determinant/scale synchronization theorem to construct the q-normalized projected-principal source internally, and exposes the corresponding signed endpoint, while exposes the* $C_\Theta \ge -1$ *companion. The corresponding side-condition-sourced public endpoints and reconstruct the named side-hypothesis record internally from* IUTStage1SourceSideConditions *before applying the same canonical-scale route. The source-gap variants and project these same sign and normalization hypotheses from* IUTStage1SourceObligationGap*, so the local-analytic scale-synchronized lower-comparison surface no longer exposes the raw side-condition record once the gap is available. Thus this new same-index route no longer has a raw* $\theta = \sum_v \mathrm{StepXI}_v \cdot (-q)$ *field at its public source boundary; it obtains that normalization from canonical* C_Θ*-scale synchronization. The constructor now also lowers the residual input of this canonical-scale source: the total Haar residual is derived from the packaged local arithmetic-degree residual source, whose pointwise Step (xi)+Haar identity is summed over the finite place index. The generic companion reads* $C_\Theta \ge -1$ *from this internally q-normalized scale-synchronized source. The concrete same-index constructor lifts this lowering through the realized-exact local-arithmetic handoff: the projected principal-product p-adic finite Step (xi) source is built from the handoff, and the scale-synchronized local-analytic source is assembled from the determinant/scale synchronization facts, the Theorem 1.10 local-analytic ledger, and the packaged local arithmetic-degree residual source. The companion endpoints and expose the signed dichotomy and* $C_\Theta \ge -1$ *without taking the scale-synchronized local-analytic source as an atomic input. The weighted-determinant refinement lowers the determinant side of this constructor one step further: the constructor-built weighted determinant equality is derived from the pointwise summand equality, anchor equality, and positive tensor-power equality against the projected localized Step (xi) determinant source. Its endpoint companions and now expose the signed dichotomy and* $C_\Theta \ge -1$ *directly from those three determinant component equalities, the canonical scale identity, the Theorem 1.10 local-analytic ledger, and the local arithmetic-degree residual payload. This is a checked constructor path, not the canonical remaining-assumption boundary: the dependency audit now records the stronger scale-synchronized source boundary, where the whole constructor-built determinant-source synchronization and the canonical* C_Θ*-scale synchronization are the source facts and the* *three component equalities are merely one way to build the determinant-source synchronization. The formula-gap companion lifts this replacement through the Theorem 1.10 formula-gap boundary: the public source carries the principal-product formula-gap residual package, projects the local-analytic ledger, and exposes the signed endpoint without a formula-gap raw theta-normalization field; its companion reads the ordered-real $C_\Theta \geq -1$ endpoint from the same internally q-normalized route. Its side-condition-sourced variants and perform the same reconstruction from* IUTStage1SourceSideConditions*, so this formula-gap canonical-scale boundary no longer exposes the side-hypothesis package as its public source input. The source-gap companions and lower the same formula-gap lower-comparison endpoints through* IUTStage1SourceObligationGap*. The case-bounded scale-synchronized constructor lowers this formula-gap boundary further: starting from the scale-synchronized local-analytic principal-product source, the valuation-ball formula-gap source, and the explicit distinguished, nondistinguished, archimedean, and distinguished $+1$ case bounds, Lean derives the formula-gap residual package through the generic local arithmetic-degree residual theorem. The corresponding signed endpoint and $C_\Theta \geq -1$ endpoint are and ; unlike the older q-normalized case-bounded surface, this route does not expose a raw same-index theta-normalization field. It is now a constructor route into the active q-normalized local-arithmetic-degree boundary, where Lean derives scale synchronization from the theta/log-volume identity and q-pilot positivity. The local-analytic case-source variant lowers the same branch one step further: its public IUT IV input is the valuation-ball local-analytic Theorem 1.10 source together with the local-analytic case split, and Lean constructs both the formula-gap matched source and the formula-gap case residual internally before invoking the scale-synchronized formula-gap route. Its signed and $C_\Theta \geq -1$ endpoints are and . The procession-bound refinement removes the expanded three-case local inequalities from this branch as well: the route now consumes the projected Theorem 1.10 procession upper bound and the distinguished $+1$ comparison, then reconstructs the local-analytic case source internally. The procession-comparison refinement lowers the same branch once more by replacing the direct Step (xi)-to-procession inequality with a procession-normalized local log-volume identification plus the upper-semi procession comparison; Lean reconstructs the procession-bound source internally before entering the same formula-gap route. The processional-estimate refinement removes that upper-semi comparison as a primitive field on the scale-synchronized endpoint: the local Step (v), Step (vi), and Step (vii) estimates derive the procession-comparison source internally. The paired diagnostic shows that deleting the distinguished $+1$ gap is a real weakening: the ordinary distinguished local upper bound can hold while the residual lower bound required by the formula-gap case source fails. The formula-gap refinement lowers the IUT IV side of this same-index source: it consumes the Theorem 1.10 formula-gap matched arithmetic-divisor source and projects the local-analytic ledger internally before invoking the same q-normalized route. Its local arithmetic-degree residual theorem derives the same residual lower bound from the pointwise $StepXI_v + Haar_v = a_v(D_v + C_v) + E_v - M_v$ identity and the total Haar-defect lower bound, rather than taking that residual inequality as an independent arithmetic fact. The reusable local payload packages this calculation away from the same-index wrapper; its endpoint theorem exposes the local arithmetic-upper identity, the arithmetic-upper finite sum as $\sum_v (StepXI_v + Haar_v + M_v)$, the main-log finite sum, and the residual lower bound from the same local degree and Haar-defect data. The packaged same-index residual constructor fills the formula-gap residual field from this reusable local source, rather than from separate raw Haar-defect and pointwise equality inputs. The same payload now constructs the localized Step (xi) local-term C_Θ source through , so that constructor path no longer takes the local main-log, Haar-defect, arithmetic-upper sum, and main-log sum data as unrelated arguments. The residual Haar summation has also been lowered by : a local lower-weight* *function of total mass* 1*, pointwise bounded by the local Haar defects, implies* $1 \leq \sum_v Haar_v$ *by finite summation. The lower-weight residual package turns this local ledger into the older residual source, and the adjusted determinant route now consumes it before deriving the q-normalized residual handoff. The adjusted-determinant route now also has the combined constructor , which takes the primitive Remark 3.9.5 localization/anchor/positive tensor-power synchronizations together with the lower-weight residual ledger and derives both the full Ob3/Ob4 equality and the global residual Haar inequality internally. Its same-gamma projected-weighted projection now derives the weighted summand, anchor, and positive tensor-power fields from that adjusted-determinant source before the projected local-analytic endpoint consumes them. The same lowering is now propagated through the principal-product scale-synchronized local-analytic source and the projected-weighted formula-gap source by , , and . Thus these projected* C_Θ *consumers no longer need the summed residual Haar inequality as the primitive residual datum. The lower-weight ledger itself is now produced by the pointwise defined-residual layer: puts weight* 1 *at the distinguished place and* 0 *elsewhere, proves the finite sum is* 1*, and converts pointwise nonnegativity plus the distinguished* $+1$ *residual into the lower-weight residual package. The finite Haar-defect source has the analogous construction , so the lower-weight ledger is no longer an independent assumption once a normalized-place Haar lower bound is available. This construction now reaches the p-adic-normalized case and arithmetic-gap sources themselves through . The stricter defined-residual layer defines the Haar defect as* $\mathrm{localArithmeticUpper}_v - \mathrm{StepXI}_v - \mathrm{mainLog}_v$*. Its theorem derives the pointwise Step (xi)+Haar arithmetic-degree identity by unfolding the IUT IV local arithmetic-upper contribution, and constructs the older packaged residual source. The corresponding constructor feeds the localized Step (xi) local-term* C_Θ *source from this defined-residual input; the remaining numerical datum is the global lower bound for the defined residual sum. The pointwise lower source lowers that datum to local nonnegativity of every defined residual plus one distinguished residual contribution at least* 1*, and derives the summed defined-residual lower bound. At the same-index formula-gap boundary, consumes the defined-residual source directly and projects internally to the packaged residual theorem. The same-index defined-residual source exposes that same lowered residual payload on the determinant/theta same-index source boundary, and projects it to the existing local arithmetic-degree route. The direct defined-residual route projections , , , and carry this lowered source through to the q-normalized signed endpoint and, via the local arithmetic-degree projection, to the scale-synchronized local-analytic endpoint without a separate canonical* C_Θ*-scale equality on the defined-residual boundary. The named public endpoint and its route/public-boundary audit aliases expose the same construction without taking the packaged residual source as a public input; the remaining local-estimate datum is the global lower bound for the defined residual sum. The pointwise same-index source lowers that boundary once more: its projection constructs the summed defined-residual source from pointwise local nonnegativity and one distinguished residual contribution. The corresponding public endpoint enters the same q-normalized signed route without exposing the global residual-sum inequality as a raw same-index input. The next local-estimate constructor now sits below this pointwise source. replaces the raw local residual nonnegativity fields by local arithmetic-gap inequalities* $\mathrm{StepXI}_v \leq \mathrm{localArithmeticUpper}_v - \mathrm{mainLog}_v$ *and one distinguished* $\mathrm{StepXI}_{v_0} + 1$ *gap. Its projection constructs the older pointwise defined-residual source by ordered-real arithmetic. The Theorem 1.10 procession-bound constructor lowers this again: the Step (xi) local term is compared with the case-defined Theorem 1.10 procession upper bound, and the existing formula arithmetic-divisor source supplies the comparison from that procession bound to* $\mathrm{localArithmeticUpper} - \mathrm{mainLog}$*. Its projection constructs the pointwise residual source internally. The valuation-ball formula-gap source now projects* *directly to the formula arithmetic-divisor source through . The valuation-ball residual constructor starts from that valuation-ball additive-Haar formula-gap data, compares localized Step (xi) with the projected case-defined procession bound, and then constructs the same pointwise defined-residual source internally. The case-split valuation-ball source lowers the same boundary further: it takes the distinguished nonarchimedean, nondistinguished zero, and archimedean local comparisons, plus one distinguished $+1$ comparison, and Lean reconstructs the projected procession-bound residual source and the pointwise residual source internally. The same source now reaches the localized Step (xi) C_Θ handoff via : the pointwise and defined residual wrappers are projected internally before applying the localized C_Θ constructor. The same case-split source now also has the closed formula-gap constructor : the local-analytic and Theorem 1.10 C_Θ trace records are projected from the valuation-ball additive-Haar formula-gap source instead of being supplied as independent constructor inputs. Its local-analytic companion constructs that formula-gap matched source from the valuation-ball additive-Haar local-analytic arithmetic-divisor source before entering the same case-bounded C_Θ handoff. The local-analytic case-source variant consumes directly: its three local cases and distinguished $+1$ inequality are stated over the valuation-ball local-analytic arithmetic-divisor source, and Lean constructs the formula-gap matched case source internally. The same case-split source also projects directly to the defined residual source and the packaged local arithmetic-degree residual source, keeping the summed Haar-defect handoff available without re-exposing the pointwise residual wrapper. The local-analytic procession-bound refinement lowers this boundary below the three explicit case inequalities. It stores the single comparison with the projected case-defined Theorem 1.10 procession upper bound, the distinguished-kind witness, and the distinguished $+1$ comparison; its projection expands the case definition and reconstructs the older local-analytic case-bounded source. The localized C_Θ constructor then consumes this processional source directly. Thus the remaining Step (xi)-to-procession comparison is still an explicit source boundary, but the distinguished, nondistinguished, and archimedean case inequalities are no longer separate inputs at the local-analytic handoff. The local-analytic procession-comparison refinement splits that comparison into the paper-facing local log-volume identification and the upper-semi procession estimate. It introduces a procession-normalized local log-volume, identifies the localized Step (xi) weighted adjusted log-volume with it, and separately records the processional upper bound against the projected Theorem 1.10 case-defined procession bound. Its projection reconstructs the older processional source, and feeds it into the localized C_Θ handoff. A further processional-estimate refinement derives the upper-semi procession estimate from the local IUT IV construction quantities: the distinguished Step (v) exact procession bound, the nondistinguished Step (vi) zero tensor bound, and the archimedean Step (vii) metric-container bound. Its projection reconstructs the comparison source, and feeds the lowered estimate into the localized C_Θ handoff. The processional arithmetic-gap variant keeps the same local processional estimates for the ordinary bounds but routes the distinguished one-unit term through the arithmetic-minus-main residual inequality. Its constructor enters the localized C_Θ handoff through the pointwise defined residual source, so the stronger distinguished $+1$-below-procession formula is no longer required on this route. The lower pointwise-residual refinement derives that distinguished arithmetic-minus-main $+1$ comparison from the pointwise defined Haar-residual inequality and the processional normalization of localized Step (xi), then projects to the arithmetic-gap source. It now also projects to the lower-weight local arithmetic-degree residual package by , so the distinguished-place indicator ledger is available on this local-estimate route. Its constructor exposes this lower handoff to the localized C_Θ route. The pointwise Haar-defect source lowers the residual payload again: it identifies the defined residual with the local Haar-normalization defect through the* *local arithmetic-degree identity, then derives the pointwise residual source from local defect nonnegativity plus one distinguished defect lower bound. Its processional wrapper projects through the same lower-weight residual package via , and the p-adic unit-ball processional Haar-defect source has the analogous projection . The wrapper and constructor feed this Haar-defect layer into the same localized C_Θ route. The same-index case-bounded source lifts this case split to the signed endpoint boundary: it keeps the determinant and theta-equality fields of the pointwise same-index route, but its residual payload is the valuation-ball Theorem 1.10 case source. Its projection constructs the pointwise residual source internally, and exposes the corresponding named public signed dichotomy. The route and public-boundary audits and record the same projection into the pointwise residual route and the resulting q-normalized comparison evidence. The companion endpoint projects through the same q-normalized route and proves $(-1 : \mathbb{R}) \le C_\Theta$. The source-obligation-gap variants and project the q-pilot positivity and normalization hypotheses from before invoking these case-bounded endpoints, so this public surface no longer names an arbitrary side-hypothesis record. The primitive-constructor side-condition variants and construct that source-obligation gap from the concrete packet's primitive constructed-SHE bundle and the Stage 1 side conditions before applying the same case-bounded route. The source-hull/determinant-obligations variants and specialize this again by projecting the Stage 1 side conditions from ; hence q-pilot positivity and source normalization are now read from the source hull/determinant obligation object at this public boundary. The constructed-hull/determinant-source variants and lower this once more: the obligations object is projected from , so the case-bounded public surface consumes the constructed Remark 3.9.5 holomorphic hull/determinant source directly. The principal-HDD variants and lower the same public surface to the principal product-hull HDD source itself: the case-bounded route reads from the source gap carried by that principal HDD source. The internal-principal-HDD variants and remove that source as a separate endpoint argument: it is reconstructed from the compact-open theta-region, pointwise determinant formula calibration, and source side hypotheses before the case-bounded route is invoked. The pointwise determinant calibration itself now has a lower checked constructor, : from the pre-ledger volume/hull formula and the summand-charted Hodge family-hull equality, Lean derives the determinant log-volume, normalized determinant, and family-hull equalities needed by the determinant-factored pointwise source. The constructor lowers the bundled case-bounded source itself: it assembles the same-index case source from the formula-gap source and the explicit valuation-ball Theorem 1.10 case residual source, with determinant synchronization and theta equality inherited from the formula-gap side. The local-analytic constructor lowers that source boundary again: the formula-gap matched Theorem 1.10 source is constructed from the valuation-ball additive-Haar local-analytic arithmetic-divisor source before the bundled case source is assembled. The parallel constructor composes the local-analytic case-source lowering through the same-index boundary: the case estimates are stated directly over the valuation-ball local-analytic arithmetic-divisor source, and Lean constructs the formula-gap matched case residual internally. The signed endpoint now composes this constructor into the internal-principal-HDD route. Its $C_\Theta \ge -1$ companion uses the same local-analytic source boundary. The projection checks the converse handoff needed by the endpoint ladder: from the same explicit case split it reconstructs the same-index local-analytic source, deriving the total Haar residual inequality through the local arithmetic-degree residual package. The composed endpoints and immediately feed this constructed case source into the internal-principal-HDD signed and $C_\Theta \ge -1$ route. The same-index local arithmetic-degree source lowers the formula-gap boundary itself: it keeps the determinant and theta-equality data but replaces the raw residual inequality field by the packaged local arithmetic-degree residual source. Its projection* *constructs the older formula-gap source, and then enters the existing q-normalized local-analytic route. Its scale-synchronized projection derives the canonical C_Θ-scale equality from the packaged theta/log-volume identity and q-pilot positivity before entering the scale-synchronized local-analytic endpoint. The direct endpoint exposes the signed Corollary 3.12 dichotomy from this local-degree source. Its route audit exposes the q-normalized endpoint, localized C_Θ audit, signed IUT IV upper bound, and dichotomy from the same local-degree source. The named preferred public endpoint exposes the signed endpoint at this lowered local-degree boundary. Its side-condition version reconstructs the named side-hypothesis record from* IUTStage1SourceSideConditions *and returns both the signed dichotomy and the companion $C_\Theta \geq -1$ bound by projecting the same local-degree source through the formula-gap and scale-synchronized routes. The source-gap version obtains the same two outputs after projecting q-pilot positivity and normalization from* IUTStage1SourceObligationGap*, so the raw side-condition record is no longer exposed on the packaged local-degree surface. The constituent endpoint goes one step lower: it takes the weighted determinant component identifications, q-normalized theta/log-volume identity, Theorem 1.10 formula-gap valuation-ball source, and local arithmetic-degree residual payload directly, constructs the local-degree source internally, and exposes the same signed and $C_\Theta \geq -1$ outputs. Its source-gap form specializes this expanded route at* IUTStage1SourceObligationGap.sideConditions*, so the constituent q-normalized boundary no longer takes the raw side-condition record when the source-obligation gap is available. Its possible-image witness refinement constructs the packet-indexed* WithSymmetryLabelTransport *from the Theorem 3.11 possible-image witness, equality-quotient possible-image family, and the symmetry-label/upper-semi/procession fiber transport laws before entering the same source-gap local-degree route. The realized-exact refinement lowers the compact-open input on this branch: Lean converts realized valuation-ball exactness to compact-open realized exactness using the local compact-open Haar-normalization law before applying the possible-image source-gap endpoint. The log-shell exact refinement lowers this once more: the public input is the pair of log-shell equalities* $\Theta_{\text{region}} = \text{realized}(\text{logShell})$ *and* $\text{logShell} = \text{valuationBall}$*, from which Lean constructs realized valuation-ball exactness and then the compact-open source. The compact-open log-Kummer map refinement lowers the same branch below these log-shell equalities: the public Kummer-map data supplies the local realization map, selected-region section, valuation-factor compatibility, and compact-open soundness; Lean derives the correspondence, image, compact-open possible-region exactness, log-shell image exactness, and finally log-shell exactness before entering the signed source-gap route. The constructor now derives that map source from the graph layer itself: graph left-totality chooses the section, graph realization identifies the map with the valuation-factor realization, and graph soundness supplies the compact-open possible-region law. The public source-gap endpoint now uses this constructor internally, so the possible-image local-degree branch no longer exposes the compact-open map package as its strongest public log-Kummer input. The same branch is lower again at and : Lean first builds the canonical graph from the compact-open image containments, then rewrites those containments from the Remark 3.9.5 possible-region equality. The same public route is now lower again at the image-law and correspondence layers: derives possible-region exactness from the two compact-open log-Kummer image laws, and extracts those image laws from compact-open log-Kummer correspondence left-totality, realization compatibility, and soundness. The local-analytic case-residual companion first constructs the source-obligation gap from the primitive constructed-SHE bundle and the constructed Remark 3.9.5 hull/determinant source. Its principal-HDD refinement reads that source gap directly from the principal product-hull CanonicalHDD direct common-container source before entering the same local-analytic comparison. The same-index case-bounded boundary* *now also consumes the q-normalized local arithmetic-degree source directly via . Lean first projects the formula-gap source from the local-degree object and then attaches the explicit Theorem 1.10 valuation-ball case residual source. Consequently the strongest case-bounded endpoint inherits the adjusted-determinant synchronization and record-family-hull q-scale lowerings already present in the local arithmetic-degree source, rather than exposing the three weighted determinant component equalities again at this boundary. The endpoint composes this case-bounded route with the record-family-hull q-scale constructor itself: the public boundary now supplies the adjusted-determinant synchronization, record-family q-scale comparison, source-obligation gap, formula-gap source, local arithmetic-degree residual source, and explicit case residual; Lean constructs the local-degree object internally before returning both the signed dichotomy and the* $C_\Theta \geq -1$ *bound. The lower scalar-boundary endpoint now bypasses that record-family q-scale input on the same case-bounded branch: it constructs the q-normalized local-degree source from full adjusted-determinant synchronization and the projected localized Step (xi) adjusted-sum q-normalization, then attaches the explicit Theorem 1.10 case residual and returns both endpoint inequalities. The derived-residual variant removes the separate local arithmetic-degree residual package from that boundary: the same valuation-ball case source first projects to , and the adjusted-sum endpoint consumes this internally derived residual. The local-analytic case-source refinement lowers this same adjusted-sum branch one more step: the formula-gap valuation-ball package and formula-gap-indexed case residual are constructed internally from the local-analytic Theorem 1.10 valuation-ball ledger and its local distinguished/nondistinguished/archimedean case inequalities. The procession-bound form lowers the same public branch to the projected case-defined Theorem 1.10 procession bound plus the distinguished one-unit gap; Lean expands these two inputs to the local-analytic case source internally before applying the adjusted-sum endpoint. The procession-comparison form lowers the same endpoint one layer further: the public input separates the localized Step (xi) equality with procession-normalized local log-volume from the upper-semi projected procession estimate, and Lean reconstructs the procession-bound source internally. The processional-estimate form removes that upper-semi estimate from the public adjusted-sum surface as well: the input is the local Step (v), Step (vi), and Step (vii) processional data, and Lean derives the projected procession comparison before applying the signed* C_Θ *route. The case-defined processional form fixes the processional local value to the local-kind case split itself and reconstructs the general processional-estimate source internally. The p-adic-normalized case-defined form makes the distinguished nonarchimedean place the normalized p-adic unit-ball Haar-defect place before projecting to the ordinary case-defined processional-estimate source. The residual-level adjusted-sum bridge is the weakest same-index adjusted-sum handoff currently used in this branch: it consumes the local arithmetic-degree residual package directly, so no case-bounded or processional-estimate residual source is exposed at that boundary. The p-adic arithmetic-gap endpoint then lowers the normalized p-adic branch below the processional-estimate assumption: the source carries the Step (v)/(vi)/(vii) case values and the arithmetic-minus-main one-unit gap at the normalized p-adic Haar-defect place, and Lean projects this data to the local arithmetic-degree residual source before entering the adjusted-sum comparison. The lower p-adic Haar-defect endpoint removes the case-defined processional arithmetic-gap package from this consumer: it takes the p-adic unit-ball Haar-defect residual source itself, projects the pointwise nonnegativity and normalized-place unit contribution to the canonical local arithmetic-degree residual package, and then enters the same adjusted-sum route. The hull/determinant-backed variant first derives the source-obligation gap from ; thus an endpoint that already carries* IUTStage1SourceHullDetObligations *no longer exposes a second assembled source-gap object. Its same-index family-hull companion replaces those weighted shadow fields by full adjusted-*

determinant synchronization and replaces the direct q-normalized theta identity by the family-hull theta comparison plus the p-adic finite hull-decomposition theorem. Its source-obligation-gap form reads q-pilot positivity and normalization from IUTStage1SourceObligationGap*, rather than from a separate side-condition record. Its record-family-hull q-scale companion lowers this public surface one step further: the unscaled* Θ_{signed}*-to-record-family-hull equality is derived from the pointwise determinant/Hodge calibration. The record-family q-scale law is now derived from the adjusted-sum q-normalization together with the determinant/Hodge calibration and the principal-product p-adic finite hull-decomposition theorem, so the unsolved scalar frontier on this branch is the paper-level derivation of that adjusted-sum q-normalization. The source-obligation-gap variant removes the raw side-condition package from this refined record-family surface as well. The p-adic unit-ball Step (xi) synchronization constructor now discharges that residual payload from the p-adic unit-ball Haar-index source and the local synchronization identity. That synchronization has now been lowered once more by the arithmetic formula matching constructor . Its source data are the two local formula equalities* $\mathrm{StepXI}_v = a_v(D_v + C_v)$ *and* $E_v = p_v^{[K_v:\mathbb{Q}_{p_v}]} + M_v$*; Lean unfolds the local arithmetic upper contribution to reconstruct both the p-adic synchronization and the expanded Haar-residual identity. The stricter source-backed bridge now feeds this canonical constructor from the valuation-ball formula-gap matched arithmetic-degree/p-adic source: the Step (xi) equality is derived from localized determinant weight/raw-log-volume calibration. The prime-error equality is no longer only a raw field at this layer: now has a constructor from the additive-Haar local analytic arithmetic-divisor evaluation source plus the p-adic identity* $E_v = \delta_v + M_v$*. Its valuation-ball refinement builds this identity into the prime-error term itself and projects to the valuation-ball formula-gap matched prime-error source. It is now also constructed from the arithmetic-degree-controlled local arithmetic source: Lean combines* $E_v = \mathrm{defect}_v + M_v$ *with the p-adic Haar-index identification* $\mathrm{defect}_v = \delta_v$*. The lower Record-Ob3/Ob5 source now consumes this valuation-ball p-adic-defect/main object directly and projects the former strict p-adic-Haar controlled source. The same constructor is now composed one layer lower from the formula-gap matched arithmetic-degree/p-adic source plus the synchronized Proposition 1.4/1.5 arithmetic-degree comparison source. Lean now also projects that comparison source from the local Step (v)/(vii) degree-identification formula source, so the p-adic split can be built from the local degree-identification data directly. The Record-Ob3/Ob5 arithmetic-divisor component route composes this with its own local-degree projection, so that p-adic split is now constructed from component data plus valuation-ball provenance. The valuation-ball Record-Ob3/Ob5 route lowers this once more by reading the component-to-valuation-ball formula equality from the valuation-ball record itself before constructing the same p-adic defect/main source and audit. This projection now factors through an explicit arithmetic-degree-controlled local arithmetic source: and the audit . Thus the finite sums, nonnegativity, formula-gap construction data, arithmetic-divisor calibration, and Proposition 1.4/1.5 comparison matching are read from the valuation-ball local-estimate package rather than restated on the p-adic split record, and consumes that constructed split together with the arithmetic-degree calibration before projecting to the canonical p-adic formula source; the formula-gap source has the corresponding constructed-prime-error constructor and audit. The canonical residual projection is now available directly from that arithmetic formula-matching source: composes the p-adic Haar-defect and Step (xi) synchronization constructors internally before producing the generic local arithmetic-degree residual source. The processional local-estimate source now reaches the same residual package directly via , and the processional p-adic unit-ball Haar source does so via . The formula-gap valuation-ball provenance is preserved. The matched local-degree arithmetic-divisor-backed source also projects to the same residual package through , so the bridge audit now records* *the Theorem 1.10 arithmetic-divisor local degree branch before entering the canonical residual consumer. The constructed Theorem 3.11 local-term source reaches the same boundary by , so the audit can now pass through the multiradial possible-image, q-pilot-alignment, determinant, and local-global C_Θ source layer before projecting to the residual source. The valuation-ball arithmetic-degree controlled local source is the strongest checked source-backed bridge in this slice: . This has now been composed through the Record-Ob3/Ob5 valuation-ball local-estimate source itself: . The component Record-Ob3/Ob5 source has the corresponding residual projection after supplying its valuation-ball formula provenance: . It also has the lower arithmetic-degree-controlled variant, in which the valuation-ball formula-gap source is reconstructed internally: . The lowest checked component variant now constructs that controlled local arithmetic source internally from the Theorem 1.10 local-analytic ledger, the p-adic Haar-index defect source, arithmetic-degree calibration, and the projected Step (v)/(vii) comparison bounds: . This residual projection now factors through the canonical p-adic arithmetic formula-matching package: , so the checked component route exposes the reconstructed $StepXI_v = a_v(D_v+C_v)$ and $E_v = \delta_v+M_v$ laws before passing to the residual source. The component-level p-adic defect/main projection has been lowered to the same arithmetic-degree-controlled local arithmetic source: reconstructs the valuation-ball formula package internally before applying the local-degree projection. For the ordinary component source the remaining synchronization at this boundary is exactly the equality identifying the Record-Ob3/Ob5 component formula with that controlled formula object. The stricter controlled component source removes that equality as a separate input: its local-degree arithmetic-divisor component is assembled over by definition, and its projection audit checks the resulting Record-Ob3/Ob5 valuation-ball source with no component formula synchronization hypothesis. The p-adic Haar controlled source lowers this once more: stores the Theorem 1.10 valuation-ball arithmetic-divisor ledger, the p-adic unit-ball Haar-index defect source, the arithmetic-degree calibration, and the projected Step (v)/(vii) comparison bounds, then reconstructs the arithmetic-degree-controlled local arithmetic source internally via . Its projection audit checks the valuation-ball Record-Ob3/Ob5 source after this internal reconstruction, so the controlled local arithmetic package is no longer an independent boundary object at the strict component layer. The same strict source now feeds the two downstream local consumers directly: projects the p-adic defect/main local-estimate source, while projects the canonical local arithmetic-degree residual package. The p-adic-defect/main controlled component source now also feeds the compact-open product-handoff formula-realized route directly by . This constructor projects the valuation-ball local analytic object, the p-adic Haar-index defect source, the arithmetic-degree calibration, the $E_v = \delta_v + M_v$ split, the distinguished/archimedean comparison bounds, and the component/formula synchronization from the single p-adic-defect/main controlled source before entering the existing product-handoff route. The same p-adic-defect/main source now also feeds the projected-local-degree formula compact-open route via . That constructor projects the local Step (v)/(vii) formula package and its comparison matching from the controlled Record-Ob3/Ob5 component chain, then uses the local-degree formula controlled source to synchronize the reconstructed local arithmetic handoff. The strict p-adic Haar controlled component source is now also derived from the local-degree formula valuation-ball comparison route, and that comparison route is derived from the controlled component layer. The remaining higher task is to construct the synchronized controlled component/local-estimate source itself, including its finite sums, nonnegativity, formula-gap inputs, component synchronization, and arithmetic-divisor calibration, from the full IUT IV local-estimate construction. The checked residual calculation is explicit: identifies the local defined residual with the p-adic unit-ball index, and sums these identities to obtain the residual lower bound consumed by the C_Θ route. The matching* *weakened-boundary diagnostic shows that the Step (xi) synchronization identity and residual/index equality alone do not force the one-unit residual lower bound; the p-adic normalized-place index law is the source-backed ingredient that excludes the zero-index model. A second diagnostic shows that* $E_v = \mathrm{defect}_v + M_v$ *does not imply the p-adic split* $E_v = \delta_v + M_v$ *unless the IUT IV arithmetic defect is identified with the p-adic Haar defect. The lower component synchronization diagnostic keeps the Theorem 1.10/p-adic Haar-index and arithmetic-degree reconstruction equations true while detaching the Record-Ob3/Ob5 component formula from the reconstructed p-adic Haar formula. The strict residual-consumer diagnostic then shows the downstream failure: the component residual and reconstructed p-adic residual can differ unless the strict controlled source carries this formula synchronization definitionally. A separate Record-Ob3/Ob4 shadow diagnostic records that equality of the weighted determinant shadow and finite-sum scale does not identify the full Ob3/Ob4 source; the anchor/localization payload may still differ. Thus the full* `recordOb3Ob4_eq_stepXI` *synchronization remains a real source obligation until it is derived from Remark 3.9.5 data. The positive criterion is now explicit: identifies two Ob3/Ob4 sources once the localization family, anchor, and positive tensor-power data agree. Lean also projects this source-level synchronization to the older weighted determinant summand, anchor, and tensor-power fields consumed by the current canonical route: . The same primitive criterion now reaches the Record-Ob3/Ob5 component boundary itself: constructs the component source without taking the full* `recordOb3Ob4_eq_stepXI` *equality as a field, and pins the constructed equality proof and projected component endpoint. The valuation-ball Record-Ob3/Ob5 source has the same lowering: composes the component constructor with the valuation-ball formula source and transports the derived Ob3/Ob4 equality across the component/formula matching. The constructor-built variant removes the separate determinant-source equality at this boundary when the constructed hull/determinant source is generated from the same Ob3/Ob4 adjusted-determinant package; Lean projects* $\mathrm{det} = \mathrm{Ob3/Ob4}$ *from the constructor-built source before entering the valuation-ball Record-Ob3/Ob5 object. Its q-normalized refinement no longer takes the scalar equality* $\mathrm{canonicalCThetaScale} = \mathrm{adjustedSummandLogVolume}$ *directly. The algebraic conversion derives that equality from the q-normalized identity* $\theta = \mathrm{adjustedSummandLogVolume} \cdot (-q)$ *and q-pilot positivity, while pins the derived scale equality and valuation-ball endpoint. The family-hull q-normalized refinement no longer asks for that adjusted-summand q-normalization directly: Lean first applies to transport the comparison across the Record-Ob3/Ob5* $\mathrm{familyHullLogVolume} = \mathrm{adjustedSummandLogVolume}$ *theorem, and its audit pins the transported q-normalization, derived canonical scale equality, and valuation-ball endpoint. At the p-adic finite-extension unit-ball boundary, Lean now derives the projected additive-Haar componentwise matching from equality of the underlying finite-extension Haar normalization factors: . This has been lowered once more to componentwise matching of the finite-extension Haar payload fields–integer unit-ball source, realization, realized-region map, compact-open radius, and Haar measure–by and . The current lower criterion pushes this through the constructed base-prime dilation/mass source: componentwise equality of the constructed local-field payloads projects to the finite-extension proper-ultrametric componentwise equality, and hence to the Ob3/Ob4 adjusted-determinant synchronization: and . The p-adic determinant criterion has now been pushed through the source-paper unit-ball Haar-character normalization as well: componentwise equality of local sources carrying* $\mu(O_v) = 1$ *and* $\mu(p_v O_v) = p_v^{-[K_v:\mathbb{Q}_p]}$ *projects through the Haar-character,* $\mathbb{R}_{\geq 0,\infty}$*-modulus, constructed dilation/mass, and proper-ultrametric layers before entering the determinant synchronization: and . The same determinant criterion is now lowered through the quotient-coset source as well. The residue-coset partition is derived from the finite additive quotient* $O_v/p_v O_v$*, then projected to the unit-ball Haar-character source before entering* *the determinant synchronization: and . The criterion is now lowered one step further to the quotient-native residue-module source: the finite residue quotient O_v/p_vO_v carries its residue-field module structure, Lean computes the quotient cardinality and representatives, and the resulting unit-ball Haar-character projection feeds the same determinant synchronization: . This has now been lowered again to the residue-submodule source. The residue-field action is carried before quotienting on the additive group of O_v, the p_vO_v image is represented by a submodule, and Lean transports the quotient module to O_v/p_vO_v before applying the residue-module determinant criterion: . The same residue-submodule object now projects directly to the unit-ball Haar-character endpoint. Lean transports the quotient module across O_v/p_vO_v and reads off $\mu(O_v) = 1$, $\mu(p_vO_v) = p_v^{-[K_v:\mathbb{Q}_p]}$, the additive Haar-character scalar, and the all-subset base-prime dilation law without reintroducing a quotient-native residue-module source: . This residue-submodule Haar endpoint now feeds the finite-place IUT IV Haar-defect boundary. The source carries a residue-submodule Haar source at every finite place and constructs the existing p-adic unit-ball Haar-index defect source by projecting each local factor to its unit-ball Haar-character normalization: . The inverse-base-prime compact-open factor is also source-backed at the finite-extension layer: Lean proves $p^{-1}O_K$ is the valuation ball of radius p, constructs the corresponding valuation-ball additive-Haar source, and transports it componentwise to the compact-open inverse-base-prime normalization via and . The named-HDD Record-Ob3/Ob5 valuation-ball boundary now also projects directly to the canonical residual package through the p-adic arithmetic formula-matching bridge: . The canonical route manifest records this residual frontier as a checked 320-entry, zero-source-obligation sub-manifest: . The newest residual-frontier entries register the Record-Ob3/Ob5 component and valuation-ball localization-anchor-tensor constructors, the constructor-built valuation-ball determinant-source lowering, the possible-image named-HDD boundary projection, the family-hull q-normalized synchronized route-input constructor, and their audits, the finite-extension inverse-base-prime valuation-ball construction and its componentwise additive-Haar synchronization, the synchronized route-input construction, the compact-open inverse-base-prime audit, the possible-image/correspondence lifts of the inverse-base-prime and product-handoff formula-realized routes, the p-adic-defect/main compact-open route-input constructor, the local-degree formula compact-open route-input constructor, the local-degree formula canonical-HDD compact-open inverse-base-prime route, and the two p-adic-defect/main compact-open product-handoff routes, the p-adic-normalized processional case local-estimate endpoint, the projection from full p-adic-normalized processional estimates to the weaker Haar-residual case source, the case-defined normalized Haar-residual case projection, the projection from normalized case-defined estimates to that Haar-residual case boundary, the normalization of a general case-defined estimate along the p-adic place alignment, packet handoff, six public signed endpoint lowerings, the case-defined reconstructed p-adic Haar-defect projection, endpoint, packet handoff, and public signed endpoint, the selected distinguished-place Theorem 1.10 projection, selected normalized-place kind theorem, selected case-defined reconstructed p-adic Haar-defect projection, endpoint, packet handoff, and public signed endpoint, plus the constructor-level p-adic-defect/main projected-weighted formula-gap residual endpoint and its projection audit, and the matched local-degree object formula/residual and case-bound projections , , , , and . A matching weakened-boundary countermodel, , records that an arbitrary detached compact open may retain unit Haar mass while the true inverse-base-prime ball has mass strictly greater than one. The checked manifest therefore tracks that the local analytic Theorem 1.10 object, Haar-defect function, pointwise equality, localized synchronization, local-degree formula projection, case-local arithmetic-degree bound recombination, and component/formula equality are bundled before the signed comparison route consumes them. The constructor-level endpoint additionally* *records that, once the arithmetic-degree calibration localized Step (xi) source is identified with the principal-product p-adic finite localized Step (xi) source, Lean projects both the formula-gap Theorem 1.10 package and the local arithmetic-degree residual package from the single p-adic-defect/main Record-Ob3/Ob5 source. The projection audit separately proves that this lower source carries the formula-gap endpoint, the valuation-ball local-analytic endpoint, and the canonical residual endpoint before the signed comparison is invoked. It classifies the p-adic-defect/main valuation-ball controlled Record-Ob3/Ob5 component source as derived rather than as a lower source obligation. The valuation-ball Theorem 1.10 local analytic source is now pinned by the source-ingredient audit , which expands it into arithmetic divisors, valuation-ball additive-Haar Proposition 1.4 constructions, additive/local analytic/proposition/formula projections, the theta-pilot local evaluation, Step (vi) zero contribution, and Step (v)/(vii) formula-to-gap inequalities. The formula-gap matched valuation-ball refinement now proves that the distinguished Proposition 1.4 valuation-ball log-shell source and the archimedean metric source carry their container bounds through the named formula-gap comparisons before projection to the additive-Haar audit. The local-degree p-adic-defect/main controlled-component audit now gathers the Step (v) distinguished different/conductor bound, the Step (vi) nondistinguished zero-gap law, the Step (vii) archimedean different/conductor identification, $E_v = \delta_v + M_v$, valuation-ball projection preservation, and the lowering into the p-adic-defect/main controlled Record-Ob3/Ob5 component. The local-degree valuation-ball projection preservation theorem transports the nondistinguished valuation-ball shell payload across $E_v = \delta_v + M_v$, so the local-degree comparison route no longer carries this preservation as a separate source obligation. The lowering constructs the canonical p-adic-defect/main controlled component source from the local-degree formula comparison source. It uses to identify the p-adic-defect/main reconstruction with the original arithmetic-degree-controlled source before transporting the controlled Record-Ob3/Ob5 component. The lowering constructs the local-degree formula comparison source from the controlled component source, and the bridge lowers that comparison layer to typed Step (v)/(vii) local-degree identification objects by projecting the formula source and valuation-ball matching proof from the object endpoint and additive formula synchronization. This matched-object source is now itself projected from the controlled Record-Ob3/Ob5 component source by , which reads the Step (v)/(vii) object data from the arithmetic-divisor component chain and reconstructs the valuation-ball formula source from the controlled local arithmetic package. The compact-open route now consumes the matched-object source directly before entering the canonical HDD inverse-base-prime handoff; its projection audit identifies this route with the older local-degree formula route after object-to-formula projection. The controlled-component variant pushes that handoff one layer lower by constructing the matched-object source internally from the synchronized controlled component. The bridge now names the direct projection from the p-adic-defect/main Record-Ob3/Ob5 boundary to that local-degree formula comparison layer. The bridge reaches the canonical residual package through this lowering. The bridge projects that source directly to the canonical residual package, reconstructing the strict p-adic-Haar controlled source, the arithmetic-degree-controlled local source, and the component/formula synchronization internally. The matched-object bridge is lower still: it reconstructs the p-adic-defect/main source from the typed Step (v)/(vii) local-degree objects and their valuation-ball formula synchronization before projecting the canonical residual package. The synchronized controlled component and p-adic-defect/main layers are now higher comparison boundaries for this derived residual route; the legacy strict p-adic-Haar residual projection remains a checked comparison route. The finite-place arithmetic-defect source is now canonicalized by : once the p-adic unit-ball Haar-index defect source is fixed, the IUT IV local arithmetic defect has no independent numerical freedom.*

This propagates to the valuation-ball arithmetic-degree local source via , which identifies the carried controlled local source with the one reconstructed from the p-adic Haar-index constructor and the same Step (xi) calibration. The strict p-adic-Haar equality and the p-adic-defect/main strict-projection equality show that this lower route also reaches the p-adic-defect/main valuation-ball local-estimate endpoint by canonical reconstruction from the arithmetic-degree-controlled local source, not by a hidden independent local-estimate package. The compound audit records the same reconstruction together with the valuation-ball endpoint, strict p-adic-Haar projection, reconstructed arithmetic-degree source, and carried controlled Record-Ob3/Ob5 component. It now also records the actual case-local Theorem 1.10 inequalities used downstream: the Step (v) distinguished exact-procession bound, Step (vi) nondistinguished zero bound, Step (vii) archimedean metric-container bound, and the recombined case-procession arithmetic-degree inequality. At the local-degree formula comparison layer, the equality shows that the p-adic-defect/main valuation-ball local-estimate object is reached by first reconstructing the arithmetic-degree-controlled local source; it is not an additional field at this comparison boundary. The same layer now exposes the nondistinguished Step (vi) zero-gap contribution through , so the local-degree audit records all three local cases before the p-adic-defect/main projection. This is backed one layer lower by the valuation-ball construction theorem , which proves the same Step (vi) zero-gap law before projecting to additive-Haar data. The now-derived strict Record-Ob3/Ob5 lowering additionally uses the stronger preservation of the valuation-ball additive-Haar construction object; the diagnostic shows the weakened-boundary failure when the reconstructed p-adic-defect/main valuation-ball shell payload is detached from the original valuation-ball shell payload. The constructed diagnostics record the p-adic defect/main split failure without arithmetic-defect identification, the ordinary component formula-matching failure, and the strict residual synchronization failure without definitional formula matching. The local-degree goal-evidence audit and public-boundary inventory recover the same relative non-vacuity and no-independent-public-input bookkeeping after internally projecting to the older local-analytic source. The endpoint-level audit exposes the same evidence under the lowered local-degree preferred public route name. The public-boundary inventory audit records, as named audit fields, that this endpoint constructs the q-normalized endpoint, localized C_Θ audit, signed IUT IV bound, and dichotomy from the same source, while no independent localized C_Θ, raw signed comparison, principal hull, theta-region principal source, measure calibration, or summand calibration remains on the public input surface. The packaged local-degree endpoint also has a source-obligation-gap form , which projects the required q-pilot sign and normalization from before applying the local arithmetic-degree source. The same lowering now reaches the expanded constituent, family-hull theta, and record-family-hull q-scale local-degree surfaces, leaving only the lower-comparison companions on this slice to pass through the named source-obligation gap. The localized-Step-(xi)-refined pointwise route goes one boundary lower: it consumes and projects the localized Ob3/Ob4 determinant source, determinant and canonical-scale identifications, local main-log terms, Haar defects, and finite-place summation identities internally. The IUT-IV-backed pointwise variant lowers the finite-place summation input again: it consumes , whose arithmetic-upper and main-log sum identities are projected from the IUT IV arithmetic-divisor local evaluation source. The Theorem 1.10 valuation-ball pointwise variant lowers that IUT IV input one step further. It consumes , whose projection obtains the arithmetic-divisor local evaluation ledger from the valuation-ball additive-Haar formula-gap source. Thus the public pointwise route now sees the distinguished Proposition 1.4 log-shell construction, nondistinguished zero case, archimedean Proposition 1.5 metric construction, and named formula-to-gap comparisons before projecting to the old IUT IV local-evaluation interface.

The constructor exposes the IPL construction, structured-SHE transport context, APT construction, certificate equality, and Theorem 3.11 subclaims before packaging the constructed qualitative input. The packet completion audit now enters that same completion route from the concrete Theorem 3.11 packet. The concrete primitive packet now projects the source spine, target-region lattice formula source, bridge-certificate constructor, and primitive constructor from explicit Hodge-theater/log-theta/log-Kummer, theta-pilot-class target-region formula, and IPL/SHE/APT bridge-component fields. In particular, Lean derives the package target-region lattice formula from before building the possible-image source; the Step (x) finite-divisor/ log-Kummer package is derived from ; the packet-derived one-sided multiradial source feeds the packet's theta-pilot possible images, vertical log-Kummer source, gluing torsor, and selected q-choice into the finite F_ℓ *procession/equality-quotient constructor; the comparison audit ties this vertical packet source to the stricter fiber* (Ind2) *action-packet transport source used by the theta-region route by proving that both constructions have the same record possible-image family and the same selected q-region after forgetting to the underlying target set; the wrapper lowers the arbitrary fiber source one step further by projecting it from a symmetry-label transport source whose indeterminacy data is the packet's vertical log-Kummer procession-normalized source; the constructor builds that wrapper from the Theorem 3.11 possible-image witness, the equality-quotient possible-image family, and explicit symmetry-label, upper-semi, and procession transports; the public route consumes this wrapper directly, so the packet-facing completion theorem no longer exposes an independent fiber* (Ind2) *action-packet transport source, while removes the wrapper itself from this theorem surface, and carries the same lowering through the principal valuation-ball theta-region source route, while carries it through the principal valuation-ball HDD-source audit, and carries it through the constructed-HDD source route, and carries it through the localized Step (xi) finite-place data route, and carries it through the localized calibration-data route, and carries it through the localized calibration-restriction route, and carries it through the principal-hull-projected local-arithmetic route, and carries it through the fiber-backed theta-region construction route, and carries it through the theta-region-defined principal valuation-ball route, and carries it through the principal pointwise metric valuation-ball route, and carries it through the pointwise calibration-projected route, and carries it through the pointwise local-arithmetic constructed route, and carries it through the localized Step (xi) source route, and carries it through the IUT IV arithmetic-divisor localized route, and carries it through the Theorem 1.10 valuation-ball IUT IV localized route, and carries it through the Hodge-formula-calibrated Theorem 1.10 route, and carries it through the local-analytic Theorem 1.10 route; the Theorem 3.11 bridge package is derived from the package-aligned , so the packet no longer supplies separate bridge source-copy or output-family alignment fields; and the Theorem 3.11 subclaim record is derived from . Its algorithmic-output certificate is assembled by first deriving structured IPL/SHE/APT witnesses from the bridge, then applying the single structured output-flag equality recorded in and consumed by , while its SHE alignment is projected from the bridge's structured SHE transport context rather than supplied as a separate packet field. The constructor pushes this to the deepest current source-spine q-pilot packet route: it accepts the structured output-flag calibration and constructs internally via . The one-sided source-spine-data route has the parallel constructor , so the lower one-sided multiradial packet path also consumes only the structured output-flag calibration instead of a standalone subclaim component. Its packet-construction audit records that this constructor preserves the source-spine selected q-choice, projected bridge component, Step (x) log-Kummer source, and target-region formula source while generating the subclaim component from the output-flag calibration. The packet route audit records that this projected constructor satisfies the primitive source audit.*

The bridge-certificate source transports the Theorem 3.11 IPL/SHE/APT bridge package across source-copy and output-family equalities to construct the primitive qualitative source; the lower bridge-certificate constructor then builds the route-facing primitive source constructor without taking that qualitative source as an independent packet field. The package-aligned bridge component now has the constructor from the lower constructed qualitative certificate source, and the primitive packet audit includes ; thus the SHE no-identification guard, the non-forbidden APT quotient transport, and the selected IPL prime-strip endpoints are visible at the concrete packet boundary. The packet constructor now consumes the constructed-qualitative bridge-alignment source directly, and the lower constructor consumes the common-container-derived alignment source. Thus the SHE/common-container equality is no longer a free packet-level alignment. The still lower constructor also fixes the packet input and selected-output prime strips as canonical projections from the constructed qualitative IPL *source. The codomain-canonical constructor additionally fixes the packet one-column Hodge theater as the constructed* SHE *codomain and the one-column history as the selected q-choice history. The q-pilot-canonical constructor constructs the selected q-choice from the constructed* SHE *q-pilot theater itself, so the codomain placement is derived by the structured* SHE *q-pilot/codomain law. The q-pilot-class constructor goes below the concrete-coordinate boundary by taking a q-pilot theta-pilot lattice class plus representative data. The component-representative constructor lowers the representative-data section to the generated component kernel, which constructs the forgotten finite label, procession state, local tensor state, and upper-semi state. The source-spine constructor goes one layer lower by reading the q-pilot history, lattice coordinate, and coric datum from the selected q-choice of the unified Theorem 3.11 source spine. The generated-full-label source-spine constructor then derives the same primitive packet from the generated full-label representative projection. The generated-q-choice constructor derives that projection from the source-provided generated q-choice plus its q-pilot theta-class and representative-label coherence. The selected-label constructor lowers this boundary again: the generated q-choice is constructed internally as the representative full-label choice over the q-pilot theta class read from the source spine, and Lean derives the projection from fieldwise source compatibility of the selected q-choice with the q-pilot theater, representative finite label, procession state, local tensor state, and upper-semi state. The component-representative selected-label constructor lowers these fieldwise compatibilities to the component representative source itself: Lean applies the source's constructed-representative theorem to the selected q-choice and transports the resulting representative across the q-pilot theater alignment. The one-sided constructor now obtains that representative source from the one-sided component package's* `volumeSource`*. Its SHE-codomain successor derives q-pilot theater placement from the source-spine one-column law plus constructed SHE codomain alignment. The source-spine-selected constructor builds the one-sided component source using the source-spine selected q-choice, making that synchronization definitional. The source-spine-data constructor also projects the theta possible-image source and gluing torsor from the source spine. Its output-flag-calibrated packet audit verifies that the constructed primitive packet keeps the same selected q-choice, projected bridge component, Step (x) log-Kummer source, and target-region formula source while deriving the subclaim component internally. The primitive-packet bridge then takes the multiradial record, source spine, record/source image agreement, and SHE codomain/one-column alignment from the same primitive packet. The component-kernel bridge constructs the one-sided component volume source from the Frobenioid divisor-column representative kernel. The generated full-label package reindex source then pulls the concrete pre-ledger/source package back along the kernel projection, assuming only a selected generated q-choice above the concrete chosen output. The remaining bridge-alignment work at this boundary is now that selected-q* *lift and generated record/image construction, rather than supplied q-pilot class fields, local states, one-column data, codomain equalities, a preassembled representative section, a direct constructed-representative equality, a direct generated-full-label representative projection, a source-provided generated q-choice, fieldwise selected-label compatibilities, an independent component representative source, a selected-q synchronization equality, an independent theta possible-image source, an independent gluing torsor, an independent record/source image-family synchronization, a monolithic one-sided component volume source, a raw q-pilot theater placement field, or an ambient coverage assertion for all concrete choices. The adjusted-log-volume direct-HDD route constructs the direct-HDD source from the record Ob3/Ob5 finite adjusted summand identity, so the normalized determinant equality is derived inside Lean rather than exposed as a separate route input. The gap-backed route lowers q-positivity and source normalization at the same boundary by projecting them from ; its direct-HDD audit records that the adjusted Ob3/Ob5 source is converted only after these side conditions have been read from the source gap. The hodge-calibrated variant also removes the direct theta/family-hull equality at this boundary: the equality is derived from for the bounded family-hull source projected from the adjusted Ob3/Ob5 determinant source. The measure-calibrated variant also removes the Ob3/Ob5-specific measure equality at this boundary: it is derived from the pre-ledger measure calibration to the product-hull holomorphic hull operator and the Ob3/Ob5/product-hull operator match. The product-hull-constructed variant lowers the latter match too: the Ob3/Ob5 adjusted source is constructed with the product-hull holomorphic hull operator, so the operator comparison is definitional instead of an independent compatibility field. The principal-product variant lowers the product-hull input itself to the Remark 3.9.5 principal presentation by scalar multiples of the local integer region; the ordinary product-hull source is projected internally. The constructor then derives the Ob3/Ob4 source and adjusted-summand equality from the theta-region-defined principal valuation-ball cover. The structured-SHE variant also derives the source-obligation gap from the structured Theorem 3.11/SHE bundle and side conditions. The summand-charted Hodge variant lowers the Hodge-family calibration boundary further: the input is the charted equality between the Hodge theta-monoid degree and the finite adjusted determinant summand sum, and the determinant, normalized determinant, and family-hull equalities are derived internally. The measure input is now lowered to , a pointwise pre-ledger calibration by the Remark 3.9.5 holomorphic-hull log-volume; Lean derives the region-measure equality used by the older constructor. The source-obligation gap has also been lowered on the qualitative side: constructs it from the strengthened structured-SHE Theorem 3.11 bundle and the side conditions, with the SHE alignment projected from the structured SHE/common-container context. Its audit records the common-container compatibility and equality with the existing structured-input obligation constructor. The remaining mathematical boundary is now the lower construction of the original common-container/SHE and log-Kummer compatibility data from primitive paper-side Hodge-theater and log-theta-lattice inputs. The norm-square boundary audit records the equality with the hardened endpoint. The closed-input audit records that this route is definitionally the underlying intrinsic route applied to the internally constructed non-Step-(xi) payload. The boundary audit records the norm-square determinant audit, the lower additive-Haar payload, the direct norm-square nonzero contradiction, and the contradiction obtained by replacing it with the legacy compact-open strict-positive unit-ball source at the same intrinsic projection, built on the underlying intrinsic route . The residue-module/unit-ball normalized restriction-calibrated source now projects to this norm-square calibration boundary via . This constructor builds the projected p-adic finite localized hull decomposition and reuses the restriction package; its remaining explicit lower inputs are the projected p-adic structure-sheaf/direct-summand synchronization* *and the selected nonzero norm-square summand. The new audit proves that this selected p-adic unit-ball norm is zero under the current normalization, so the projection is diagnostic; the additive-Haar inverse-dilation branch is not silently identified with the intrinsic unit-ball branch. The residue restriction-calibrated source now also projects to the unified inverse-dilation localized-adjusted source via . Its audit records the residue transported-factor handoff, the unified inverse-dilation boundary, strict selected Haar mass, positive normalized compact-open log-volume, and nonzero direct-summand norm on the inverse-base-prime additive-Haar side. This audit is now threaded through , so it is part of the restriction-calibrated public route evidence rather than a detached side certificate. Finite summation derives the Ob3-1 inequalities comparing the structure-sheaf log-volume with the localized bundle log-volume; the subtraction formula derives adjusted raw local positivity; positive determinant weights derive localized adjusted summand positivity; finite summation gives localized family-hull positivity. The theta scalar at the strict boundary is the environment theta divisor log-volume by definition. The public route now constructs the older Hodge-summand, Gaussian/Hodge, Hodge-evaluation, and Hodge-scalar sources internally from the Hodge canonical-degree divisor construction, audited evaluation, and localized adjusted-sum calibration, then derives the environment-theta/localized adjusted-sum equality; Lean derives the equality with the family-hull log-volume from the localized Ob3/Ob5 identity. This derived equality implies the family-hull/global calibration via theta/ordinary realified Frobenioid compatibility, and the localized determinant/family-hull equality then gives the determinant/global equality needed for the Gaussian-global nonzero proof. The Gaussian local-global Frobenioid restriction identity then proves the global product-formula equality before the product-formula audit is invoked. The strictest finite-extension public boundary now constructs the Step (xi)/Remark 3.9.5 paper trace from this same finite-extension, localized determinant, Hodge-evaluation, theta/family-hull, and Ob7 source chain, rather than accepting a standalone paper-trace input. Its calibrated variant also replaces the raw Hodge theta-monoid/localized-adjusted sum equality by the Hodge/family-hull equality, then derives the adjusted-sum equality through the localized Ob3/Ob5 identity. Its restriction-calibrated variant also replaces the placewise Hodge canonical-degree/local-prime equalities by an anchored local-global restriction source plus one global Hodge canonical-degree calibration. The newest additive-Haar strict-factor variant derives the selected nonzero norm from the concrete inverse-dilation Haar factor $p^{-1}O_K$, and its cover wrapper derives the localized hull-containment and volume-sum fields from a cover identity. Its normalized-structure-sheaf variant derives the direct-summand handoff from the additive-Haar compact-open normalized log-volume identity. The intrinsic finite-extension and residue-module variants now preserve this lowered handoff through the unit-ball Haar-character and residue quotient constructors; the unit-ball compact-open factor variant separately identifies the distinguished structure-sheaf summand with its valuation-unit-ball normalized compact-open factor before invoking the older restriction-calibrated route. The principal-product direct-HDD route audit now also exposes , keeping the same restriction-calibrated goal evidence visible at the current strongest product-hull public surface. The structure-sheaf-positive, raw-positive, summand-positive, weighted-partition, explicit raw-ratio, and one-summand routes remain as compatibility layers. The Step (xi) source and obligation package are projected internally, so the public boundary no longer takes a standalone `StepXIHullDeterminantObligations`, a preassembled paper-derived Step (xi) source, or an independent normalized-determinant upper-bound estimate. Its principal-product hull audit now exposes the bounded-family upper-semi quotient source: nonempty quotient-indexed possible regions have the same collapsed quotient image, and equality-orbit choices retain the matching region, log-volume, quotient image, and hull-containment endpoint. Its determinant audit also threads the localized Ob5* *quotient/determinant bridge, so this quotient source and the bounded-family determinant endpoint are checked against the same constructed family-hull data. Separately, the target-charted Hodge/*IPL *constructor-built route now has its own Ob7 product-formula handoff: the weighted determinant prime-strip product formula is synchronized with the same constructor-built Step (xi) determinant source, Lean constructs the coric log-Kummer source, and the resulting audit packages Ob5, exact-Ξ Ob6, and Ob7 at one route boundary. At the localized Ob7 constructor boundary, the tensor-power upper bound is derived from the theta/family-hull equality and localized determinant comparison, not supplied beside the product-formula source. The public product-formula route now also has a determinant- calibrated theorem whose audit boundary uses this constructor directly; the older tensor-bound theorem remains as a compatibility layer. Its evidence audit exposes the finite summand product formula, the determinant/global prime-strip equality, the theta-derived tensor-power bound, and the retained source/target/log-Kummer-column identities at this same public boundary. The downstream raw-adjusted, common-place, diagonal restriction, and environment-theta public wrappers likewise invoke this determinant-calibrated constructor directly, so their Ob7 trace no longer carries a separate inline tensor-power proof. The local product-formula and coric prime-strip data remain explicit lower source inputs. The looser route theorems remain as compatibility layers underneath this promoted boundary. The C_Θ handoff is now tied to the constructed Step (xi) source as well: identifying the local canonical Step (xi) scale with the finite-place scale in the IUT IV summation source lets Lean derive the global C_Θ dominance from the IUT IV plus-one cancellation, including the constructor-built exact-theta source.*

Theorem 5.19 (Extend SHE through analytic Step (xi))**.** *The legacy finite model checks a* 1*-column transport law named* SHE*. The source-faithful* SHE *algorithm and its extension through the common-container, holomorphic-hull, and determinant objects remain open.*

Proof. The source proof must show that the relevant operations are valid while the q-pilot log-volume associated to the input prime strip remains fixed. □

Theorem 5.20 (Extend IPL through analytic Step (xi))**.** *The legacy finite model keeps two typed histories distinct. The source theta link, its prime strips, the* IPL *algorithm, and their extension through the Step (xi) analytic objects remain open.*

Proof. The value-group and unit portions of the link must be tracked together. Using different lattice arrows for the two portions would destroy the simultaneous comparison needed in Step (xi). □

Theorem 5.21 (Extend APT to analytic descent)**.** *The legacy finite model checks translations, a quotient, and a preorder. The source* APT *algorithm, genuine indeterminacy actions, and analytic descent to the reorganized "3.11.5" input remain open.*

Proof. This is the proof content of the first, second, and third triangles in Mochizuki's 2026 decomposition of the Stage 1 implication. □

Theorem 5.22 (Construct hull plus determinant)**.** *The paper-derived Step (xi) source should be lowered further to actual holomorphic-hull and determinant operations corresponding to Remark 3.9.5 and Step (xi) of IUT III, including the analytic construction of the source-level possible-image witnesses from the multiradial/log-theta construction rather than taking them as lower source fields.*

Proof. This must turn localizations of arithmetic vector bundles into determinant line data whose log-volume can be compared with the q-pilot line-bundle log-volume. □

Theorem 5.23 (Construct log-Kummer upper-semi compatibility). *The log-Kummer correspondence in the* 1*-column should supply the nonarchimedean upper-semi compatibility currently consumed by the strongest packet-calibrated route.*

Proof. The current Lean route already uses a finite Kummer-plus-forgetting chain to derive the source-side equality. The remaining task is to replace the packet-normalized target calibrations by source-level log-Kummer constructions.

The current CI audit freezes the active source-reconstruction trust boundary as . Its fifteen entries are exact IUT I–III construction obligations consumed by the assembled source Theorem 3.11 column boundary. Lean also proves : the active manifest contains none of the old packet, `sourceWithSymmetry`, `compactOpenRealizedExactSource`, or `hodgeEvaluation` adapter inputs. The same-index q-normalized finite route remains separately recorded as ; it is a downstream regression and Corollary 3.12 bookkeeping surface, not evidence for M1–M3 paper clauses. The p-adic finite localized hull-vector-bundle source is no longer an independent boundary item on this route: Lean derives it from the finite-extension restriction-calibrated local arithmetic handoff before comparing the weighted determinant and IUT IV residual data. The Theorem 3.11 source audit now also records the concrete Ind2 local-tensor transport source: and . These declarations expose the fixed Hodge-theater/history/theta-pilot coordinate, procession and upper-semi transports, direct-summand equality, normalized log-volume preservation, equality-quotient compatibility, and theta-pilot class invariance. The canonical C_Θ-scale equality is also no longer exposed as a public field at this boundary; Lean derives it from the q-normalized theta/log-volume identity and q-pilot positivity. At the constructor-built valuation-ball Record-Ob3/Ob5 boundary, the same scalar conversion is now checked against the adjusted-summand log-volume, while the determinant equality remains projected from the constructor-built Ob3/Ob4 source. The family-hull q-normalized source now also builds the synchronized named-HDD route input from the possible-image witness before the compact-open routes consume it. The residual-frontier manifest contains 320 checked entries: 314 derived entries and 6 constructed countermodel diagnostics. It now records the source-level principal pointwise log-shell global C_Θ audit and its exact-Ξ consequences, which derive the family-hull log-volume equality and no-Ind3-generator conclusion without exposing the compact-open log-Kummer correspondence at that consequence boundary. It also records the pointwise log-Kummer projection-preservation audits: the compact-open map, correspondence, image, possible-region, and reconstructed pointwise sources keep the same principal valuation-ball source, theta labels, and cover places. It also records the direct pointwise construction of the compact-open realized-exact source and compact-open map source via the log-shell exact source and compact-open Haar normalization, together with the preservation audit for the realized theta-region equality and selected principal hull. It also records the determinant-factored pointwise projection audit, which reconstructs the older summand formula from determinant/Hodge calibration and the record-native determinant decomposition into adjusted summands. It also records the lower determinant-free pointwise measure/summand source audit and the full summand-charted determinant-source audit ; together these expose the Step (xi) measure/log-volume identity, charted Hodge theta equality, finite adjusted-summand identity, determinant log-volume equality, normalized determinant equality, family-hull equality, and theta-signed equality at the pointwise calibration boundary. It also records the reusable Step (xi)-to-case-procession arithmetic-degree bridge: when both sides equal $a_v(D_v + C_v)$, Lean derives the direct local comparison. It further records the p-adic cancellation constructor deriving arithmetic formula matching from Step (xi) synchronization and $E_v = \delta_v + M_v$. The selected p-adic endpoint is now lower than that surface: the public selected endpoint consumes a p-adic-normalized case-defined arithmetic-gap source carrying the Step (xi)-to-case-procession equality and the normalized-place arithmetic-minus-main +1 inequality, while the selected Theorem 1.10 ledger fixes the normalized p-adic place as distinguished. The selected case-defined Haar-defect source now derives this selected arithmetic-gap source directly: the selected equality gives $StepXI_v = caseProcession_v$, the p-adic unit-ball arithmetic identity gives $upper_v = StepXI_v + defect_v + M_v$, and the normalized Haar defect gives $1 \leq defect_v$. The stronger normalized processional-estimate package is routed through this lowering after producing the selected case-defined equality. The p-adic-normalized case source used by the Haar-residual route is now also projected from the lower processional pointwise-residual source after identifying its distinguished residual place with the p-adic normalized place. Thus the unit margin is supplied by the residual lower-bound theorem instead of the stronger processional-estimate ledger. The packet-facing p-adic constructor composes this projection internally, so the route can consume the pointwise local-analytic IUT IV source, expanded p-adic Haar residual identity, processional pointwise-residual source, and normalized-place alignment directly. The same p-adic-normalized case source is first projected from the lower processional arithmetic-gap theorem by retaining its Step (v)/(vi)/(vii) normalized local-log-volume identities and transporting the distinguished place to the p-adic normalized place. This alignment is now internalized in a p-adic-normalized arithmetic-gap source indexed over the reconstructed Haar-residual object itself, so the distinguished place is definitionally the p-adic normalized finite place. The arithmetic-gap input is threaded through the p-adic unit-ball localized public route without exposing a separate distinguished-place equality: Lean derives the pointwise local-analytic localized source and expanded p-adic Haar-residual identity from the p-adic unit-ball source, constructs the p-adic-normalized case source from the p-adic-normalized arithmetic-gap source, and then enters the signed endpoint. The canonical selected p-adic unit-ball/local-analytic public endpoint is now lower than the prime-error-split/case-local formula handoff, the older reconstructed Haar-defect handoff, the direct Step (xi)-to-case-procession handoff, and the stronger normalized-estimate package: Lean consumes the selected p-adic arithmetic-gap source and derives the normalized-place kind from the selected Theorem 1.10 ledger. The earlier arithmetic-degree route remains checked as a comparison path: it derives the same selected case-defined equality from the p-adic arithmetic formula $StepXI_v = a_v(D_v + C_v)$, the three Theorem 1.10 Step (v)/(vi)/(vii) local arithmetic-degree clauses, and the local-evaluation alignment. Lean first recombines the distinguished exact-procession, nondistinguished zero, and archimedean metric-container clauses into the all-place case-procession arithmetic-degree formula, then reconstructs the selected p-adic source internally. Separately, the general valuation-ball Theorem 1.10 source now proves the corresponding arithmetic-degree upper-bound route: the distinguished exact procession, nondistinguished zero, and archimedean metric-container bounds recombine to $\mathrm{caseProcession}_v \leq a_v(D_v + C_v)$, and the stronger equality source maps to this estimate source by *le_of_eq*. The older projected-procession/arithmetic-minus-main comparison remains the independent upper-bound chain, while this new source records the paper-local arithmetic-degree estimate directly. The additive-Haar valuation-ball arithmetic-degree-controlled local source now projects to this canonical bound source; the distinguished and archimedean comparisons are reused, and the nondistinguished zero bound is derived from nonnegativity of $D_v + C_v$ and the arithmetic-degree coefficient. This bound now lowers through the Record-Ob3/Ob5 valuation-ball source, the local-degree formula source, the controlled component source, and the p-adic-defect/main controlled source, so the pointwise Theorem 1.10 bound and residual package live at the same checked lower source boundary. The same source now projects directly to the canonical local arithmetic-degree residual package: the Step (xi) calibration gives $StepXI_v = a_v(D_v+C_v)$, the IUT IV/p-adic defect source gives $E_v = \delta_v + M_v$, and Lean derives $StepXI_v + \delta_v = a_v(D_v + C_v) + E_v - M_v$ together with the finite Haar-defect lower bound. The p-adic-normalized arithmetic-gap branch has now been lowered to the same case-defined local value: Lean reconstructs the ordinary normalized arithmetic-gap source, normalized case source, lower-weight residual package, Haar-residual handoff, and pointwise/p-adic unit-ball packet constructors from a single Step (xi) equality against the case-defined procession plus the normalized-place arithmetic-minus-main +1 inequality. The same case-defined arithmetic-gap source now reaches the signed p-adic unit-ball/local-analytic public endpoint by reconstructing the older normalized arithmetic-gap record internally. When the stronger case-defined p-adic-normalized processional estimate source is available, Lean derives that arithmetic-minus-main +1 inequality by composing its projected-procession +1 bound with Theorem 1.10's local arithmetic-divisor gap estimate. The case-defined estimate source itself can now be constructed from the lower local-analytic procession-bound residual source plus the case-defined Step (xi) equality: Lean transports the lower Step (xi) +1 procession-bound inequality across that equality, rather than taking the case-procession +1 field independently. The same case-defined equality now reconstructs the lower processional pointwise-residual source from the pointwise defined residual package: Lean projects the three Step (v)/(vi)/(vii) procession-normalization identities from the case function, while the distinguished unit margin remains the pointwise residual theorem. The Haar-defect layer is lowered in the same way: from the pointwise Haar-defect residual source and the same case-defined equality, Lean reconstructs the processional Haar-defect residual source and its downstream lower-weight/local arithmetic-degree endpoints, without carrying the three identity families independently. The local C_Θ constructor now consumes this lower pointwise-Haar/case-procession package directly: Lean rebuilds the processional Haar-defect source internally before entering the established localized C_Θ handoff. The selected p-adic public branch now consumes the selected p-adic-normalized case-defined arithmetic-gap source and reconstructs the selected case-defined Haar-defect source before invoking the same endpoint. The p-adic public branch consumes the p-adic unit-ball localized Step (xi) source, a selected distinguished-place local-analytic Theorem 1.10 ledger at the p-adic Haar normalized place, the local-evaluation equality, and a selected case-defined reconstructed p-adic Haar-defect source. Lean derives the p-adic normalized-place kind from the selected local-kind construction, then reconstructs the specialized p-adic-normalized case-defined estimate source, expanded p-adic Haar-residual identity, p-adic Step (xi) synchronization, weaker case-defined case source, older normalized-case record, pointwise residual theorem, generic Haar-defect source, and pointwise local-analytic localized source internally. The same normalized case source is now also checked as the weakening of the full local-analytic processional estimate source: after normalized-place alignment, Lean forgets the distinguished +1 processional upper-bound field and retains the Step (v)/(vi)/(vii) normalized local-volume identities. The case-defined processional source now fixes that local value directly from the Theorem 1.10 case split, so the ordinary and p-adic-normalized processional estimate constructors no longer expose the three Step (v)/(vi)/(vii) identity families as independent fields on this branch. The normalized Haar-residual case source has now been lowered to the same case-defined value: Lean reconstructs the older distinguished, nondistinguished, and archimedean identity families from a single Step (xi) equality against the case-defined value plus the normalized-place kind. A reconstructed p-adic Haar-defect variant now indexes the processional residual data over the localized p-adic unit-ball packet and matching Theorem 1.10 local-evaluation equality, and the selected case-defined reconstructed variant now indexes the valuation-ball local-analytic source by the p-adic Haar normalized place itself. The selected local-kind construction proves that this place is distinguished, so the current public frontier exposes only the equality against the Theorem 1.10 case-defined local value; normalized-place distinguishedness and local case expansion are reconstructed instead of public equality-family inputs. The stronger case-bounded IUT IV source is now a constructor into the local arithmetic-degree residual package, not the canonical public residual input. The audit includes the one-place diagnostic , which shows that a detached p-adic finite scale can satisfy neither the handoff-derived theta identity nor the endpoint comparison automatically. It also includes , showing that the signed $q \leq \theta$ comparison is essential for the $C_\Theta \geq -1$ lower-bound projection: q-positivity plus the IUT IV-style upper bound alone permit $C_\Theta < -1$. Finally, shows why the local arithmetic-degree residual lower bound cannot simply be deleted. In the current manifest this package is discharged by the p-adic-defect/main Record-Ob3/Ob5 valuation-ball route, which projects directly to the canonical residual package through . The remaining IUT IV source obligation at this boundary is the lower construction of that valuation-ball Record-Ob3/Ob5 local-estimate source from the full paper-level local-estimate proof. The next source-level replacements should remove further entries from this manifest rather than introducing more public endpoint variants. The public `Iut.Stage1.IUTStage1StepXI` module is now a re-export facade over `Iut.Stage1.IUTStage1StepXI.Core`, so further canonical-route slices can be split out while preserving existing names. □

Theorem 5.24 (Preserve comparison levels)**.** *The source construction must preserve the distinction between representative j^2-coordinates, sign-quotient full labels, averaged finite Gaussian data, ordered real-line transports, and the final hull/log-volume endpoint.*

Proof. This is the formal defense against an illicit collapse of comparison levels. It is also the place where the Scholze–Stix objection should be tested most carefully in Lean. □

Chapter 6

After Corollary 3.12

6.1 IUT IV and arithmetic consequences

Theorem 6.1 (IUT IV consumes the Corollary 3.12 input)**.** *The log-volume estimates in IUT IV should consume a Corollary 3.12-shaped bound and turn it into the later Diophantine inequalities.*

Proof. The repository treats this as conditional real algebra for now. This blueprint does not yet expand the IUT IV dependency graph. □

Corollary 6.2 (High-level arithmetic consequences)**.** *The claimed downstream consequences include Vojta-type inequalities for hyperbolic curves, abc, and Szpiro-type statements.*

Proof. These are roadmap nodes only. They should remain high-level until the Stage 1 source route has been made substantially less conditional. □

6.2 Engineering next steps

Proposition 6.3 (Refactor the large source module)**.** *The large Stage 1 source file should be split into reviewable modules for finite labels and Gaussian data, Step (xi) hull and determinant endpoints, Step (xii) local Frobenioid shifts, and downstream ordered-real algebra.*

Proof. This should be a mechanical refactor preserving theorem names where possible and rebuilding after each move. □

Proposition 6.4 (Keep the blueprint synchronized)**.** *Each time a source-facing record is replaced by a construction, the corresponding blueprint node should move from to a Lean-tagged formal statement, and the dependency graph should be rebuilt.*

Proof. The blueprint should become the public map of which mathematical obligations are still assumptions and which have been discharged in Lean. □

Chapter 7

References

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